CPYR Agent Communication

JULY 3

CPYR

V1.1

CPYR Agent Streams

Stream definition

Communication on the Ethernet are defined by stream numbers at the 19th byte of the Header, This is a special tailored Header that the CPYR Agent uses in order to communicate with the CPYR Box.

CPYR Agents collect the data from different SW components allocates them to different Streams. To enable any future expansion stream version id has been given, and due to the limitation of the UDP frame size limitation 1490 Byte, stream chunk id was given in order to enable fragmentation of the messages.
Most important stream is the SW version info stream which preserves all the SW components which will be sent every 1 Sec.

SW_Version_Info(0x102)- ST Number (72) - ST Version (1)

Signal Name	Description	Startbit	Length [Bit]	Value Type	Factor	Offset	Minimum	Maximum	Unit
CPYR agent_Verison_Variant	Major release variant to main processor	0	8	Unsigned	1	0	0	255	N/A
CPYR agent_Version_Major	Major updates which will not support backwards compatibility (ie. SPI updates)	8	8	Unsigned	1	0	0	255	N/A
CPYR agent_Version_Minor	Incremental updates	16	8	Unsigned	1	0	0	255	N/A
CPYR agent_Version_Field	Minor updates and re-releases. >100 decimal indicates a US release.	24	8	Unsigned	1	0	0	255	N/A
Algorunner_Verison_Variant	Major release variant to Algo processor	32	8	Unsigned	1	0	0	255	N/A
Algorunner_Version_Major	Major updates which will not support backwards compatibility (ie. SPI updates)	40	8	Unsigned	1	0	0	255	N/A
Algorunner_Version_Minor	Incremental updates	48	8	Unsigned	1	0	0	255	N/A
Algorunner_Version_Field	Minor updates and re-releases.	56	8	Unsigned	1	0	0	255	N/A
ETSEL_Version_Major	ETSEL Major Version	64	8	Unsigned	1	0	0	255	N/A
ETSEL_Version_Minor	ETSEL Minor Version	72	8	Unsigned	1	0	0	255	N/A
ETSEL_Version_Revision	ETSEL Revision Version	80	8	Unsigned	1	0	0	255	N/A
TAPS_Version	OCTAP Version	88	32	IEEE Float	1	0	0	3.40E+38	N/A
Tracker_Version_Major	Tracker Major Version	120	8	Unsigned	1	0	0	255	N/A
Tracker_Version_Minor	Tracker Minor Version	128	8	Unsigned	1	0	0	255	N/A
Tracker_Version_Revision	Tracker Revision Version	136	8	Unsigned	1	0	0	255	N/A
SPP_Version_Major	SPP Major Version	144	8	Unsigned	1	0	0	255	N/A
SPP_Version_Minor	SPP Minor Version	152	8	Unsigned	1	0	0	255	N/A
SPP_Version_Revision	SPP Revision Version	160	8	Unsigned	1	0	0	255	N/A
OTP_Version_Major	OTP Major Version	168	8	Unsigned	1	0	0	255	N/A
OTP_Version_Minor	OTP Minor Version	176	8	Unsigned	1	0	0	255	N/A
OTP_Version_Revision	OTP Revision Version	184	8	Unsigned	1	0	0	255	N/A
VISIONCO_SPI_API_Version	VISIONCO SPI API Version	192	32	IEEE Float	1	0	0	3.40E+38	N/A
ME_SW_Version	Vision Software Version	224	32	IEEE Float	1	0	0	3.40E+38	N/A
MAINRADAR_SW_Release_Revision	hardware version change	256	8	Unsigned	1	0	0	255	N/A
MAINRADAR_SW_Promote_Revision	milestone updates for a certain hardware	264	8	Unsigned	1	0	0	255	N/A
MAINRADAR_SW_Field_Revision	stage change in a milestone	272	8	Unsigned	1	0	0	255	N/A
MAINRADAR_SW_Field	minor change in a stage	280	8	Unsigned	1	0	0	255	N/A
MAINRADAR_PBL_Release_Revision	milestone updates	288	8	Unsigned	1	0	0	255	N/A
MAINRADAR_PBL_Promote_Revision	stage change	296	8	Unsigned	1	0	0	255	N/A
MAINRADAR_PBL_Field_Revision	minor updates	304	8	Unsigned	1	0	0	255	N/A
VSE_CS_Version_Major	Curvature and Sideslip Estimation Algorithm Major Version	312	8	Unsigned	1	0	0	255	N/A
VSE_CS_Version_Minor	Curvature and Sideslip Estimation Algorithm Minor Version	320	8	Unsigned	1	0	0	255	N/A
VSE_CS_Version_Field	Curvature and Sideslip Estimation Algorithm Field Version	328	8	Unsigned	1	0	0	255	N/A
VSE_YRE_Version_Major	Yaw Rate Estimation Algorithm Major	336	32	Unsigned	1	0	0	2.15E+09	N/A

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Data Link and Network Layer

Instrumented ECUs with an installed Ethernet Adapter may be used for Ethernet communication.

The CPYR agent instrumentation shall communicates at BRR 100 Mbps Full-Duplex or 1Gbit at the (RJ45)

Inst_Bus. The Ethernet CPYR agent network specification shall be defined as below:

- Internet Protocol version 4.
- Subnet mask for the network is 255.255.255.0.
- External Device has a fixed IP address of 10.2.0.10

A Category 5e or better cable must be used when connecting nodes with a maximum length of 100 m.

Transport Layer

For the instrumentation messages defined in this document, The CPYR agent shall use the User Datagram Protocol Transport (UDP) for sends and receives. CPYR agent is responsible for adding the UDP header and send the packets.

The Source Port field identifies the sender of the message.
The Destination Port field identifies the receiver of the message. See Application Layer section for definition of allowed ports.
The Length field indicates the length of the datagram including the header bytes.
The Checksum field is used for error checking and includes the header and all data bytes.
The Data field shall contain the Application message.

User Datagram Protocol (UDP)

An Ethernet device driver shall be implemented and provide an abstraction of the User Datagram Protocol (UDP).
UDP device driver configuration shall support setting the UDP broadcast port used for broadcasting UDP datagrams.
The UDP device driver shall allow transmission of broadcast datagrams after the driver is initialized.
An API service shall be implemented that allows the UDP server to send a datagram to the broadcast port. The server may reject service requests to send a broadcast datagram when the previous broadcast datagram has not been sent.
It shall be possible to re-initialize the UDP device driver. Re-initialization may be used when attempting to recover from a fault condition.

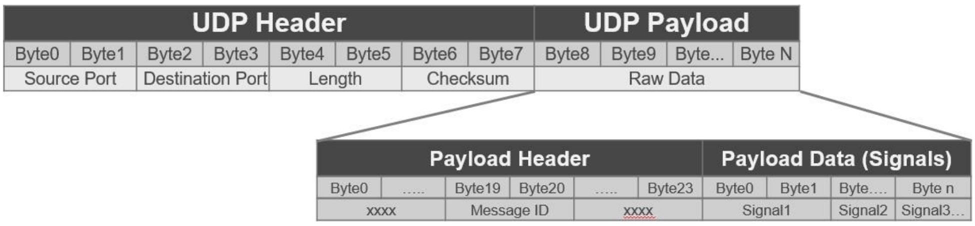
Application Layer

Communication from the CPYR agent application software is accomplished by using the

port. The table below specifies the Port address used.

Port Name	Port Address
CPYR agent Port	50014 C35E
External Port(Server)	50015 C35F

The content of the UDP data payload is likely to have two parts, header block and the data block.
The payload header block contains the message ID (Byte19 and Byte20) Stream Number and Stream Version and the data block contains the message signals as per the Inst_Ethernet dbc.[3]



All data stored in the header shall be in little Endian format (i.e., Least Significant Byte (LSB) first).
All signals data in the messages shall be in little Endian format (i.e., Least Significant Byte (LSB) first).

Payload Header

Applicationmessageshavea24-byteheaderwiththefollowingformat:LayerByte... **Application messages have a 24-byte header with the following format:**

Layer	Byte Offset	Size (Bits)	Signal	Description
Application	0	16	versionInfo	Version of this header specification. The lower byte contains the size the message header. The upper byte contains the record version number.
	2	16	sourceTxCnt	Source Transmit Counter. Incremented on every transmission from the source Application.
	4	32	sourceTxTime	Source Transmit Time. Timestamp (ms) when packed was queued for transmission.
	8	8	sourceInfo	A constant identifying the sending application [0:63].
	9	8	reservedSrc1	Reserved for future use. Set to 0.
	10	8	reservedSrc2	Reserved for future use. Set to 0.
	11	8	reservedSrc3	Reserved for future use. Set to 0.
Process	12	32	streamIndex	Stream-specific index used for data retrieval (e.g. Look Index for radar).
	16	16	streamDataLen	Stream data payload size (bytes). Size for each stream must be constant.
	18	8	streamTxCnt	Stream Transmit Counter. Increments on each transmission of associated stream number.
	19	8	streamNumber	Stream number One source may transmit multiple, asynchronous streams.
	20	8	streamVersion	Stream data structure format/version (how to interpret data).
	21	8	streamChunks	A Stream may be transmitted from the process layer as M equal-size chunks that can be reconstructed on the receive side. Set to 0 if the transmission is not part of a series of chunks (normal mode). Set to M (>=2) to indicate stream is sent as a series of M chunks.
	22	8	streamChunkIdx	Set to 0 when streamChuns is 0; otherwise, represents this chnk' s position in the reconstruction queue [0:(streamChunks-1)].

	23	8	reservedStr3	Reserved for future use. Set to 0.

Ethernet Application Messages

The following table lists all the Ethernet application messages specified in this document.

Message Name	Stream Number	DLC(Byte)	Source Node	Receiver Node	Update Rate (ms)	Description
VISIONCO_BASE_DIAG_STATUS_MSG	40	91	AGENT	External	55.5	VisionCo4 base diagnostic status message.
VISIONCO_APP_DIAG_STATUS_MSG	41	52	AGENT	External	55.5	VisionCo4 application diagnostic status message.
VISIONCO_DYNAMIC_CAL_MSG	42	46	AGENT	External	55.5	VisionCo4 Service Point Calibration (SPC) and AutoFix output.

VISIONCO_TAC_CAL_MSG	43	45	AGENT	External	55.5	VisionCo4 Target Auto-Calibration (TAC) output.
VISIONCO_VIS_ROAD_DATA_MSG	44	524	AGENT	External	55.5	VisionCo4 road detection data message.
VISIONCO_VIS_LIGHT_SENSOR_DATA_MSG	45	844	AGENT	External	55.5	VisionCo4 AHBC function data message.
VISIONCO_VIS_TRAFFIC_SIGNS_MSG	46	524	AGENT	External	55.5	Traffic signs as detected by vision subsystem.
VISIONCO_VIS_AEB_INFO_MSG	47	36	AGENT	External	55.5	AEB decisions (for braking or warning) based on vision.
VISIONCO_VIS_OBSTACLES_MSG	48	3320	AGENT	External	55.5	Vision objects as detected by vision subsystem.
VISIONCO_VIS_FAILSAFE_MSG	49	34	AGENT	External	55.5	Vision failsafes
VISIONCO_VIS_SRP_MSG	50	37	AGENT	External	55.5	Suspension Road Preview
VISIONCO_DIAGNOSTIC_MSG	51	920	AGENT	External	55.5	The application diagnostics protocol is a message that contains information about the current status of the application (internal states, diagnostics, errors, etc.). The message is being sent from VisionCo at the end of every image processing loop.
VISIONCO_BOOT_STATUS_MSG	52	44	AGENT	External	Boot Time	Boot Status message
CPYR AGENT_VEH_STATE_MSG	53	65	AGENT	External	15	CPYR agent vehicle state message.
VISIONCO_TIMESYNC_MSG (0xE8)	54					
VISIONCO_SAFETY_MSG (0xE7)	55					
Init_Params_Vision (0x92)	56					
Intrinsic Calibration (0x84)	57					
Init_IPB (0x78)	58					
Init_PCW (0x79)	59					
BT_TPP_SPP	60	48				OBJECTSENSE Signals Received from Algorunner
BT_TPP_TAD	61	460				OBJECTSENSE Signals Received from Algorunner
BT_TPP_AutoHitch	62	64				OBJECTSENSE Signals Received from Algorunner
BT_TPP_DriveHistory	63	40				OBJECTSENSE Signals Received from Algorunner
BT_TPP_Localization_Results	64	32				OBJECTSENSE Signals Received from Algorunner
System_Power_Status	70	125	AGENT	External	5	Reports the power status that happen at the 5 ms rate
Electrical_Status	71	24	AGENT	External	25	Reports the status of SW & LEDs and the RAW Data of ADC Inputs
_SW_Version_Info	72	22	AGENT	External	1000	Reports SW version numbers
CFG_Parameters	73	148	AGENT	External	50	Reports CFG Parameters
Veh_Estimation_Compensation	74	19	AGENT	External	10	Reports information used for resim
Active_Fault_Status	75	24	AGENT	External	50	Active fault status bytes 0 to 31
Latched_Fault_Status	76	24	AGENT	External	50	Latched fault status bytes 0 to 31
History_Fault_Status	77	24	AGENT	External	50	History fault status bytes 0 to 31
CPYR AGENT_SPI_Debug	78	8	AGENT	External	1000	Contains debug information from the SPI driver.
_Temperature	79	10	AGENT	External	100	Message with the CPYR AGENT sensors temperatures: imager, MIPS, VMP, Imager, VisionCo4, IMAGE2P, Algorunner and thermal status
OBJECTSENSE_Signals	80	43	AGENT	External	25	Reports information used for HIL testing

81	SOD_Signals	81	25	AGENT	External	50	Reports information used for HIL testing
82	Reserved						
..	Reserved						
99	Reserved						
100	cpu core throughput	100					cpu core throughput metric
101	ARTM Module	101					Status of tasks periodicity and scheduling
110	AFL	110				40	feature debug streams
111	LKS	111				40	feature debug streams
112	L25	112				40	feature debug streams
113	TSR, WWA, IACC	113				20	feature debug streams
114	TA_Veh	114				40	feature debug streams
115	FSM	115				40	feature debug streams
116	ACC	116				40	feature debug streams
117	HA	117				40	feature debug streams
118	CHA	118				40	feature debug streams

CPYR AGENT Application Messages

System_Power_Status

-System_Power_Status(0x100)-STNumber(70)-STVersion(1) **This message will be used to test functionalities.**

Message System_Power_Status (0x100) shall be implemented as follows:

SignalName	State Description	StartBit	Size	Sign	Factor	Offset	Min	Max
Rear_USSs_Iout	Current representative of sensed voltage of Rear USS	0	32	@1+	1	0	0	0
Heater_Iout	Current representative of sensed voltage of Heater	32	32	@1+	1	0	0	0
RVC_Iout	Current representative of sensed voltage of RVC	64	32	@1+	1	0	0	2000
POC_LDO1_FVC_Current_SENSE_Voltage	Scaled engineering value of FVC LDO1 regulater current sense voltage	96	32	@1+	1	0	0	25
POC_LDO1_RVC_Current_SENSE_Voltage	Scaled engineering value of RVC LDO1 regulater current sense voltage	128	32	@1+	1	0	0	25
POC_LDO2_RH_Current_SENSE_Voltage	Scaled engineering value of RH Mirror LDO2 regulater current sense voltage	160	32	@1+	1	0	0	25
POC_LDO2_LH_Current_SENSE_Voltage	Scaled engineering value of LH Mirror LDO2 regulater current sense voltage	192	32	@1+	1	0	0	25
POC_LDO3_FWC_Current_SENSE_Voltage	Scaled engineering value of FWC LDO3 regulater current sense voltage	224	32	@1+	1	0	0	25
LKS_SW_Voltage	Lane Keeping Switch Voltage	256	32	@1+	1	0	0	25

APA_SW_Voltage_AN1	Active Park Assist Switch Voltage from ADC AN1	288	32	@1+	1	0	0	25
APA_SW_Voltage_AN24	Active Park Assist Switch Voltage from ADC AN24	320	32	@1+	1	0	0	25
BPA_SW_Voltage	Base Park Aid Switch Voltage	352	32	@1+	1	0	0	25
View_SW_Voltage	Surround View Switch Voltage	384	32	@1+	1	0	0	25
L_BLIS_OUT_Voltage	Left BLIS_OUT LED Voltage	416	32	@1+	1	0	0	25
R_BLIS_OUT_Voltage	Right BLIS_OUT LED Voltage	448	32	@1+	1	0	0	25
PARK_AID_OUT_Voltage	PARK AID_OUT LED Voltage	480	32	@1+	1	0	0	25
SIDERADAR_RR_Current_SENSE_Voltage	Scaled engineering value of RR SIDERADARs current sense voltage.	512	32	@1+	1	0	0	3.3
SIDERADAR_LR_Current_SENSE_Voltage	Scaled engineering value of LR SIDERADARs current sense voltage.	544	32	@1+	1	0	0	3.3
SIDERADAR_LF_Current_SENSE_Voltage	Scaled engineering value of LF SIDERADARs current sense voltage.	576	32	@1+	1	0	0	3.3
SIDERADAR_RF_Current_SENSE_Voltage	Scaled engineering value of RF SIDERADARs current sense voltage.	608	32	@1+	1	0	0	3.3
FLR_Current_SENSE_Voltage	Scaled engineering value of Forward Looking Radar current sense voltage.	640	32	@1+	1	0	0	3.3
Heater_Current_SENSE_Voltage	Scaled engineering value of Heater current sense voltage.	672	32	@1+	1	0	0	3.3
SIDERADAR_RR_PWR_Voltage	SIDERADAR RR power Voltage	704	32	@1+	1	0	0	25
SIDERADAR_LR_PWR_Voltage	SIDERADAR LR power Voltage	736	32	@1+	1	0	0	25
ALGORUNNER_3V3_Voltage	ALGORUNNER 3V3 Voltage Supply	768	32	@1+	1	0	0	6.6
ALGORUNNER_1V8_Voltage	ALGORUNNER 1V8 Voltage Supply	800	32	@1+	1	0	0	1.8
ALGORUNNER_1V0_Voltage	ALGORUNNER 1V0 Voltage Supply	832	32	@1+	1	0	0	3.3
ALGORUNNER_1V35_Voltage	ALGORUNNER 1V35 Voltage Supply	864	32	@1+	1	0	0	3.3
VISIONCO_3V3_Voltage	VISIONCO 3V3 Voltage	896	32	@1+	1	0	0	6.6
VISIONCO_1V0_Voltage	VISIONCO_1V0_Voltage	928	32	@1+	1	0	0	6.6
VISIONCO_1V1_Voltage	VISIONCO 1V1 Voltage	960	32	@1+	1	0	0	6.6
VISIONCO_1V8_Voltage	VISIONCO 1V8 Voltage	992	32	@1+	1	0	0	6.6
SIDERADAR_LF_PWR_Voltage	SIDERADAR LF Power Voltage	1024	32	@1+	1	0	0	25
SIDERADAR_RF_PWR_Voltage	SIDERADAR RF Power Voltage	1056	32	@1+	1	0	0	25
FLR_PWR_Voltage	FLR Power Voltage	1088	32	@1+	1	0	0	25
HEATER_PWR_Voltage	HEATER Power Voltage	1120	32	@1+	1	0	0	25
HSD_R_ULTRASONIC_Current_SENSE_Voltage	Scaled engineering value of HSD Rear ULTRASONIC current sense voltage.	1152	32	@1+	1	0	0	3.3
HSD_F_ULTRASONIC_Current_SENSE_Voltage	Scaled engineering value of HSD Front ULTRASONIC current sense voltage.	1184	32	@1+	1	0	0	3.3
NTSC_ENCODER_P_Voltage	NTSC ENCODER P Voltage	1216	32	@1+	1	0	0	25
NTSC_ENCODER_N_Voltage	NTSC ENCODER N Voltage	1248	32	@1+	1	0	0	25
VBattU	VBattU	1280	32	@1+	1	0	0	25.5
K_3V3_Decay_Thresh	#N/A	1312	32	@1+	1	0	0	65535
k_Switches_Open_Max_Volt_Level	k_Switches_Open_Max_Volt_Level Max voltage that can be available at the ADC if switch is not pressed/Open assuming worst case vin=19v is 2.428 V	1344	32	@1+	1	0	0	19
k_LED_DRIVER_ON_Volt_Level	erence Voltage of LED ON Voltage	1376	32	@1+	1	0	0	25
k_LED_DRIVER_min_batt_volt	erence to Minimum Battery Voltage	1408	32	@1+	1	0	0	25

VISIONCO4_off_wait_ms	Running timer of VISIONCO4_OFF state.	1440	16	@1+	1	0	0	1000
VISIONCO4_core_dump_wait_ms	Running timer of VISIONCO4_core_dump state.	1456	16	@1+	1	0	0	1000
IMAGE2P_off_wait_ms	IMAGE2P_OFF state timeout timer	1472	16	@1+	1	0	0	1000
ALGORUNNER_off_wait_ms	ALGORUNNER_OFF state timeout timer	1488	16	@1+	1	0	0	1000
MAINRADAR_off_wait_ms	MAINRADAR_OFF state timeout timer	1504	16	@1+	1	0	0	1000
SIDERADAR_RFRONT_off_wait_ms	SIDERADAR_RFRONT_OFF state timeout timer	1520	16	@1+	1	0	0	1000
SIDERADAR_RREAR_off_wait_ms	SIDERADAR_RREAR_OFF state timeout timer	1536	16	@1+	1	0	0	1000
SIDERADAR_LFRONTInitTimeout	The CPYR agent doesn't receive a correctly formatted application CAN-FD messages in K_Max_Time_In_SIDERADAR_LFRONT_Init.	1552	16	@1+	1	0	0	1000
SIDERADAR_LFRONT_off_wait_ms	SIDERADAR_LFRONT_OFF state timeout timer	1568	16	@1+	1	0	0	1000
SIDERADAR_LREAR_off_wait_ms	SIDERADAR_LREAR_OFF state timeout timer	1584	16	@1+	1	0	0	1000
Front_USSs_Iout	Current representative of sensed voltage of Front USS	1600	16	@1+	1	0	0	2000
SIDERADAR_LR_Iout	Current representative of sensed voltage of LR SIDERADAR	1616	16	@1+	1	0	0	2000
SIDERADAR_LF_Iout	Current representative of sensed voltage of LF SIDERADAR	1632	16	@1+	1	0	0	2000
SIDERADAR_RR_Iout	Current representative of sensed voltage of RR SIDERADAR	1648	16	@1+	1	0	0	2000
SIDERADAR_RF_Iout	Current representative of sensed voltage of RF SIDERADAR	1664	16	@1+	1	0	0	2000
MAINRADAR_Iout	Current representative of sensed voltage of MAINRADAR	1680	16	@1+	1	0	0	2000
LH_MIRROR_Iout	Current representative of sensed voltage of LH_MIRROR	1696	16	@1+	1	0	0	2000
FVC_Iout	Current representative of sensed voltage of FVC	1712	16	@1+	1	0	0	2000
RH_MIRROR_Iout	Current representative of sensed voltage of RH_MIRROR	1728	16	@1+	1	0	0	2000
FWC_MIRROR_Iout	Current representative of sensed voltage of FWC_MIRROR	1744	16	@1+	1	0	0	2000
K_Max_Time_In_VISIONCO4_Init	Maximum time to remain in VISIONCO4_Init mode before resetting VISIONCO4 by going to VISIONCO4_OFF.	1760	16	@1+	1	0	0	65535
K_Min_Time_In_VISIONCO4_Off_ms	Minimum time in which the VISIONCO4 must remain off.	1776	16	@1+	1	0	0	65535
K_Max_Time_In_VISIONCO4_Core_Dump	Maximum time to remain in VISIONCO4_Core_Dump mode before resetting VISIONCO4 by going to VISIONCO4_OFF. The default value is set to 60 seconds to conform with the BPP spec from ME.	1792	16	@1+	1	0	0	65535
K_Max_Time_In_VISIONCO4_SOFT_RESET	#N/A	1808	16	@1+	1	0	0	65535
K_IMAGE2P_Retry_Counter_Max	Maximum number of retrys for initialization the IMAGE2P.	1824	16	@1+	1	0	0	65535
K_Max_Time_In_IMAGE2P_Init	Maximum time to remain in IMAGE2P_Init mode before resetting IMAGE2P by going to IMAGE2P_OFF.	1840	16	@1+	1	0	0	65535

K_Min_Time_In_IMAGE2P_Off_ms	Minimum time in which the IMAGE2P must remain off.	1856	16	@1+	1	0	0	65535
K_ALGORUNNER_Retry_Counter_Max	Maximum number of retrys for initialization the ALGORUNNER.	1872	16	@1+	1	0	0	65535

K_Max_Time_In_ALGORUNNER_Init	Maximum time to remain in ALGORUNNER_Init mode before resetting ALGORUNNER by going to ALGORUNNER_OFF. Currently it's set to 60000ms to satisfy DV testing.	1888	16	@1+	1	0	0	65535
K_Min_Time_In_ALGORUNNER_Off_ms	Minimum time in which the ALGORUNNER must remain off.	1904	16	@1+	1	0	0	65535
K_MAINRADAR_Retry_Counter_Max	Maximum number of retrys for initialization the MAINRADAR.	1920	16	@1+	1	0	0	65535
K_Max_Time_In_MAINRADAR_Init	Maximum time to remain in MAINRADAR_Init mode before resetting MAINRADAR by going to MAINRADAR_OFF.	1936	16	@1+	1	0	0	65535
K_Min_Time_In_MAINRADAR_Off_ms	Minimum time in which the MAINRADAR must remain off.	1952	16	@1+	1	0	0	65535
K_SIDERADAR_RFRONT_Retry_Counter_Max	Maximum number of retrys for initialization the SIDERADAR_RFRONT.	1968	16	@1+	1	0	0	65535
K_Max_Time_In_SIDERADAR_RFRONT_Init	Maximum time to remain in SIDERADAR_RFRONT_Init mode before resetting SIDERADAR_RFRONT by going to SIDERADAR_RFRONT_OFF.	1984	16	@1+	1	0	0	65535
K_Min_Time_In_SIDERADAR_RFRONT_Off_ms	Minimum time in which the SIDERADAR_RFRONT must remain off.	2000	16	@1+	1	0	0	65535
K_SIDERADAR_RREAR_Retry_Counter_Max	Maximum number of retrys for initialization the SIDERADAR_RREAR.	2016	16	@1+	1	0	0	65535
K_Max_Time_In_SIDERADAR_RREAR_Init	Maximum time to remain in SIDERADAR_RREAR_Init mode before resetting SIDERADAR_RREAR by going to SIDERADAR_RREAR_OFF.	2032	16	@1+	1	0	0	65535
K_Min_Time_In_SIDERADAR_RREAR_Off_ms	Minimum time in which the SIDERADAR_RREAR must remain off.	2048	16	@1+	1	0	0	65535
K_SIDERADAR_LFRONT_Retry_Counter_Max	Maximum number of retrys for initialization the SIDERADAR_LFRONT.	2064	16	@1+	1	0	0	65535
K_Max_Time_In_SIDERADAR_LFRONT_Init	Maximum time to remain in SIDERADAR_LFRONT_Init mode before resetting SIDERADAR_LFRONT by going to SIDERADAR_LFRONT_OFF.	2080	16	@1+	1	0	0	65535
K_Min_Time_In_SIDERADAR_LFRONT_Off_ms	Minimum time in which the SIDERADAR_LFRONT must remain off.	2096	16	@1+	1	0	0	1
K_SIDERADAR_LREAR_Retry_Counter_Max	Maximum number of retrys for initialization the SIDERADAR_LREAR.	2112	16	@1+	1	0	0	65535
K_Max_Time_In_SIDERADAR_LREAR_Init	Maximum time to remain in SIDERADAR_LREAR_Init mode before resetting SIDERADAR_LREAR by going to SIDERADAR_LREAR_OFF.	2128	16	@1+	1	0	0	65535
K_Min_Time_In_SIDERADAR_LREAR_Off_ms	Minimum time in which the SIDERADAR_LREAR must remain off.	2144	16	@1+	1	0	0	65535
k_Therm_A2D_Hi	Limits on highest temp value for the Thermistor out of range DTC. Note: Thermistor counts to temp have inverse relationship.	2160	16	@1+	1	0	0	4096
k_Therm_A2D_Lo	Limits on lowest temp value for the Thermistor out of range DTC. Note: Thermistor counts to temp have inverse relationship.	2176	16	@1+	1	0	0	4096
k_battery_supply_time_constant	Time constant for calculating filtered battery supply "VBattU" from UnFilterVBattU	2192	16	@1+	1	0	1	1000

SIDERADAR_RREAR_Init_wait_ms	Right Rear SIDERADAR Init timer	2208	16	@1+	1	0	0	65535

SIDERADAR_LFRONT_Init_wait_ms	From SIDERADAR Init Timer	2224	16	@1+	1	0	0	65535
SIDERADAR_RFRONT_Off_wait_ms	SIDERADAR_RFRONT_OFF state timeout timer	2240	16	@1+	1	0	0	65535
SIDERADAR_RFRONT_Init_wait_ms	From SIDERADAR Init Timer	2256	16	@1+	1	0	0	65535
MAINRADAR_Off_wait_ms	MAINRADAR_OFF state timeout timer	2272	16	@1+	1	0	0	65535
MAINRADAR_Init_wait_ms	MAINRADAR Init Timer	2288	16	@1+	1	0	0	65535
VISIONCO4_Off_wait_ms	Running timer of VISIONCO4_OFF state.	2304	16	@1+	1	0	0	65535
VISIONCO4_Init_wait_ms	VISIONCO4 Init timer	2320	16	@1+	1	0	0	65535
ALGORUNNER_Off_wait_ms	ALGORUNNER_OFF state timeout timer	2336	16	@1+	1	0	0	65535
ALGORUNNER_Init_Retry_Counter	Counter of the retrys for initialization the ALGORUNNER.	2352	16	@1+	1	0	0	65535
ALGORUNNER_Init_wait_ms	ALGORUNNER INIT timeout	2368	16	@1+	1	0	0	65535
VISIONCO4_Init_Retry_Counter	Counter for the retrys of VISIONCO4 initialization.	2384	16	@1+	1	0	0	65535
VISIONCO4_Core_Dump_wait_ms	VisionCo core dump wait timer	2400	16	@1+	1	0	0	65535
VISIONCO4_Core_Dump_Complete_wait_ms	VisionCo core dump complete timer	2416	16	@1+	1	0	0	65535
VISIONCO4_Init_Retry_Counter	Counter for the retrys of VISIONCO4 initialization.	2432	8	@1+	1	0	0	255
IMAGE2P_Init_Retry_Counter	Counter of the retrys for initialization the IMAGE2P.	2440	8	@1+	1	0	0	255
ALGORUNNER_Init_Retry_Counter	Counter of the retrys for initialization the ALGORUNNER.	2448	8	@1+	1	0	0	255
MAINRADAR_Init_Retry_Counter	Counter of the retrys for initialization the MAINRADAR.	2456	8	@1+	1	0	0	255
SIDERADAR_RFRONT_Init_Retry_Counter	Counter of the retrys for initialization the SIDERADAR_RFRONT.	2464	8	@1+	1	0	0	255
SIDERADAR_RREAR_Init_Retry_Counter	Counter of the retrys for initialization the SIDERADAR_RREAR.	2472	8	@1+	1	0	0	255
SIDERADAR_LFRONT_Init_Retry_Counter	Counter of the retrys for initialization the SIDERADAR_LFRONT.	2480	8	@1+	1	0	0	255
SIDERADAR_LREAR_Init_Retry_Counter	Counter of the retrys for initialization the SIDERADAR_LREAR.	2488	8	@1+	1	0	0	255
Bypass_ADC_conversion	Defines which ADC channel shall be Bypassed in ADC conversion	2496	8	@1+	1	0	0	0
CPYR agentOnBdT	CPYR agent on Board Temperature	2504	8	@1+	1	0	-40	125
VisionCo4OnBdT	VisionCo4 on Board Temperature	2512	8	@1+	1	0	-40	125
IMAGE2POnBdT	IMAGE2P on Board Temperature	2520	8	@1+	1	0	-40	125
AlgorunnerOnBdT	Algorunner on Board Temperature	2528	8	@1+	1	0	-40	125
CPYR agent_Rx_Valid_VisionCo4_SPI_Msgs	EyqQ4 to CPYR agent Valid SPI flags Ture False	2536	8	@1+	1	0	0	0
CPYR agent_Rx_Valid_ALGORUNNER_Ethernet_Msgs	Algorunner to CPYR agent Ethernet messages flag: true false	2544	8	@1+	1	0	0	0
CPYR agent_Rx_Valid_SIDERADAR_RFRONT_CAN_Msgs	Internal PMM variable set/cleared according to the status reported from the E2E.	2552	8	@1+	1	0	0	0
CPYR agent_Rx_Valid_SIDERADAR_RREAR_CAN_Msgs	Right Rear SIDERADAR to CPYR agent can messages flag TRUE FALSE	2560	8	@1+	1	0	0	0

CPYR agent_Rx_Valid_SIDERADAR_LFRONT_CAN_Msgs	left front SIDERADAR to CPYR AGENT can messages Vaild flag TRUE	2568	8	@1+	1	0	0	0

	FALSE							
CPYR agent_Rx_Valid_SIDERADAR_LREAR_CAN_Msgs	Left read SIDERADAR to CPYR agent valid can mesages flag TRUE FALSE	2576	8	@1+	1	0	0	0
VisionCo4_SPI_Rolling_Count_Err	VisionCo4 SPI to CPYR agent rolling counter error flag TRUE FALSE	2584	8	@1+	1	0	0	0
ALGORUNNER_ETHERNET_Rolling_Count_Err	this flag is set/clear when the number of attempts for communicate with the Algo runner through the Ethernet TRUE FALSE	2592	8	@1+	1	0	0	0
VisionCo4_SPI_Lost_Comm	This flag is set/clear regarding the status of the the communication with the VisionCo4 TRUE FALSE	2600	8	@1+	1	0	0	0
ALGORUNNER_ETHERNET_Lost_Comm	This flag is set/clear regarding the Ethernet connection status between the CPYR agent and the Algo runner TRUE FALSE	2608	8	@1+	1	0	0	0
Com_with_MAINRADAR_Is_Dropped	This flag is set/clear regarding the communication with the MAINRADAR active or dropped TRUE FALSE	2616	8	@1+	1	0	0	0
Com_with_SIDERADAR_RFRONT_Is_Dropped	This flag is set/clear regarding the communication with the front SIDERADAR active or dropped TRUE FALSE	2624	8	@1+	1	0	0	0
Com_with_SIDERADAR_RREAR_Is_Dropped	This flag is set/clear regarding the communication with the right rear SIDERADAR active or dropped TRUE FALSE	2632	8	@1+	1	0	0	0
Com_with_SIDERADAR_LFRONT_Is_Dropped	This flag is set/clear regarding the communication with the left front SIDERADAR active or dropped TRUE FALSE	2640	8	@1+	1	0	0	0
Com_with_SIDERADAR_LREAR_Is_Dropped	This flag is set/clear regarding the communication with the left rear SIDERADAR active or dropped TRUE FALSE	2648	8	@1+	1	0	0	0
k_Max_Time_CPYR agent_Thermal_Shutdown_ms	0	2656	8	@1+	1	0	0	500
k_Max_Time_IMAGE2P_Off_IMAGE2P_Thermal_Shutdown_ms	0	2664	8	@1+	1	0	0	500
k_Max_Time_IMAGE2P_CAMs_VBattState_Not_In_NORM_VOLT_ms	0	2672	8	@1+	1	0	0	500
k_Max_Time_USSs_VBattState_Not_In_NORM_VOLT_ms	0	2680	8	@1+	1	0	0	500
k_Max_Time_USSs_Power_Failure_ms	0	2688	8	@1+	1	0	0	500
k_Max_Time_CAMs_Power_Failure_ms	0	2696	8	@1+	1	0	0	500
K_VisionCo4_Retry_Counter_Max	Maximum number of retrys for initialization the VisionCo4.	2704	8	@1+	1	0	0	255
K_XCP_Force_VISIONCO_Core_Dump_Failure	if the flag K_XCP_Force_VISIONCO_Core_Dump_Failure = true then VISIONCO shall be in Core dump	2712	8	@1+	1	0	0	1

	state.							
k_VISIONCO4_cycle_counter	#N/A	2720	8	@1+	1	0	1	255
k_ALGORUNNER_cycle_counter	#N/A	2728	8	@1+	1	0	1	255
k_MAINRADAR_cycle_counter	#N/A	2736	8	@1+	1	0	1	255
k_SIDERADAR_RFRONT_cycle_counter	#N/A	2744	8	@1+	1	0	1	255
k_SIDERADAR_RREAR_cycle_counter	#N/A	2752	8	@1+	1	0	1	255
k_SIDERADAR_LFRONT_cycle_counter	#N/A	2760	8	@1+	1	0	1	255
k_SIDERADAR_LREAR_cycle_counter	#N/A	2768	8	@1+	1	0	1	255
K_ThermRecoverDelta	Current representative of sensed voltage of Rear USS	2776	8	@1+	1	0	0	255
K_ThermRecoverDelta	Defines the amount of temp needed to cool down from over temp recovery	2784	8	@1+	1	0	0	255
K_ThermRecoverDelta	Defines the amount of temp needed to cool down from over temp recovery	2792	8	@1+	1	0	0	255
k_OverTempEnterThresh	Thermistor Upper temp threshold of Normal temp band. Min range -50C to +150C	2800	8	@1+	1	0	-50	150
k_TR_OverTempEnterThresh	Thermistor Upper temp threshold of Normal temp band. Min range -50C to +150C	2808	8	@1+	1	0	-50	150
k_VisionCoOverTempEnterThresh	VisionCo MIPS, VMP upper temp threshold of Normal temperature band.Min range -50C to +150C	2816	8	@1+	1	0	-50	150
k_IMAGE2P_OverTempEnterThresh	Thermistor Upper temp threshold of Normal temp band. Min range -50C to +150C	2824	8	@1+	1	0	-50	150
k_ImagerOverTempEnterThresh	Imager Upper temp threshold of Normal temp band. Min range -50C to +150C	2832	8	@1+	1	0	-50	150
k_ThermTemp_Inc	Thermistor temperature out of range increment filter	2840	8	@1+	1	0	-128	127
k_ThermTemp_Dec	Thermistor temperature out of range decrement filter	2848	8	@1+	1	0	-128	127
k_ThermRange_Inc	Thermistor out of range increment filter	2856	8	@1+	1	0	-128	127
k_ThermRange_Dec	Thermistor out of range decrement filter	2864	8	@1+	1	0	-128	127
k_Switches_Pressed_Volt_Level	k_Switches_Pressed_Volt_Level	2872	8	@1+	1	0	1	25
k_Switches_Vmin	k_Switches_Vmin	2880	8	@1+	1	0	10	25
k_Switches_Vmax	k_Switches_Vmax	2888	8	@1+	1	0	6	25
k_Switches_Debounce_Counts	k_Switches_Debounce_Counts	2896	8	@1+	1	0	0	200
k_LED_Driver_debounce_counts	erence Debounce time for LED driver	2904	8	@1+	1	0	0	100
k_BootOverTemp_Inc	k_BootOverTemp_Inc	2912	8	@1+	1	0	0	127
k_BootOverTemp_Dec	k_BootOverTemp_Dec	2920	8	@1+	1	0	0	127
k_VISIONCOBoot_BIST_Inc	k_VISIONCOBoot_BIST_Inc	2928	8	@1+	1	0	0	127
k_VISIONCOBoot_BIST_Dec	0	2936	8	@1+	1	0	0	127
k_VISIONCOBoot_CS_Inc	k_VISIONCOBoot_CS_Inc	2944	8	@1+	1	0	0	127
k_VISIONCOBoot_CS_Dec	k_VISIONCOBoot_CS_Dec	2952	8	@1+	1	0	0	127
k_VISIONCOBoot_DDR_Inc	k_VISIONCOBoot_DDR_Inc	2960	8	@1+	1	0	0	127
k_VISIONCOBoot_DDR_Dec	k_VISIONCOBoot_DDR_Dec	2968	8	@1+	1	0	0	127
k_VISIONCOBoot_LoadFlash_Inc	k_VISIONCOBoot_LoadFlash_Inc	2976	8	@1+	1	0	0	127
k_VISIONCOBoot_LoadFlash_Dec	k_VISIONCOBoot_LoadFlash_Dec	2984	8	@1+	1	0	0	127
k_VISIONCOBoot_AppCS_Inc	k_VISIONCOBoot_AppCS_Inc	2992	8	@1+	1	0	0	127

k_VISIONCOBoot_AppCS_Dec	k_VISIONCOBoot_AppCS_Dec	3000	8	@1+	1	0	0	127
k_VISIONCOBoot_Cal_CS_Inc	k_VISIONCOBoot_Cal_CS_Inc	3008	8	@1+	1	0	0	127
k_VISIONCOBoot_Cal_CS_Dec	k_VISIONCOBoot_Cal_CS_Dec	3016	8	@1+	1	0	0	127
k_VISIONCOBoot_MEST_CS_Inc	k_VISIONCOBoot_MEST_CS_Inc	3024	8	@1+	1	0	0	127
k_VISIONCOBoot_MEST_CS_Dec	k_VISIONCOBoot_MEST_CS_Dec	3032	8	@1+	1	0	0	127
k_LIN_Bus_Fault_Inc_Cnt	k_LIN_Bus_Fault_Inc this is the Increment step size counter for LIN Rear Camera Bus Signal / Message Failure fault	3040	8	@1+	1	0	0	255
k_LIN_Bus_Fault_De_Cnt	k_LIN_Bus_Fault_Dec this is the decrement step size counter for LIN Rear Camera Bus Signal / Message Failure fault	3048	8	@1+	1	0	0	255
k_LIN_RearCamera_Fault_Inc_Cnt	#N/A	3056	8	@1+	1	0	0	255
k_LIN_RearCamera_Fault_Dec_Cnt	#N/A	3064	8	@1+	1	0	0	255
k_LIN_R_Cam_Nocfg_Fault_Inc_Cnt	#N/A	3072	8	@1+	1	0	0	255
k_LIN_R_Cam_Nocfg_Fault_Dec_Cnt	#N/A	3080	8	@1+	1	0	0	255
k_SPI_Rolling_Count_Delta	it is the acceptable value of missing SPI MSGs between CPYR agent & VISIONCO4 and vice versa.	3088	8	@1+	1	0	0	255
k_Ethernet_Rolling_Count_Delta	it is the acceptable value of missing Ethernet MSGs between CPYR agent & ALGORUNNER and vice versa.	3096	8	@1+	1	0	0	255
SIDERADAR_LREAR_Power_Mode_State	Left Rear SIDERADAR power mode status: Init Norm Standby off	3104	8	@1+	1	0	0	255
SIDERADAR_RREAR_Power_Mode_State	Right Rear SIDERADAR power mode status	3112	8	@1+	1	0	0	255
SIDERADAR_LFRONT_Power_Mode_State	Left Front SIDERADAR Power Mode status : Init, Norm, Stand by, OFF	3120	8	@1+	1	0	0	255
SIDERADAR_RFRONT_Power_Mode_State	Front SIDERADAR power mode status	3128	8	@1+	1	0	0	255
MAINRADAR_Power_Mode_State	MAINRADAR_Power_Mode_State	3136	8	@1+	1	0	0	255
IMAGE2P_Power_Mode_State	IMAGE2P Power Mode Status	3144	8	@1+	1	0	0	255
ALGORUNNER_Power_Mode_State	ALGORUNNER Power Mode Status	3152	8	@1+	1	0	0	255
VISIONCO4_Power_Mode_State	VisionCo Power mode state var	3160	8	@1+	1	0	0	255
CPYR agentApplicationFilePresent	The primary boot loader detected a loadable/correct CPYR agent application files.	3168	1	@1+	1	0	0	1
ProgrammingMsgReceived	TRUE if a CAN programming message is received while CPYR agent is running in PBL_INIT, FALSE otherwise.	3169	1	@1+	1	0	0	1
VisionCo4InitTimeout	The CPYR agent doesn't receive a correctly formatted application SPI messages in K_Max_Time_In_VISIONCO4_Init.	3170	1	@1+	1	0	0	1
IMAGE2PInitTimeout	The CPYR agent doesn't receive a correctly formatted application Ethernet messages in K_Max_Time_In_IMAGE2P_Init.	3171	1	@1+	1	0	0	1
ALGORUNNERInitTimeout	The CPYR agent doesn't receive a correctly formatted application Ethernet messages in K_Max_Time_In_ALGORUNNER_Init.	3172	1	@1+	1	0	0	1
MAINRADARInitTimeout	The CPYR agent doesn't receive a correctly formatted application CAN-FD messages in K_Max_Time_In_MAINRADAR_Init.	3173	1	@1+	1	0	0	1

SIDERADAR_RFRONTInitTimeout	The CPYR agent doesn't receive a correctly formatted application CAN-FD messages in K_Max_Time_In_SIDERADAR_RFRONT_Init.	3174	1	@1+	1	0	0	1

SIDERADAR_RREARInitTimeout	The CPYR agent doesn't receive a correctly formatted application CAN-FD messages in K_Max_Time_In_SIDERADAR_RREAR_Init.	3175	1	@1+	1	0	0	1
SIDERADAR_LREARInitTimeout	The CPYR agent doesn't receive a correctly formatted application CAN-FD messages in K_Max_Time_In_SIDERADAR_LREAR_Init.	3176	1	@1+	1	0	0	1
LKA_SW_Status	LKA_SW_Status 1 (PRESSED). 0 (NOT_PRESSED).	3177	1	@1+	1	0	0	1
View_SW_Status	View_SW_Status 1 (PRESSED or ACTIVE). 0 (NOT_PRESSED or INACTIVE).	3178	1	@1+	1	0	0	1
BPA_SW_Status	BPA_SW_Status 1 (PRESSED or ACTIVE). 0 (NOT_PRESSED or INACTIVE).	3179	1	@1+	1	0	0	1
APA_SW_Status	APA_SW_Status 1 (PRESSED or ACTIVE). 0 (NOT_PRESSED or INACTIVE).	3180	1	@1+	1	0	0	1
VisionCo4_SPI_E2E_PROFILE_ERR	#N/A	3181	1	@1+	1	0	0	0
VisionCo4_ERROUT	0	3182	1	@1+	1	0	0	0
TR_OTA_Ongoing	Internal PMM flag set/cleared according to the status reported from the OTA_Mgr.	3183	1	@1+	1	0	0	0
TR_OTA_Diag_Failure	Internal PMM flag set/cleared according to the status reported from the OTA diagnostics.	3184	1	@1+	1	0	0	0
VisionCo4_OTA_Ongoing	Internal PMM flag set/cleared according to the status reported from the OTA_Mgr.	3185	1	@1+	1	0	0	0
VisionCo4_OTA_Diag_Failure	Internal PMM flag set/cleared according to the status reported from the OTA diagnostics.	3186	1	@1+	1	0	0	0
K_Enable_Lost_Comm_VISIONCO4_Reset	If TRUE, then lost of communication from the VISIONCO4 while in VISIONCO4_NORM or VISIONCO4_OFF states will cause a reset of the VISIONCO4.	3187	1	@1+	1	0	0	1
K_Ignore_VISIONCO4_Diag_Shutdown	Set to TRUE so that VISIONCO4 does not reset if this flag is TRUE.	3188	1	@1+	1	0	0	1
K_Enable_Lost_Comm_IMAGE2P_Reset	If TRUE, then lost of communication from the IMAGE2P while in IMAGE2P_NORM or IMAGE2P_OFF states will cause a reset of the IMAGE2P.	3189	1	@1+	1	0	0	1
K_Ignore_IMAGE2P_Diag_Shutdown	Set to TRUE so that IMAGE2P does not reset if this flag is TRUE.	3190	1	@1+	1	0	0	1
K_Enable_Lost_Comm_ALGORUNNER_Reset	If TRUE, then lost of communication from the ALGORUNNER while in ALGORUNNER_NORM or ALGORUNNER_OFF states will cause a reset of the ALGORUNNER.	3191	1	@1+	1	0	0	1
K_Ignore_ALGORUNNER_Diag_Shutdown	Set to TRUE so that ALGORUNNER does not reset if this flag is TRUE.	3192	1	@1+	1	0	0	1
K_Enable_Lost_Comm_MAINRADAR_Reset	If TRUE, then lost of communication from the MAINRADAR while in MAINRADAR_NORM or MAINRADAR_OFF states will cause a reset of the MAINRADAR.	3193	1	@1+	1	0	0	1

K_Ignore_MAINRADAR_Diag_Shutdown	Set to TRUE so that MAINRADAR does not reset if this flag is TRUE.	3194	1	@1+	1	0	0	1
K_Enable_Lost_Comm_SIDERADAR_RFRONT_Reset	If TRUE, then lost of communication	3195	1	@1+	1	0	0	1

	from the SIDERADAR_RFRONT while in SIDERADAR_RFRONT_NORM or SIDERADAR_RFRONT_OFF states will cause a reset of the SIDERADAR_RFRONT.							
K_Ignore_SIDERADAR_RFRONT_Diag_Shutdown	Set to TRUE so that SIDERADAR_RFRONT does not reset if this flag is TRUE.	3196	1	@1+	1	0	0	1
K_Enable_Loss_Comm_SIDERADAR_RREAR_Reset	If TRUE, then lost of communication from the SIDERADAR_RREAR while in SIDERADAR_RREAR_NORM or SIDERADAR_RREAR_OFF states will cause a reset of the SIDERADAR_RREAR.	3197	1	@1+	1	0	0	1
K_Ignore_SIDERADAR_RREAR_Diag_Shutdown	Set to TRUE so that SIDERADAR_RREAR does not reset if this flag is TRUE.	3198	1	@1+	1	0	0	1
K_Enable_Lost_Comm_SIDERADAR_LFRONT_Reset	If TRUE, then lost of communication from the SIDERADAR_LFRONT while in SIDERADAR_LFRONT_NORM or SIDERADAR_LFRONT_OFF states will cause a reset of the SIDERADAR_LFRONT.	3199	1	@1+	1	0	0	1
K_Ignore_SIDERADAR_LFRONT_Diag_Shutdown	Set to TRUE so that SIDERADAR_LFRONT does not reset if this flag is TRUE.	3200	1	@1+	1	0	0	1
K_Enable_Lost_Comm_SIDERADAR_LREAR_Reset	If TRUE, then lost of communication from the SIDERADAR_LREAR while in SIDERADAR_LREAR_NORM or SIDERADAR_LREAR_OFF states will cause a reset of the SIDERADAR_LREAR.	3201	1	@1+	1	0	0	1
K_Ignore_SIDERADAR_LREAR_Diag_Shutdown	Set to TRUE so that SIDERADAR_LREAR does not reset if this flag is TRUE.	3202	1	@1+	1	0	0	1
K_Ignore_ULTRASONIC_FRONT_Diag_Shutdown	Set to TRUE so that ULTRASONIC_FRONT does not reset if this flag is TRUE.	3203	1	@1+	1	0	0	1
K_Ignore_ULTRASONIC_REAR_Diag_Shutdown	Set to TRUE so that ULTRASONIC_REAR does not reset if this flag is TRUE.	3204	1	@1+	1	0	0	1
k_Ignore_RH_MIRR_CAM_Diag_Shutdown	Set to TRUE so that RH_MIRR_CAM does not reset due to diag reasons	3205	1	@1+	1	0	0	1
k_Ignore_LH_MIRR_CAM_Diag_Shutdown	Set to TRUE so that LH_MIRR_CAM does not reset due to diag reasons.	3206	1	@1+	1	0	0	1
k_Ignore_FVC_Diag_Shutdown	Set to TRUE so that FVC CAM does not reset due to diag reasons.	3207	1	@1+	1	0	0	1
k_Ignore_RVC_Diag_Shutdown	Set to TRUE so that RVC CAM does not reset due to diag reasons.	3208	1	@1+	1	0	0	1
FVC_Power_Mode_State	FCV power mode status: on off	3209	1	@1+	1	0	0	1
SIDERADAR_RREAR_Reset_Request	Right Rear Radar Reset Request	3210	1	@1+	1	0	0	1
RVC_Power_Mode_State	RVC power mode status: on off	3211	1	@1+	1	0	0	1
LH_MIRR_CAM_Power_Mode_State	Left hand mirror cam power mode status : on off	3212	1	@1+	1	0	0	1
RH_MIRR_CAM_Power_Mode_State	Right Hand Mirror cam power mode status: on off	3213	1	@1+	1	0	0	1

REAR_USSs_Power_Mode_State	Rear Ultrasonic power mode status: on off	3214	1	@1+	1	0	0	1
FRONT_USSs_Power_Mode_State	Front ultrasonics power mode status on	3215	1	@1+	1	0	0	1

	off							
SIDERADAR_RFRONT_Reset_Request	Front SIDERADAR Reset Request	3216	1	@1+	1	0	0	1
SIDERADAR_RFRONTInitTimeout	SIDERADAR Init Sequnce time out timer flag	3217	1	@1+	1	0	0	1
MAINRADAR_Reset_Request	MAINRADAR Reset Request	3218	1	@1+	1	0	0	1
ALGORUNNER_Reset_Request	ALGORUNNER reset request	3219	1	@1+	1	0	0	1
VISIONCO4SoftResetTimeout	Internal PMM flag set/cleared according to Soft reset Time Out event	3220	1	@1+	1	0	0	1

Electrical_Status

Electrical_Status(0x101)-STNumber(71)-STVersion(1) **This message provides the data of all input signals.**

Message Electrical_Status (0x101) shall be implemented as follows:

SignalName	State Description	StartBit	Size	Sign	Offset	Factor	Min	Max
LKS_SW_Voltage	ADC Physical value	0	32	@1+	1	0	0	25
APA_SW_Voltage	ADC Physical value	32	32	@1+	1	0	0	25
View_SW_Voltage	ADC Physical value	64	32	@1+	1	0	0	25
BPA_SW_Voltage	ADC Physical value	96	32	@1+	1	0	0	25
POC_LDO1_RVC_FVC_SENSE_Current_Voltage	ADC Physical value	128	32	@1+	1	0	0	25
POC_LDO2_LH_RH_SENSE_Current_Voltage	ADC Physical value	160	32	@1+	1	0	0	25
POC_LDO3_FWC_SENSE_Current_Voltage	ADC Physical value	192	32	@1+	1	0	0	25
SIDERADAR_LR_RR_SENSE_Current_Voltage	ADC Physical value	224	32	@1+	1	0	0	25
SIDERADAR_RF_LF_SENSE_Current_Voltage	ADC Physical value	256	32	@1+	1	0	0	25
HEATER_FLR_SENSE_Current_Voltage	ADC Physical value	288	32	@1+	1	0	0	25
HSD_ULTRASONIC_SENSE_Current_Voltage	ADC Physical value	320	32	@1+	1	0	0	25
L_BLIS_OUT_Voltage	ADC Physical value	352	32	@1+	1	0	0	25
R_BLIS_OUT_Voltage	ADC Physical value	384	32	@1+	1	0	0	25
PARK_AID_OUT_Voltage	ADC Physical value	416	32	@1+	1	0	0	25
ALGORUNNER_3V3_Voltage	ADC Physical value	448	32	@1+	1	0	0	25
ALGORUNNER_1V8_Voltage	ADC Physical value	480	32	@1+	1	0	0	25
NTSC_ENCODER_P_Voltage	ADC Physical value	512	32	@1+	1	0	0	25
SIDERADAR_LR_PWR_Voltage	ADC Physical value	544	32	@1+	1	0	0	25
ALGORUNNER_1V0_Voltage	ADC Physical value	576	32	@1+	1	0	0	25
ALGORUNNER_1V35_Voltage	ADC Physical value	608	32	@1+	1	0	0	25
NTSC_ENCODER_N_Voltage	ADC Physical value	640	32	@1+	1	0	0	25
VBattU	ADC Physical value	672	32	@1+	1	0	0	25
VISIONCO_3V3_Voltage	ADC Physical value	704	32	@1+	1	0	0	25
VISIONCO_1V0_Voltage	ADC Physical value	736	32	@1+	1	0	0	25
VISIONCO_1V1_Voltage	ADC Physical value	768	32	@1+	1	0	0	25
VISIONCO_1V8_Voltage	ADC Physical value	800	32	@1+	1	0	0	25
SIDERADAR_LF_PWR_Voltage	ADC Physical value	832	32	@1+	1	0	0	25
SIDERADAR_RF_PWR_Voltage	ADC Physical value	864	32	@1+	1	0	0	25
FLR_PWR_Voltage	ADC Physical value	896	32	@1+	1	0	0	25
HEATER_PWR_Voltage	ADC Physical value	928	32	@1+	1	0	0	25
SIDERADAR_RR_PWR_Voltage	ADC Physical value	960	32	@1+	1	0	0	25

AURIX_Rev	ADC Physical value	992	16	@1+	1	0	0	65535
_HW_Variant	ADC Physical value	1008	16	@1+	1	0	0	65535
RawNTSC_ENCODER_AURIX_ADC_P	NA	1024	12	@1+	1	0	0	4095
RawNTSC_ENCODER_AURIX_ADC_N	NA	1036	12	@1+	1	0	0	4095
VisionCo4OnBdT	ADC Physical value	1048	8	@1-	1	0	-40	125
IMAGE2POnBdT	ADC Physical value	1056	8	@1-	1	0	-40	125
AlgorunnerOnBdT	ADC Physical value	1064	8	@1-	1	0	-40	125
CPYR agentOnBdT	ADC Physical value	1072	8	@1-	1	0	-40	125
RVC_Iout	RVC POC Current Sense	1080	8	@1+	0.05	0	0	12.5
FVC_Iout	FVC POC Current Sense	1088	8	@1+	0.05	0	0	12.5
LH_MIRROR_Iout	LH Mirror POC Current Sense	1096	8	@1+	0.05	0	0	12.5
RH_MIRROR_Iout	RH Mirror POC Current Sense	1104	8	@1+	0.05	0	0	12.5
FWC_MIRROR_Iout	FWC POC Current Sense	1112	8	@1+	0.05	0	0	12.5
SIDERADAR_LR_Iout	SIDERADAR_LR Current Sense	1120	8	@1+	0.05	0	0	12.5
SIDERADAR_RR_Iout	SIDERADAR_RR Current Sense	1128	8	@1+	0.05	0	0	12.5
SIDERADAR_LF_Iout	SIDERADAR_LF Current Sense	1136	8	@1+	0.05	0	0	12.5
SIDERADAR_RF_Iout	SIDERADAR_RF Current Sense	1144	8	@1+	0.05	0	0	12.5
MAINRADAR_Iout	MAINRADAR Current Sense	1152	8	@1+	0.05	0	0	12.5
Heater_Iout	Heater Current Sense	1160	8	@1+	0.05	0	0	12.5
Rear_USSs_Iout	Rear Ultrasonics Current Sense	1168	8	@1+	0.05	0	0	12.5
Front_USSs_Iout	Front Ultrasonics Current Sense	1176	8	@1+	0.05	0	0	12.5
SIDERADAR_LF_PWR_Current_Voltage	SIDERADAR_LF_PWR_Current_Voltage	1184	8	@1+	0.08	0	0	20.4
SIDERADAR_RF_PWR_Current_Voltage	SIDERADAR_RF_PWR_Current_Voltage	1192	8	@1+	0.08	0	0	20.4
SIDERADAR_LR_PWR_Current_Voltage	SIDERADAR_LR_PWR_Current_Voltage	1200	8	@1+	0.08	0	0	20.4
SIDERADAR_RR_PWR_Current_Voltage	SIDERADAR_RR_PWR_Current_Voltage	1208	8	@1+	0.08	0	0	20.4
FLR_PWR_Current_Voltage	FLR_PWR_Current_Voltage	1216	8	@1+	0.08	0	0	20.4
HEATER_PWR_Current_Voltage	HEATER_PWR_Current_Voltage	1224	8	@1+	0.08	0	0	20.4
LKA_SW_Status	Lane Keeping Switch	1232	1	@1+	1	0	0	1
APA_SW_Status	Active Park Assist Switch	1233	1	@1+	1	0	0	1
View_SW_Status	Surround View Switch	1234	1	@1+	1	0	0	1
BPA_SW_Status	Base ParkAid Switch	1235	1	@1+	1	0	0	1
L_BLIS_LED_Status	L_BLIS_LED_Status	1236	1	@1+	1	0	0	1
R_BLIS_LED_Status	R_BLIS_LED_Status	1237	1	@1+	1	0	0	1
PARK_AID_LED_Status	PARK_AID_LED_Status	1238	1	@1+	1	0	0	1
AURIX_HSD3_HEATER_EN	Heater driver enable status 0x0 = Turn Off 0x1 = Turn On	1239	1	@1+	1	0	0	1
HeaterOutputShortCircuit	Heater short circuit fault 0x0 = No fault 0x1 = Fault present	1240	1	@1+	1	0	0	1
HeaterOutputOpenCircuit	Heater open circuit fault 0x0 = No fault 0x1 = Fault present	1241	1	@1+	1	0	0	1
HeaterShortAcrossCycles	Indicates a Short To Ground (STG) condition across a large number of ignition cycles. 0x0 = No fault 0x1 = Fault present	1242	1	@1+	1	0	0	1

	Revision								
RL_SW_Release_Revision	Rear Left SIDERADAR5 Hardware revision (ex: A3	720	8	Unsigned	1	0	0	255	N/A
RL_SW_Promote_Revision	Rear Left Updated every stage with Major updates	728	8	Unsigned	1	0	0	255	N/A
RL_SW_Field_Revision	Rear Left Updated for minor updates	736	8	Unsigned	1	0	0	255	N/A
RL_SW_Field	Rear Left Updated for small updates such as Bug fix	744	8	Unsigned	1	0	0	255	N/A
VCAN_DBC_Version_Major	VCAN DBC Version Major	752	8	Unsigned	1	0	0	255	N/A
VCAN_DBC_Version_Minor	VCAN DBC Version Minor	760	8	Unsigned	1	0	0	255	N/A
VCAN_DBC_Version_Revision	VCAN DBC Version Revision	768	8	Unsigned	1	0	0	255	N/A
MAINRADAR_PCAN_DBC_Version_Major	MAINRADAR PCAN DBC Version Major	776	8	Unsigned	1	0	0	255	N/A
MAINRADAR_PCAN_DBC_Version_Minor	MAINRADAR PCAN DBC Version Minor	784	8	Unsigned	1	0	0	255	N/A
MAINRADAR_PCAN_DBC_Version_Revision	MAINRADAR PCAN DBC Version Revision	792	8	Unsigned	1	0	0	255	N/A
SIDERADAR_PCAN_DBC_Version_Major	SIDERADAR PCAN DBC Version Major	800	8	Unsigned	1	0	0	255	N/A
SIDERADAR_PCAN_DBC_Version_Minor	SIDERADAR PCAN DBC Version Minor	808	8	Unsigned	1	0	0	255	N/A
SIDERADAR_PCAN_DBC_Version_Revision	SIDERADAR PCAN DBC Version Revision	816	8	Unsigned	1	0	0	255	N/A
_Application_Version	will provide interface "pi__Lib_SW_Version" to sent SW library version DE__Lib_Version Where DE__Lib_Version is 16 byte ASCII formatted with unused bytes set to NULL	824	128	Unsigned	1	0	0		NA

CFG_Parameters

CFG_Parameters(0x103)-STNumber(73)-STVersion(1) **Message CFG_Parameters (0x103) shall be implemented as follows:**

gnalName	Discription	StartBit	Size	Sign	Offset	Factor	Min	Max
SensorCfg_FWC_SerNum	It is an array of 16 bytes [For Stage-F : It is an array of 32 bytes]	0	64	@1+	0	1	0	1.84E+19
VehicleCfg_SelfSteerGradient	Module Configuration Mode Feature	64	16	@1+	0	1	0	65536
VehicleCfg_OverallLength	D1	80	16	@1+	0	1	0	65536
VehicleCfg_Wheelbase	VehicleCfg_OverallLength = D3	96	16	@1+	0	1	0	65536
VehicleCfg_FrontAxleToBumper	VehicleCfg_FrontAxleToBumper = D4	112	16	@1+	0	1	0	65536
VehicleCfg_RearAxleToBumper	VehicleCfg_RearAxleToBumper = D5	128	16	@1+	0	1	0	65536
VehicleCfg_Width	VehicleCfg_Width = D6	144	16	@1+	0	1	0	65536
VehicleCfg_WidthWithMirror	VehicleCfg_WidthWithMirror = D7	160	16	@1+	0	1	0	65536
VehicleCfg_FrntTrckWidthCenter	VehicleCfg_FrntTrckWidthCenter	176	16	@1+	0	1	0	65536
VehicleCfg_FrntTrckWidthOutside	VehicleCfg_FrntTrckWidthOutside	192	16	@1+	0	1	0	65536

VehicleCfg_RearTrckWidthCenter	VehicleCfg_RearTrckWidthCenter	208	16	@1+	0	1	0	65536
VehicleCfg_RearTrckWidthOutside	VehicleCfg_RearTrckWidthOutside	224	16	@1+	0	1	0	65536
SensorCfg_DPAC_FrontSerNum	Sensor Configuration DPAC Front serial numbers LVDSSerialNum_Front[] from will	240	16	@1+	0	1	0	65536

	be stored in this parameter.							
SensorCfg_DPAC_LeftSerNum	DPAC Left Camera Serial Number	256	16	@1+	0	1	0	65536
SensorCfg_DPAC_RightSerNum	DPAC Right Camera Serial Number	272	16	@1+	0	1	0	65536
SensorCfg_DPAC_RearSerNum	DPAC Rear Camera Serial Number	288	16	@1+	0	1	0	65536
SensorCfg_FWC_ZGndToWhlHouse_Right	Z-axis height measurement from ground to the top center of the wheelhouse (including wheel lip molding if applicable)	304	16	@1+	0	1	0	65536
SensorCfg_FWC_ZGndToWhlHouse_Left	Z-axis height measurement from ground to the top center of the wheelhouse (including wheel lip molding if applicable)	320	16	@1+	0	1	0	65536
FeatureCfg_LINSelLINRVC	Unsigned integer index into the method 3 LIN Slave Configuration Array for the LIN RVC	336	8	@1+	0	1	0	255
VehicleCfg_LINSel_RearCamera2	VehicleCfg_LINSel_RearCamera2	344	8	@1+	0	1	0	255
MarketCfg_Country	Vehicle market country configuration parameter. 0x0 (Undefined) 0x1(Rest of World) 0x2 (UK) 0x3 (Ireland) 0x4 (Canada) 0x5 (China) 0x6 (Japan) 0x7 (USA) 0x8 (South_Africa) 0x9 (South_Korea) 0xA (Australia) 0xB (Oman) 0xC (Bahrain) 0xD (UAE)\ 0xE (Qatar) 0xF- 0xFF (Not_Used) For "Not Used" values respond with NRC 0x31 for write DID request.	352	8	@1+	0	1	0	255
VehicleCfg_Engine	0x0 (Undefined) 0x1 (I3_NA) 0x2 (I3_TC_or_SC) 0x3 (I4_NA) 0x4 (I4_TC_or_SC) 0x5 (I5_NA) 0x6 (I5_TC_or_SC) 0x7 (V6_NA) 0x8 (V6_TC_or_SC) 0x9 (V8_NA) 0xA (V8_TC_or_SC) 0xB (HEV) 0xC (PHEV) 0xD (BEV) 0xE-0xFF(Not Used) For "Not Used" values respond with NRC 0x31 for write DID request.	360	8	@1+	0	1	0	255
VehicleCfg_SteeringRatio	0.1 – 25.59, w/ 0.01 resolution This is obsolete workitem. Please er updated	368	16	@1+	0	1	0	255
VehicleCfg_VIN1	Vehicle Identification Number 1	384	8	@1+	0	1	0	255
VehicleCfg_VIN2	Vehicle Identification Number 2	392	8	@1+	0	1	0	255
VehicleCfg_VIN3	Vehicle Identification Number 3	400	8	@1+	0	1	0	255

VehicleCfg_VIN4	Vehicle Identification Number 4	408	8	@1+	0	1	0	255
VehicleCfg_VIN5	Vehicle Identification Number 5	416	8	@1+	0	1	0	255
VehicleCfg_VIN6	Vehicle Identification Number 6	424	8	@1+	0	1	0	255
VehicleCfg_VIN7	Vehicle Identification Number 7	432	8	@1+	0	1	0	255
VehicleCfg_VIN8	Vehicle Identification Number 8	440	8	@1+	0	1	0	255
VehicleCfg_VIN9	Vehicle Identification Number 9	448	8	@1+	0	1	0	255
VehicleCfg_VIN10	Vehicle Identification Number 10	456	8	@1+	0	1	0	255
VehicleCfg_VIN11	Vehicle Identification Number 11	464	8	@1+	0	1	0	255
VehicleCfg_VIN12	Vehicle Identification Number 12	472	8	@1+	0	1	0	255
VehicleCfg_VIN13	Vehicle Identification Number 13	480	8	@1+	0	1	0	255
VehicleCfg_VIN14	Vehicle Identification Number 14	488	8	@1+	0	1	0	255
VehicleCfg_VIN15	Vehicle Identification Number 15	496	8	@1+	0	1	0	255
VehicleCfg_VIN16	Vehicle Identification Number 16	504	8	@1+	0	1	0	255
VehicleCfg_VIN17	Vehicle Identification Number 17	512	8	@1+	0	1	0	255
VehicleCfg_FWC_VehicleType	Used to configure the FWC-relevant Method 3 parameters	520	8	@1+	0	1	0	255
VehicleCfg_DPAC_VehicleType	Used to configure the DPAC-relevent Method 3 parameters 0x00=DPAC_VehicleType	528	8	@1+	0	1	0	255
ModuleCfg_DiagResetDelay	0-12750mS, with 50mS resolution Unit is mili-Second	536	8	@1+	0	1	0	255
VehicleCfg_VehicleType	Used to select feature-relevent calibration tables 0x00=Undefined	544	8	@1+	0	1	0	255
SensorCfg_FWC_Pitch	it is an array of 4 elements each of 1 byte.	552	8	@1+	0	1	0	255
SensorCfg_FWC_Yaw	it is an array of 4 elements each of 1 byte.	560	8	@1+	0	1	0	255
SensorCfg_FWC_Roll	it is an array of 4 elements each of 1 byte.	568	8	@1+	0	1	0	255
CustomerSetting_OSW1OffsetIndex_Pers1	0x0 (Off) 0x1 (On)	576	6	@1+	0	1	0	64
CustomerSetting_OSW1OffsetIndex_Pers2	0x0 (Off) 0x1 (On)	582	6	@1+	0	1	0	64
CustomerSetting_OSW1OffsetIndex_Pers3	0x0 (Off) 0x1 (On)	588	6	@1+	0	1	0	64
CustomerSetting_OSW1OffsetIndex_Pers4	0x0 (Off) 0x1 (On)	594	6	@1+	0	1	0	64
CustomerSetting_OSW1OffsetIndex_Vehicle	0x0 (Off) 0x1 (On)	600	6	@1+	0	1	0	64
Padding	#N/A	606	2	@1+	0	1	0	1
CustomerSetting_IACCOffset_Pers1	Index into IACCOffset array	608	6	@1+	0	1	0	64
CustomerSetting_IACCOffset_Pers2	Index into IACCOffset array	614	6	@1+	0	1	0	64
CustomerSetting_IACCOffset_Pers3	Index into IACCOffset array	620	6	@1+	0	1	0	64
CustomerSetting_IACCOffset_Pers4	Index into IACCOffset array	626	6	@1+	0	1	0	64
CustomerSetting_IACCOffset_Vehicle	Index into IACCOffset array	632	6	@1+	0	1	0	64
Padding	#N/A	638	2	@1+	0	1	0	1
FeatureCfg_DPAC_TopViewIconColorIndex	Index into top view color array 0-15	640	4	@1+	0	1	0	15
MarketCfg_Region	0x0 (Undefined) 0x1 (Eu) 0x2 (NA) 0x3 (SA) 0x4 (APA_China) 0x5 (APA) 0x6 (Africa)	644	4	@1+	0	1	0	15

	0x7 (GCC) 0x8 (Australia_NZ) 0x9-0xF (NotUsed) For "Not Used" values respond with NRC 0x31 for write DID request.							
VehicleCfg_ECEVehicleCategory	0x0 = Undefined 0x1 = M1 0x2 = M2 0x3 = M3 0x4 = N1 0x5 = N2 0x6 = N3 0x7-0xF = Not Used For "Not Used" values respond with NRC 0x31 for write DID request.	648	4	@1+	0	1	0	15
ModuleFeatureCfg_LKS	Lane Keeping System 0x0 (OFF) 0x1 (LKAlert) 0x2 (LKAlert+LKAid) 0x3 (LKAlert+LKAid+LCWA) 0x4-0x7 (Not Used) For "Not Used" values respond with NRC 0x31 for write DID request.	652	3	@1+	0	1	0	7
ModuleFeatureCfg_TSR	Traffic Sign Recognition 0x0 (Off) 0x1 (SLOIF) 0x2 (SLIF) 0x3-0x8 (Unused) For "Not Used" values respond with NRC 0x31 for write DID request.	655	3	@1+	0	1	0	7
FeatureCfg_TSRMode	0x0 (Undefined) 0x1 (OFF) 0x2 (CameraOnlyOn) 0x3 (FusionOn) 0x4-0x7 (Not Used)	658	3	@1+	0	1	0	7
FeatureCfg_DPAC_ParkAssistType	0 = None 1 = Four_CH 2 = Eight_CH 3 = Ten_CH 4 = Twelve_CHFLK 0x5 – 0x7 = Not Used For "Not Used" values respond with NRC 0x31 for write DID request.	661	3	@1+	0	1	0	7
Padding	#N/A	664	8	@1+	0	1	0	1
CustomerSetting_ACCGapSetting_Pers1	0x0 = Not_Used 0x1 = Gap1 0x2 = Gap2 0x3 = Gap3 0x4 = Gap4 0x5 = Gap5 0x6 = Not_Used 0x7 = Not_Used For "Not Used" values respond with NRC 0x31 for write DID request.	672	3	@1+	0	1	0	7
CustomerSetting_ACCGapSetting_Pers2	0x0 = Not_Used 0x1 = Gap1 0x2 = Gap2 0x3 = Gap3 0x4 = Gap4 0x5 = Gap5 0x6 = Not_Used 0x7 = Not_Used	675	3	@1+	0	1	0	7

	For "Not Used" values respond with NRC 0x31 for write DID request.							
CustomerSetting_ACCGapSetting_Pers3	0x0 = Not_Used 0x1 = Gap1 0x2 = Gap2 0x3 = Gap3 0x4 = Gap4 0x5 = Gap5 0x6 = Not_Used 0x7 = Not_Used For "Not Used" values respond with NRC 0x31 for write DID request.	678	3	@1+	0	1	0	7
CustomerSetting_ACCGapSetting_Pers4	0x0 = Not_Used 0x1 = Gap1 0x2 = Gap2 0x3 = Gap3 0x4 = Gap4 0x5 = Gap5 0x6 = Not_Used 0x7 = Not_Used For "Not Used" values respond with NRC 0x31 for write DID request.	681	3	@1+	0	1	0	7
CustomerSetting_ACCGapSetting_Vehicle	0x0 = Not_Used 0x1 = Gap1 0x2 = Gap2 0x3 = Gap3 0x4 = Gap4 0x5 = Gap5 0x6 = Not_Used 0x7 = Not_Used For "Not Used" values respond with NRC 0x31 for write DID request.	684	3	@1+	0	1	0	7
FeatureCfg_PA_ObjectDetectCamViewSrc	0x00 - PAM 0x01 - CAM 0x02 - R_Gear 0x03 - PAM_CAM	687	3	@1+	0	1	0	7
VehicleCfg_APASwitchSrcModule	0 = None 1 = FCDIM 2 = SYNC_Gen2 3 = SYNC_Gen3 4 = ACM	690	3	@1+	0	1	0	7
VehicleCfg_WindshieldType	0x0 (Base) 0x1 (Acoustic) 0x2 (IR_Coated) 0x3-0x7 (NotUsed) For "Not Used" values respond with NRC 0x31 for write DID request.	693	3	@1+	0	1	0	7
VehicleCfg_LKSActuator	0x0 (HapticMotor) 0x1 (EPAS) 0x2 (AFS) 0x3 (Audio) 0x4-0x7 (NotUsed) For "Not Used" values respond with NRC 0x31 for write DID request.	696	3	@1+	0	1	0	7
VehicleCfg_HeadlampType	0x0 (Halogen) 0x1 (Xenon) 0x2 (LED) 0x3-0x7 (NotUsed) For "Not Used" values respond with	699	3	@1+	0	1	0	7

	NRC 0x31 for write DID request.							
Padding	#N/A	702	2	@1+	0	1	0	1
ModuleFeatureCfg_FcwHUD	Forward Collision Warning Heads-Up 0 = Disabled 1 = FCW_HUD 2 = Advanced_HUD 3 = Not Used For "Not Used" values respond with NRC 0x31 for write DID request.	704	2	@1+	0	1	0	4
ModuleFeatureCfg_APA	0 = None 1= SAP 2 = FAP 3 = FAP+RePA	706	2	@1+	0	1	0	4
ModuleFeatureCfg_TAD	Trailer Backing Assist 0 = Targetless TAD 1 = Target TAD 2 = Not Applicable 3 = Also Not Applicable For "Not Applicable" values respond with NRC 0x31 for write DID request.	708	2	@1+	0	1	0	4
ModuleFeatureCfg_HighBeam	Auto High Beam 0 = Disabled 1 = AHBC 2 = GFHB_ADB 3 = Not Used For "Not Used" values respond with NRC 0x31 for write DID request.	710	2	@1+	0	1	0	4
ModuleFeatureCfg_RBA	0 = Disabled 1 = RBA 2 = RPAwBRK 3 = CTAwBrk	712	2	@1+	0	1	0	4
FeatureCfg_LKSMyKey	0x0 = Undefined 0x1 = OFF 0x2 = ON 0x3 = Not Used For "Not Used" values respond with NRC 0x31 for write DID request.	714	2	@1+	0	1	0	4
FeatureCfg_LKSStrategy	Module configuration Strategy 0x0 = Undefined 0x1 = Non_EuNCAP 0x2 = EuNCAP 0x3 = Not Used For "Not Used" values respond with NRC 0x31 for write DID request.	716	2	@1+	0	1	0	4
FeatureCfg_DPAC_RearOverlayType	0 = Disabled 1 = Static 2 = NotUsed 3 = Full For "Not Used" values respond with NRC 0x31 for write DID request.	718	2	@1+	0	1	0	4
FeatureCfg_DPAC_FrontOverlayType	0 = Disabled 1 = Static 2 = NotUsed 3 = Full For "Not Used" values respond with NRC 0x31 for write DID request.	720	2	@1+	0	1	0	4
FeatureCfg_DPAC_RearCamType	[NONE|LIN|LVDS] 0 = None 1 = LIN 2 = LVDS 3 = Not Used	722	2	@1+	0	1	0	4

	For "Not Used" values respond with NRC 0x31 for write DID request.							
FeatureCfg_PA_Channels	0 = CH4 1 = CH8 2 = CH12 3 = Unused For "Unused" values respond with NRC 0x31 for write DID request.	724	2	@1+	0	1	0	4
FeatureCfg_SlaveExistsLINRVC	0 = Absent 1 = Present	726	2	@1+	0	1	0	4
FeatureCfg_SlaveExists_RearCamera2	This is the CHMSL camera 0 = Absent 1 = Present	728	2	@1+	0	1	0	4
MarketCfg_DrivingSide	For Stage F and later releases, the enumeration in OBJECTSENSE is OPPOSITE of the enumeration in / SPI messages. Convert the enumerations as shown here: OBJECTSENSE Definition: 0x0 = RightSide 0x1 = LeftSide / Definition: 0x0 = LeftSide 0x1 = RightSide Default Value = RightSide Previously the definition was: Side of the road on which the CPYR agent travel lane resides (eg. US = RightSide, UK = LeftSide). 0x0 (Undefined) 0x1 (RightSide) 0x2 (LeftSide) 0x3 (NotUsed) For "Not Used" values respond with NRC 0x31 for write DID request.	730	2	@1+	0	1	0	4
MarketCfg_SpeedLimitSignUnit	0x0 (Undefined) 0x1 (KPH) 0x2 (MPH) 0x3 (Not Used) For "Not Used" values respond with NRC 0x31 for write DID request.	732	2	@1+	0	1	0	4
CustomerSetting_FCWSensitivity_Pers1	0x0 = Not_Used 0x1 = FCW_Sensitivity_1 (Short Setting) 0x2 = FCW_Sensitivity_2 (Normal Setting) 0x3 = FCW_Sensitivity_3 (Long Setting) For "Not Used" values respond with NRC 0x31 for write DID request.	734	2	@1+	0	1	0	4
CustomerSetting_FCWSensitivity_Pers2	0x0 = Not_Used 0x1 = FCW_Sensitivity_1 (Short Setting) 0x2 = FCW_Sensitivity_2 (Normal Setting) 0x3 = FCW_Sensitivity_3 (Long Setting) For "Not Used" values respond with NRC 0x31 for write DID request.	736	2	@1+	0	1	0	4
CustomerSetting_FCWSensitivity_Pers3	0x0 = Not_Used 0x1 = FCW_Sensitivity_1 (Short Setting) 0x2 = FCW_Sensitivity_2 (Normal Setting) 0x3 = FCW_Sensitivity_3 (Long Setting) For "Not Used" values respond with	738	2	@1+	0	1	0	4

	NRC 0x31 for write DID request.							
CustomerSetting_FCWSensitivity_Pers4	0x0 = Not_Used 0x1 = FCW_Sensitivity_1 (Short Setting) 0x2 = FCW_Sensitivity_2 (Normal Setting) 0x3 = FCW_Sensitivity_3 (Long Setting) For "Not Used" values respond with NRC 0x31 for write DID request.	740	2	@1+	0	1	0	4
CustomerSetting_FCWSensitivity_Vehicle	0x0: Not_Used FCW_Sensitivity_1 (0x1): Short Setting FCW_Sensitivity_2 (0x2): Normal Setting FCW_Sensitivity_3 (0x3): Long Setting	742	2	@1+	0	1	0	4
CustomerSetting_LKAlertIntensity_Pers1	0x0 = OFF 0x1 = Low 0x2 = Medium 0x3 = High	744	2	@1+	0	1	0	4
CustomerSetting_LKAlertIntensity_Pers2	0x0 = OFF 0x1 = Low 0x2 = Medium 0x3 = High	746	2	@1+	0	1	0	4
CustomerSetting_LKAlertIntensity_Pers3	0x0 = OFF 0x1 = Low 0x2 = Medium 0x3 = High	748	2	@1+	0	1	0	4
CustomerSetting_LKAlertIntensity_Pers4	0x0 = OFF 0x1 = Low 0x2 = Medium 0x3 = High	750	2	@1+	0	1	0	4
CustomerSetting_LKAlertIntensity_Vehicle	0x0 = OFF 0x1 = Low 0x2 = Medium 0x3 = High	752	2	@1+	0	1	0	4
CustomerSetting_LKSSetting_Pers1	Lane Keeping System Mode Selection 0x0 = Not_Used 0x1 = LKAlert 0x2 = LKAid 0x3 = LKAlert+LKAid For "Not Used" values respond with NRC 0x31 for write DID request.	754	2	@1+	0	1	0	4
CustomerSetting_LKSSetting_Pers2	Lane Keeping System Mode Selection 0x0 = Not_Used 0x1 = LKAlert 0x2 = LKAid 0x3 = LKAlert+LKAid For "Not Used" values respond with NRC 0x31 for write DID request.	756	2	@1+	0	1	0	4
CustomerSetting_LKSSetting_Pers3	Lane Keeping System Mode Selection 0x0 = Not_Used 0x1 = LKAlert 0x2 = LKAid 0x3 = LKAlert+LKAid For "Not Used" values respond with NRC 0x31 for write DID request.	758	2	@1+	0	1	0	4
CustomerSetting_LKSSetting_Pers4	Lane Keeping System Mode Selection 0x0 = Not_Used 0x1 = LKAlert 0x2 = LKAid 0x3 = LKAlert+LKAid	760	2	@1+	0	1	0	4

	For "Not Used" values respond with NRC 0x31 for write DID request.							
CustomerSetting_LKSSetting_Vehicle	Lane Keeping System Mode Selection 0x0 = Not_Used 0x1 = LKAlert 0x2 = LKAid 0x3 = LKAlert+LKAid For "Not Used" values respond with NRC 0x31 for write DID request.	762	2	@1+	0	1	0	4
CustomerSetting_LKSLKAMode_Pers1	0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)	764	2	@1+	0	1	0	4
CustomerSetting_LKSLKAMode_Pers2	0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)	766	2	@1+	0	1	0	4
CustomerSetting_LKSLKAMode_Pers3	0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)	768	2	@1+	0	1	0	4
CustomerSetting_LKSLKAMode_Pers4	0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)	770	2	@1+	0	1	0	4
CustomerSetting_LKSLKAMode_Vehicle	0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)	772	2	@1+	0	1	0	4
VehicleCfg_PA_LED	0 = Yes 1 = No 2 = Dimmable 3 = Not Used For "Not Used" values respond with NRC 0x31 for write DID request.	774	2	@1+	0	1	0	4
ModuleFeatureCfg_RBAAlertType	0 = Off 1 = Graphic 2 = Text 3 = Both	776	2	@1+	0	1	0	4
VehicleCfg_FuelType	0 = Gasoline 1 = Diesel 2 = Hybrid 3 = NotUsed For "Not Used" values respond with NRC 0x31 for write DID request.	778	2	@1+	0	1	0	4
ModuleFeatureCfg_PA_UseWheelTicks	0 = Four 1 = Two 2 = Zero	780	2	@1+	0	1	0	4
VehicleCfg_LksPcaSwitch	0 = Hardwire_LKS 1 = Hardwire_PCA 2 = NoHardwareSwitchPresent 3 = Not Used For "Not Used" values respond with NRC 0x31 for write DID request.	782	2	@1+	0	1	0	4
VehicleCfg_ApaBaSwitch	0 = Hardwire_APA 1 = Network_APA 2 = NoSwitchPresent 3 = Hardwire_BA	784	2	@1+	0	1	0	4
ModuleFeatureCfg_CANPA_SwitchCPYR agent	0 = FCIMA 1 = FCIMB 2 = NONE 3 = ATCM	786	2	@1+	0	1	0	4
VehicleCfg_LaneAssist_SwType	VehicleCfg_LaneAssist_SwType	788	2	@1+	0	1	0	4
VehicleCfg_PrkAidSwitch	0 = HARDWIRE 1 = SOFT 2 = CAN 3 = NONE	790	2	@1+	0	1	0	4

VehicleCfg_PCAType	0x0 = (OFF) 0x1 = (RadarFusion) 0x2 = (CameraOnly) 0x3 = (NotUsed) For "Not Used" values respond with NRC 0x31 for write DID request. For Stage-F, Please er below values 0x0 = OFF 0x1 = RadarFusion 0x2 = CameraOnly 0x3 = FrontCornerRadarFusion For Stage-G, Please er below values 0x0 = OFF 0x1 = CameraOnly_52deg 0x2 = CameraOnly_100deg 0x3 = RadarFusion_52deg 0x4 = RadarFusion_100deg 0x5 = FrontCornerRadarFusion_52deg 0x6 = FrontCornerRadarFusion_100deg 0x7= NotUsed	792	2	@1+	0	1	0	4
VehicleCfg_ACCType	0x0 (OFF) 0x1 (RadarFusion) 0x2-0x3 (Not_Used) For "Not Used" values respond with NRC 0x31 for write DID request.	794	2	@1+	0	1	0	4
VehicleCfg_GearShiftByWire	0 = Not_By_Wire 1 = Shift_By_Wire 2 = Range_By_Wire 3 = Not Used For "Not Used" values respond with NRC 0x31 for write DID request.	796	2	@1+	0	1	0	4
VehicleCfg_WindshieldHeaterType	0x0 = (Direct) 0x1 = (Network) 0x2 = (NoHeaterPresent) 0x3 = (NotUsed) For "Not Used" values respond with NRC 0x31 for write DID request.	798	2	@1+	0	1	0	4
VehicleCfg_DCUIndicator	0x0 = None 0x1 = Driver Door 0x2 = Passenger Door 0x3 = Both	800	2	@1+	0	1	0	4
VehicleCfg_StartStop	0 = Not Present 1 = Present	802	2	@1+	0	1	0	4
VehicleInfo_TowBar	0 = Tow_Bar_Not_Present 1 = Tow_Bar_Present	804	2	@1+	0	1	0	4
VehicleInfo_TrailerTowPresent	0 = No 1 = Yes	806	2	@1+	0	1	0	4
ModuleFeatureCfg_CamSNCheck	0 = DISABLED 1 = ENABLED	808	2	@1+	0	1	0	4
ModuleFeatureCfg_DistAlert	Distance Alert 0 = Disabled 1 = Enabled	810	1	@1+	0	1	0	1
ModuleFeatureCfg_DAS	Driver Alert System 0 = Disabled 1 = Enabled	811	1	@1+	0	1	0	1
ModuleFeatureCfg_IACC	Intelligent Ade Cruise Control 0 = Disabled 1 = Enabled	812	1	@1+	0	1	0	1
ModuleFeatureCfg_TJA	0x0 (Disabled) 0x1 (Enabled)	813	1	@1+	0	1	0	1

ModuleFeatureCfg_WWA	Wrong-Way Alert 0 = Disabled 1 = Enabled	814	1	@1+	0	1	0	1
ModuleFeatureCfg_SBLM	Sign-Based Light Modes 0 = Disabled 1 = Enabled	815	1	@1+	0	1	0	1
ModuleFeatureCfg_SRP	Suspension Road Preview 0 = Disabled 1 = Enabled	816	1	@1+	0	1	0	1
ModuleFeatureCfg_BTT	0x0 (Disabled) 0x1 (Enabled)	817	1	@1+	0	1	0	1
ModuleFeatureCfg_ATD	0x0 (Disabled) 0x1 (Enabled)	818	1	@1+	0	1	0	1
ModuleFeatureCfg_ODCV	0 = Disabled 1 = Enabled	819	1	@1+	0	1	0	1
ModuleFeatureCfg_BLIS	0x0 (Enabled) 0x1 (Disabled)	820	1	@1+	0	1	0	1
ModuleFeatureCfg_CTA	0x0 (Enabled) 0x1 (Disabled)	821	1	@1+	0	1	0	1
ModuleFeatureCfg_BoundaryAlert	0 = Disabled 1 = Enabled Variable Name For Stage-D = ModuleFeatureCfg_BoundaryAlert For Stage-E = ModuleFeatureCfg_l For Stage-F = ModuleFeatureCfg_BoundaryAlert	822	1	@1+	0	1	0	1
ModuleFeatureCfg_VOD	0x0 (Disabled) 0x1 (Enabled)	823	1	@1+	0	1	0	1
ModuleFeatureCfg_AH	0x0 (Disabled) 0x1 (Enabled)	824	1	@1+	0	1	0	1
ModuleFeatureCfg_TrailerAssist	Trailer Assist 0 = Disabled 1 = Enabled	825	1	@1+	0	1	0	1
ModuleFeatureCfg_HwyAssist	0x0 (Disabled) 0x1 (Enabled)	826	1	@1+	0	1	0	1
ModuleFeatureCfg_VideoOverEther	0x0 (Disabled) 0x1 (Enabled)	827	1	@1+	0	1	0	1
ModuleFeatureCfg_VideoOutput	0x0 (Analog) 0x1 (Digital)	828	1	@1+	0	1	0	1
FeatureCfg_DPAC_WrdRemoteCamera	Wired Remote Camera 0 = Absent 1 = Present	829	1	@1+	0	1	0	1
FeatureCfg_DPAC_OffroadFrontCam	Off-Road Front Camera Behavior 0 = Disabled 1 = Enabled	830	1	@1+	0	1	0	1
FeatureCfg_DPAC_LowPowerMode	Low Power Mode for 360° cameras 0 = Disabled 1 = Enabled	831	1	@1+	0	1	0	1
FeatureCfg_DPAC_ParkAssistDisplay	Visual Park Aid in DPAC 0 = Disabled 1 = Enabled	832	1	@1+	0	1	0	1
FeatureCfg_DPAC_DoorAjarInd	0 = Disabled 1 = Enabled	833	1	@1+	0	1	0	1
FeatureCfg_DPAC_TailgateAjarInd	0 = Disabled 1 = Enabled	834	1	@1+	0	1	0	1
FeatureCfg_DPAC_LongTermCalibration	Enable the long-term Adjustment 0 = Disabled 1 = Enabled	835	1	@1+	0	1	0	1
FeatureCfg_DPAC_360Present	0 = False 1 = True	836	1	@1+	0	1	0	1
FeatureCfg_DPAC_FrontCamPresent	0 = False 1 = True	837	1	@1+	0	1	0	1

FeatureCfg_DPAC_SideCamsPresent	0 = False 1 = True	838	1	@1+	0	1	0	1
FeatureCfg_DPAC_RearSplit	0 = Off 1 = On	839	1	@1+	0	1	0	1
FeatureCfg_TrailerAssistType	Trailer Backing vs. Reverse guidance 0 = TBA 1 = TRG Only	840	1	@1+	0	1	0	1
FeatureCfg_DPAC_DirtyLensDetect	Enable Dirty Lens Detection 0 = Disabled 1 = Enabled	841	1	@1+	0	1	0	1
FeatureCfg_DPAC_WrdRemoteCameraInd	Control display of Aux Camera Disconnected Icon 0 = Disabled 1 = Enabled	842	1	@1+	0	1	0	1
FeatureCfg_APAF_6pt_Fusion	0 = Disabled 1 = Enabled	843	1	@1+	0	1	0	1
FeatureCfg_LKSIgnOnStrategy	Lane keep Igition On Strategy 0 = Off 1 = On	844	1	@1+	0	1	0	1
FeatureCfg_BLIS_Left_DCU_Cfg	0x0 (Hardwire) 0x1 (Network)	845	1	@1+	0	1	0	1
FeatureCfg_BLIS_Right_DCU_Cfg	0x0 (Hardwire) 0x1 (Network)	846	1	@1+	0	1	0	1
FeatureCfg_DPAC_OffsetView_Enable_Cfg	0x0 (Disabled) 0x1 (Enabled)	847	1	@1+	0	1	0	1
FeatureCfg_DPAC_CornerView_Enable_Cfg	0x0 (Disabled) 0x1 (Enabled)	848	1	@1+	0	1	0	1
FeatureCfg_FWC_CameraFoV	FWC camera type connected with ECU. 0x0 = Normal 0x1 = Wide Note : is same as this Data-dictionary variable	849	1	@1+	0	1	0	1
FeatureCfg_BA_Locking	0x0 (Disabled) 0x1 (Enabled)	850	1	@1+	0	1	0	1
FeatureCfg_BA_Windows	0x0 (Disabled) 0x1 (Enabled)	851	1	@1+	0	1	0	1
FeatureCfg_BA_UserMode	0x0 (Disabled) 0x1 (Enabled)	852	1	@1+	0	1	0	1
FeatureCfg_BA_DefaultMode	0x0 (Disabled) 0x1 (Enabled)	853	1	@1+	0	1	0	1
FeatureCfg_BLIS_RearRadar	0x0 (Absent) 0x1 (Present)	854	1	@1+	0	1	0	1
FeatureCfg_BLIS_FrontRadar	0x0 (Absent) 0x1 (Present)	855	1	@1+	0	1	0	1
FeatureCfg_CCM_CenterRadar	0x0 (Absent) 0x1 (Present)	856	1	@1+	0	1	0	1
VehicleCfg_SteWhlSide	Side of the CPYR agent vehicle on which the steering wheel resides 0 = Lefthand_Drive 1 = Righthand_Drive	857	1	@1+	0	1	0	1
CustomerSetting_FCWOnOff_Pers1	0x0 = Off 0x1 = On	858	1	@1+	0	1	0	1
CustomerSetting_FCWOnOff_Pers2	0x0 = Off 0x1 = On	859	1	@1+	0	1	0	1
CustomerSetting_FCWOnOff_Pers3	0x0 = Off 0x1 = On	860	1	@1+	0	1	0	1
CustomerSetting_FCWOnOff_Pers4	0x0 = Off 0x1 = On	861	1	@1+	0	1	0	1
CustomerSetting_FCWOnOff_Vehicle	0x0 = Off 0x1 = On	862	1	@1+	0	1	0	1
CustomerSetting_FCWDistanceOnOff_Pers1	0x0 = Off 0x1 = On	863	1	@1+	0	1	0	1
CustomerSetting_FCWDistanceOnOff_Pers2	0x0 = Off 0x1 = On	864	1	@1+	0	1	0	1
CustomerSetting_FCWDistanceOnOff_Pers3	0x0 = Off	865	1	@1+	0	1	0	1

	0x1 = On							
CustomerSetting_FCWDistanceOnOff_Pers4	0x0 = Off 0x1 = On	866	1	@1+	0	1	0	1
CustomerSetting_FCWDistanceOnOff_Vehicle	0x0 = Off 0x1 = On	867	1	@1+	0	1	0	1
CustomerSetting_ACCSelected_Pers1	Ade Vs Manual Cruise Control 0x0 = No 0x1 = Yes	868	1	@1+	0	1	0	1
CustomerSetting_ACCSelected_Pers2	0x0 = No 0x1 = Yes	869	1	@1+	0	1	0	1
CustomerSetting_ACCSelected_Pers3	0x0 = No 0x1 = Yes	870	1	@1+	0	1	0	1
CustomerSetting_ACCSelected_Pers4	0x0 = No 0x1 = Yes	871	1	@1+	0	1	0	1
CustomerSetting_ACCSelected_Vehicle	0x0 = No 0x1 = Yes	872	1	@1+	0	1	0	1
CustomerSetting_LKSOnOff_Pers1	0x0 Off 0x1 On	873	1	@1+	0	1	0	1
CustomerSetting_LKSOnOff_Pers2	0x0 Off 0x1 On	874	1	@1+	0	1	0	1
CustomerSetting_LKSOnOff_Pers3	0x0 Off 0x1 On	875	1	@1+	0	1	0	1
CustomerSetting_LKSOnOff_Pers4	0x0 Off 0x1 On	876	1	@1+	0	1	0	1
CustomerSetting_LKSOnOff_Vehicle	0x0 Off 0x1 On	877	1	@1+	0	1	0	1
CustomerSetting_LKSSensitivity_Pers1	0x0 Normal 0x1 Level2	878	1	@1+	0	1	0	1
CustomerSetting_LKSSensitivity_Pers2	0x0 (Normal) 0x1 (Level2)	879	1	@1+	0	1	0	1
CustomerSetting_LKSSensitivity_Pers3	0x0 Normal 0x1 Level2	880	1	@1+	0	1	0	1
CustomerSetting_LKSSensitivity_Pers4	0x0 (Normal) 0x1 (Level2)	881	1	@1+	0	1	0	1
CustomerSetting_LKSSensitivity_Vehicle	0x0 Normal 0x1 Level2	882	1	@1+	0	1	0	1
CustomerSetting_LCASelected_Pers1	0x0 (Off) 0x1 (On)	883	1	@1+	0	1	0	1
CustomerSetting_LCASelected_Pers2	0x0 (Off) 0x1 (On)	884	1	@1+	0	1	0	1
CustomerSetting_LCASelected_Pers3	0x0 (Off) 0x1 (On)	885	1	@1+	0	1	0	1
CustomerSetting_LCASelected_Pers4	0x0 (Off) 0x1 (On)	886	1	@1+	0	1	0	1
CustomerSetting_LCASelected_Vehicle	0x0 (Off) 0x1 (On)	887	1	@1+	0	1	0	1
CustomerSetting_AHBOnOff_Pers1	0x0 = Off 0x1 = On	888	1	@1+	0	1	0	1
CustomerSetting_AHBOnOff_Pers2	0x0 = Off 0x1 = On	889	1	@1+	0	1	0	1
CustomerSetting_AHBOnOff_Pers3	0x0 = Off 0x1 = On	890	1	@1+	0	1	0	1
CustomerSetting_AHBOnOff_Pers4	0x0 = Off 0x1 = On	891	1	@1+	0	1	0	1
CustomerSetting_AHBOnOff_Vehicle	0x0 = Off 0x1 = On	892	1	@1+	0	1	0	1
CustomerSetting_DASOnOff_Pers1	0x0 = Off 0x1 = On	893	1	@1+	0	1	0	1
CustomerSetting_DASOnOff_Pers2	0x0 = Off 0x1 = On	894	1	@1+	0	1	0	1

CustomerSetting_DASOnOff_Pers3	0x0 = Off 0x1 = On	895	1	@1+	0	1	0	1
CustomerSetting_DASOnOff_Pers4	0x0 = Off 0x1 = On	896	1	@1+	0	1	0	1
CustomerSetting_DASOnOff_Vehicle	0x0 = Off 0x1 = On	897	1	@1+	0	1	0	1
CustomerSetting_WWAOnOff_Pers1	0x0 = Off 0x1 = On	898	1	@1+	0	1	0	1
CustomerSetting_WWAOnOff_Pers2	0x0 = Off 0x1 = On	899	1	@1+	0	1	0	1
CustomerSetting_WWAOnOff_Pers3	0x0 = Off 0x1 = On	900	1	@1+	0	1	0	1
CustomerSetting_WWAOnOff_Pers4	0x0 = Off 0x1 = On	901	1	@1+	0	1	0	1
CustomerSetting_WWAOnOff_Vehicle	0x0 = Off 0x1 = On	902	1	@1+	0	1	0	1
CustomerSetting_TSRChimeOnOff_Pers1	0x0 (Off) 0x1 (On)	903	1	@1+	0	1	0	1
CustomerSetting_TSRChimeOnOff_Pers2	0x0 (Off) 0x1 (On)	904	1	@1+	0	1	0	1
CustomerSetting_TSRChimeOnOff_Pers3	0x0 (Off) 0x1 (On)	905	1	@1+	0	1	0	1
CustomerSetting_TSRChimeOnOff_Pers4	0x0 (Off) 0x1 (On)	906	1	@1+	0	1	0	1
CustomerSetting_TSRChimeOnOff_Vehicle	0x0 (Off) 0x1 (On)	907	1	@1+	0	1	0	1
CustomerSetting_OSWOnOff_Pers1	0x0 (Off) 0x1 (On)	908	1	@1+	0	1	0	1
CustomerSetting_OSWOnOff_Pers2	0x0 (Off) 0x1 (On)	909	1	@1+	0	1	0	1
CustomerSetting_OSWOnOff_Pers3	0x0 (Off) 0x1 (On)	910	1	@1+	0	1	0	1
CustomerSetting_OSWOnOff_Pers4	0x0 (Off) 0x1 (On)	911	1	@1+	0	1	0	1
CustomerSetting_OSWOnOff_Vehicle	0x0 (Off) 0x1 (On)	912	1	@1+	0	1	0	1
CustomerSetting_ACCAutoSetSpeedOnOff_Pers1	ACC On/Off Setting 0x0 = Off 0x1 = On	913	1	@1+	0	1	0	1
CustomerSetting_ACCAutoSetSpeedOnOff_Pers2	0x0 = Off 0x1 = On	914	1	@1+	0	1	0	1
CustomerSetting_ACCAutoSetSpeedOnOff_Pers3	0x0 = Off 0x1 = On	915	1	@1+	0	1	0	1
CustomerSetting_ACCAutoSetSpeedOnOff_Pers4	0x0 = Off 0x1 = On	916	1	@1+	0	1	0	1
CustomerSetting_ACCAutoSetSpeedOnOff_Vehicle	0x0 Off 0x1 On	917	1	@1+	0	1	0	1
CustomerSetting_TJAOnOff_Pers1	Traffic Jam Assist On/Off Setting 0x0 = Off 0x1 = On	918	1	@1+	0	1	0	1
CustomerSetting_TJAOnOff_Pers2	Traffic Jam Assist On/Off Setting 0x0 = Off 0x1 = On	919	1	@1+	0	1	0	1
CustomerSetting_TJAOnOff_Pers3	Traffic Jam Assist On/Off Setting 0x0 = Off 0x1 = On	920	1	@1+	0	1	0	1
CustomerSetting_TJAOnOff_Pers4	Traffic Jam Assist On/Off Setting 0x0 = Off 0x1 = On	921	1	@1+	0	1	0	1
CustomerSetting_TJAOnOff_Vehicle	Traffic Jam Assist On/Off Setting 0x0 = Off 0x1 = On	922	1	@1+	0	1	0	1
CustomerSetting_SoSelected_Pers1	0x0 (Off) 0x1 (On)	923	1	@1+	0	1	0	1
CustomerSetting_SoSelected_Pers2	0x0 (Off) 0x1 (On)	924	1	@1+	0	1	0	1

CustomerSetting_SoSelected_Pers3	0x0 (Off) 0x1 (On)	925	1	@1+	0	1	0	1
CustomerSetting_SoSelected_Pers4	0x0 (Off) 0x1 (On)	926	1	@1+	0	1	0	1
CustomerSetting_SoSelected_Vehicle	0x0 (Off) 0x1 (On)	927	1	@1+	0	1	0	1
FeatureCfg_PA_LineBasedParking	0 = Disabled 1 = Enabled	928	1	@1+	0	1	0	1
FeatureCfg_PA_DrvTube	0 = Disabled 1 = Enabled	929	1	@1+	0	1	0	1
FeatureCfg_PA_VisualDrvTube	0 = Disabled 1 = Enabled	930	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_Front	Enable Front Park Assist 0 = Off 1 = On	931	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_Rear	Enable Rear Park Assist 0 = Off 1 = On	932	1	@1+	0	1	0	1
FeatureCfg_PA_MapTracking	0 = Off 1 = On	933	1	@1+	0	1	0	1
FeatureCfg_PA_MyKey	0 = Disabled 1 = Enabled	934	1	@1+	0	1	0	1
FeatureCfg_PA_SIPa	0 = Off 1 = On	935	1	@1+	0	1	0	1
FeatureCfg_PA_SIPe	0 = Off 1 = On	936	1	@1+	0	1	0	1
FeatureCfg_PA_SnsBlockDetect	0 = Disabled 1 = Enabled	937	1	@1+	0	1	0	1
FeatureCfg_PA_TTC_Front_Warn	0 = Disabled 1 = Enabled	938	1	@1+	0	1	0	1
FeatureCfg_PA_TTC_Rear_Warn	0 = Disabled 1 = Enabled	939	1	@1+	0	1	0	1
FeatureCfg_PA_TTC_Side_Warn	0 = Disabled 1 = Enabled	940	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_SlotFoundSuppression	0x0 (Disable) 0x1 (Enable)	941	1	@1+	0	1	0	1
FeatureCfg_PA_FilterActive	0 = Off 1 = On	942	1	@1+	0	1	0	1
FeatureCfg_PA_WaitForSteering_Display	0 = Disabled 1 = Enabled	943	1	@1+	0	1	0	1
VehicleCfg_BidirectionalWhlSensor	0 = Not Present 1 = Present	944	1	@1+	0	1	0	1
VehicleCfg_DriveType	0 = FWD 1 = RWD	945	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_FL_WheelTick_Standstill	0 = Not Present 1 = Present	946	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_FR_WheelTick_Standstill	0 = Not Present 1 = Present	947	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_RL_WheelTick_Standstill	0 = Not Present 1 = Present	948	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_RR_WheelTick_Standstill	0 = Not Present 1 = Present	949	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_SOPa	0 = Off 1 = On	950	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_SPA	0 = Off 1 = On	951	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_VehCondCalc	0 = Disabled	952	1	@1+	0	1	0	1

	1 = Enabled							
ModuleFeatureCfg_PA_Lim_Blk_Fun	0 = Disabled 1 = Enabled	953	1	@1+	0	1	0	1
VehicleCfg_WhlSensorAxle_Cfg	0x0 (Front) 0x1 (Rear)	954	1	@1+	0	1	0	1
VehicleCfg_TrlrLghtModulePresent	0 = Not Present 1 = Present	955	1	@1+	0	1	0	1
VehicleCfg_TrlrBrkModulePresent	0 = Not Present 1 = Present	956	1	@1+	0	1	0	1
VehicleCfg_AutoTransStyle	0 = Oldstyle 1 = Newstyle	957	1	@1+	0	1	0	1
VehicleCfg_ParkBrkType	0 = Manual 1 = EPB	958	1	@1+	0	1	0	1
VehicleCfg_TransmissionType	0 = Automatic 1 = Manual	959	1	@1+	0	1	0	1
VehicleCfg_DPAC_FrontCameraSw	0 = Network 1 = Hardwire	960	1	@1+	0	1	0	1
ModuleFeatureCfg_OATSignalCPYR agent	0 = FCIMA 1 = HVAC	961	1	@1+	0	1	0	1
ModuleFeatureCfg_PA_Hotkey	0 = Not Present 1 = Present	962	1	@1+	0	1	0	1
VehicleCfg_ActiveSuspension	0x0 = Not Present 0x1 = Present	963	1	@1+	0	1	0	1
VehicleCfg_APAInterface	0 = Oldstyle 1 = Newstyle	964	1	@1+	0	1	0	1
VehicleInfo_TrailerLightModule	0x0 (Not Present) 0x1 (Present)	965	1	@1+	0	1	0	1
VehicleInfo_TrailerBrakeModule	0x0 (Not Present) 0x1 (Present)	966	1	@1+	0	1	0	1
DPAC_EOLAlignmentComplete_Cfg	0x0 (False) 0x1 (True)	967	1	@1+	0	1	0	1
DPAC_SVCAlignmentComplete_Cfg	0x0 (False) 0x1 (True)	968	1	@1+	0	1	0	1
TADAlignmentComplete_Cfg	It represents the Alignment completion status of the Trailer Camera i.e. TAD	969	1	@1+	0	1	0	1
VehicleCfg_CCM_VehicleType	Used to select CCM feature-relevant calibration tables	970	1	@1+	0	1	0	1
TrailerAngleProtectionBit	0x0 (NULL) 0x1 (One_bit)	971	1	@1+	0	1	0	1

Veh_Estimation_Compensation

Veh_Estimation_Compensation-STNumber(12)-STVersion(7)

Message Veh_Estimation_Compesation shall be implemented as follows:

SignalName	Value Type	Start Bit	Length
timestamp_us	unsigned long long int	0	64
veh_index	unsigned long int	64	32
raw_speed_mps	float	96	32
raw_yaw_rate_rps	float	128	32
raw_lat_accel	float	160	32
raw_long_accel	float	192	32
raw_steering_angle_deg	float	224	32

comp_steering_angle_deg	float	256	32
comp_yaw_rate_unfiltered	float	288	32
accel_ped_pos_pct	float	320	32
comp_yaw_rate_filtered	float	352	32
yaw_rate_bias	float	384	32
filt_veh_speed_over_ground	float	416	32
sensor_sideslip	float	448	32
sensor_long_velocity	float	480	32
sensor_lat_velocity	float	512	32
sideslip_rear_axle	float	544	32
vcs_lat_velocity	float	576	32
curvature_rear_axle	float	608	32
yaw_rate_sa	float	640	32
vcs_sideslip	float	672	32
vcs_long_velocity	float	704	32
raw_speed_qf	unsigned char	736	8
raw_yaw_rate_qf	unsigned char	744	8
raw_lat_accel_qf	unsigned char	752	8
raw_long_accel_qf	unsigned char	760	8
wheel_slip	unsigned char	768	8
comp_yaw_rate_qf	enum	776	8
comp_steering_angle_qf	unsigned char	784	8
yaw_rate_bias_qf	enum	792	8
yaw_rate_sa_qf	unsigned char	800	8
stationary	unsigned char	808	8
reverse_status	unsigned char	816	8
high_beam_status	unsigned char	824	8
front_wiper_cmd	unsigned char	832	8
brake_pedal_status	unsigned char	840	8
turn_signal_status	unsigned char	848	8
acc_braking_status	unsigned char	856	8
wiper_speed	unsigned char	864	8
validity_flags	unsigned char	872	8
padding	unsigned short int	880	16

StePinComp_An_Est	float	896	32
StePinRelInit_An_Sns	float	928	32
VehLat2_A_Actl	float	960	32
VehLong2_A_Actl	float	992	32
VehOverGnd_V_Est	float	1024	32
VehYaw_W_Actl	float	1056	32
ApedPos_Pc_ActlArb	float	1088	32
GearLvrPos_D_Actl	float	1120	8
GearRvrse_D_Actl	float	1128	8
StePinCompAnEst_D_Qf	unsigned char	1136	8
VehLatAActl_D_Qf	unsigned char	1144	8
VehLongAActl_D_Qf	unsigned char	1152	8
VehYawWActl_D_Qf	unsigned char	1160	8
ApedPosPcActl_D_Qf	unsigned char	1168	8
WhlDirRl_D_Actl	unsigned char	1176	8
WhlDirRr_D_Actl	unsigned char	1184	8
WhlDirFl_D_Actl	unsigned char	1192	8
WhlDirFr_D_Actl	unsigned char	1200	8
VehStop_D_Stat	unsigned char	1208	8
TracCtlPtActv_B_Actl	unsigned char	1216	8
AbsActv_B_Actl	unsigned char	1224	8
AccBrkTotTqMn_B_Actl	unsigned char	1232	8
HeadLghtHiOn_B_Stat	unsigned char	1240	8
BpedDrvAppl_D_Actl	unsigned char	1248	8
TurnLghtSwtch_D_Stat	unsigned char	1256	8
WiprFront_D_Stat	unsigned char	1264	8
VehicleStationary	unsigned char	1272	8
periodic_thread_proc	unsigned char	1280	32
periodic_thread_max	unsigned int	1312	32
maxThreadPeriodicity	unsigned int	1344	32
Latency_Veh_Velocity_time	unsigned int	1376	32

Latency_Yaw_Rate_time	unsigned int	1408	32
threadPeriodicityErrorCount	unsigned int	1440	16
padding	unsigned short int	1456	16
YawRateBias_Compensation_Major	unsigned short int	1472	32
YawRateBias_Compensation_Minor	float	1504	32
YawRateBias_Compensation_Field	float	1536	32
yaw_rate_estimation_major	float	1568	32
yaw_rate_estimation_minor	int	1600	32
yaw_rate_estimation_field	int	1632	32
curvature_and_sideslip_estimation_Major	int	1664	8
curvature_and_sideslip_estimation_Minor	unsigned char	1672	8
curvature_and_sideslip_estimation_Field	unsigned char	1680	8

Active_Fault_Status

Active_FaultStatus-STNumber(75)-STVersion(1) **This message is used to report fault status information.**

Message Active_Fault_Status shall be implemented as fault table defined in [5].

Signal Name	Description	StartBit	Size	Sign	Offset	Factor	Min	Max
VisionCo_3_3V_Supply_Out_Of_Range	0	0	1	@1+	0	1	0	1
VisionCo_1_8V_Supply_Out_Of_Range	0.1	1	1	@1+	0	1	0	1
VisionCo_1_1V_Supply_Out_Of_Range	0.2	2	1	@1+	0	1	0	1
VisionCo_1_0V_Supply_Out_Of_Range	0.3	3	1	@1+	0	1	0	1
TR_3_3V_Supply_Out_Of_Range	0.4	4	1	@1+	0	1	0	1
TR_1_8V_Supply_Out_Of_Range	0.5	5	1	@1+	0	1	0	1
TR_1_35V_Supply_Out_Of_Range	0.6	6	1	@1+	0	1	0	1
TR_1_0V_Supply_Out_Of_Range	0.7	7	1	@1+	0	1	0	1
IMAGE2P_PGOOD_Fail_Fault	1	8	1	@1+	0	1	0	1
FVC_CAMPWR_LDO_FailFRONT_CAMERA_FAULT_CKT_ShortToGnd	1.1	9	1	@1+	0	1	0	1
RVC_CAMPWR_LDO_FailREAR_CAMERA_FAULT_CKT_ShortToGnd	1.2	10	1	@1+	0	1	0	1
LH_Mirror_CAMPWR_LDO_FailLEFT_CAMERA_FAULT_CKT_ShortToGnd	1.3	11	1	@1+	0	1	0	1
RH_Mirror_CAMPWR_LDO_FailRIGHT_CAMERA_FAULT_CKT_ShortToGnd	1.4	12	1	@1+	0	1	0	1
FWC_CAMPWR_LDO_FailFWC_CAMERA_FAULT_CKT_ShortToGnd	1.5	13	1	@1+	0	1	0	1
Reserved	1.6	14	1	@1+	0	1	0	1
Reserved	1.7	15	1	@1+	0	1	0	1
Reserved	2	16	1	@1+	0	1	0	1
Reserved	2.1	17	1	@1+	0	1	0	1
Reserved	2.2	18	1	@1+	0	1	0	1
MAINRADAR_Output_Open_Circuit	2.3	19	1	@1+	0	1	0	1
FET_Fault_Across_cycles	2.4	20	1	@1+	0	1	0	1
FET_Fail_Across_Cycles	2.5	21	1	@1+	0	1	0	1
TR_SerialFlash_PowerUp_Fail	2.6	22	1	@1+	0	1	0	1

System_Over_Temperature	2.7	23	1	@1+	0	1	0	1
CPYR agent_On_Board_Thermistor_Over_Temp	3	24	1	@1+	0	1	0	1
VisionCo_On_Board_Thermistor_Over_Temp	3.1	25	1	@1+	0	1	0	1
TR_On_Board_Thermistor_Over_Temp	3.2	26	1	@1+	0	1	0	1
IMAGE2P_On_Board_Thermistor_Over_Temp	3.3	27	1	@1+	0	1	0	1
VisionCo_MIPS_Over_Temp	3.4	28	1	@1+	0	1	0	1
VisionCo_VMP_Over_Temp	3.5	29	1	@1+	0	1	0	1
VisionCo_Imager_Over_Temp	3.6	30	1	@1+	0	1	0	1
VisionCo_Boot_OverTemp_Fault	3.7	31	1	@1+	0	1	0	1
VisionCo_DDR_Over_Temp	4	32	1	@1+	0	1	0	1
CPYR agent_On_Board_Thermistor_Out_Of_Range	4.1	33	1	@1+	0	1	0	1
VisionCo_On_Board_Thermistor_Out_Of_Range	4.2	34	1	@1+	0	1	0	1
IMAGE2P_On_Board_Thermistor_Out_Of_Range	4.3	35	1	@1+	0	1	0	1
TR_On_Board_Thermistor_Out_Of_Range	4.4	36	1	@1+	0	1	0	1
BatteryLowFault	4.5	37	1	@1+	0	1	0	1
Reserved	4.6	38	1	@1+	0	1	0	1
Reserved	4.7	39	1	@1+	0	1	0	1
DTC_F00049_AURIX_Fault	5	40	1	@1+	0	1	0	1
DTC_F00042_AURIX_Fault	5.1	41	1	@1+	0	1	0	1
Reserved	5.2	42	1	@1+	0	1	0	1
Heater_Short_Across_Cycles	5.3	43	1	@1+	0	1	0	1
Heater_Output_Short_Circuit	5.4	44	1	@1+	0	1	0	1
Heater_Output_Open_Circuit	5.5	45	1	@1+	0	1	0	1
Heater_Short_Fail_Across_Cycles	5.6	46	1	@1+	0	1	0	1
CHMSL_camera_DEC_amplifier_ShortToBatt	5.7	47	1	@1+	0	1	0	1
AUX_camera_DEC_amplifier_ShortToBatt	6	48	1	@1+	0	1	0	1
LKS_Switch_Stuck_On_Fault	6.1	49	1	@1+	0	1	0	1
APA_Switch_Stuck_On_Fault	6.2	50	1	@1+	0	1	0	1
_360_Switch_Stuck_On_Fault	6.3	51	1	@1+	0	1	0	1
BPA_Switch_Stuck_On_Fault	6.4	52	1	@1+	0	1	0	1
L_BLIS_LED_ShortToBatt	6.5	53	1	@1+	0	1	0	1
R_BLIS_LED_ShortToBatt	6.6	54	1	@1+	0	1	0	1
PARK_AID_LED_ShortToBatt	6.7	55	1	@1+	0	1	0	1
L_BLIS_LED_ShortToGnd	7	56	1	@1+	0	1	0	1
R_BLIS_LED_ShortToGnd	7.1	57	1	@1+	0	1	0	1
PARK_AID_LED_ShortToGnd	7.2	58	1	@1+	0	1	0	1
Reserved	7.3	59	1	@1+	0	1	0	1
AlgoRunner_Error_Fault	7.4	60	1	@1+	0	1	0	1
Reserved	7.5	61	1	@1+	0	1	0	1
Reserved	7.6	62	1	@1+	0	1	0	1
Pmic_Image2_Reset_Fault	7.7	63	1	@1+	0	1	0	1
CPYR agent_ENET_Message_CRC_Err	8	64	1	@1+	0	1	0	1
CPYR agent_ENET_Missing_Message_Err	8.1	65	1	@1+	0	1	0	1
CPYR agent_ENET_Missing_Data_Err	8.2	66	1	@1+	0	1	0	1
CPYR agent_ENET_MessageReception_Err	8.3	67	1	@1+	0	1	0	1

CPYR agent_ENET_VersionMismatch_Err	8.4	68	1	@1+	0	1	0	1
CPYR agent_Application_CRC_Fault	8.5	69	1	@1+	0	1	0	1
CPYR agent_Callibration_CRC_Fault	8.6	70	1	@1+	0	1	0	1
CPYR agent_Cal_Complete_And_Compatible	8.7	71	1	@1+	0	1	0	1
VisionCo_Cal_Complete_And_Compatible_SWP3	9	72	1	@1+	0	1	0	1
VisionCo_MESP_Coding_failure	9.1	73	1	@1+	0	1	0	1
VisionCo_Application_Complete_And_Compatible	9.2	74	1	@1+	0	1	0	1
VisionCo_Application_Checksum	9.3	75	1	@1+	0	1	0	1
VisionCo_Calibration_Checksum	9.4	76	1	@1+	0	1	0	1
IMAGE2P_Appl_Complete_And_Compatible	9.5	77	1	@1+	0	1	0	1
CPYR agent_Spi_Frame_Format_Error	9.6	78	1	@1+	0	1	0	1
CPYR agent_Spi_Msg_Format_Error	9.7	79	1	@1+	0	1	0	1
CPYR agent_spi_msg_timeout	a.0	80	1	@1+	0	1	0	1
CPYR agent_Spi_Tx_Error	a.1	81	1	@1+	0	1	0	1
CPYR agent_Spi_CRC_Error	a.2	82	1	@1+	0	1	0	1
Spi_version_mismatch	a.3	83	1	@1+	0	1	0	1
Spi_Ipc_Comm_Reset	a.4	84	1	@1+	0	1	0	1
VisionCo_Spi_Frame_Format_Error	a.5	85	1	@1+	0	1	0	1
VISIONCO_Spi_Msg_Format_Error	a.6	86	1	@1+	0	1	0	1
VisionCo_Spi_Msg_Timeout	a.7	87	1	@1+	0	1	0	1
VisionCo_SPI_CRC_ERROR	b.0	88	1	@1+	0	1	0	1
VisionCo_Serial_Flash_Power_Up_Fail	b.1	89	1	@1+	0	1	0	1
TR_Aurix_FirstTimePowerUpFlashCheck	b.2	90	1	@1+	0	1	0	1
Aurix_OTA_Flash_Fail	b.3	91	1	@1+	0	1	0	1
KAM_Design_Read_Fail_Fault	b.4	92	1	@1+	0	1	0	1
KAM_Design_Write_Fail_Fault	b.5	93	1	@1+	0	1	0	1
VisionCo_DDR_Fault	b.6	94	1	@1+	0	1	0	1
Reserved	b.7	95	1	@1+	0	1	0	1
VisionCo_Boot_Fault	c.0	96	1	@1+	0	1	0	1
VisionCo_Core_Fault	c.1	97	1	@1+	0	1	0	1
VisionCo_Bist_Memory_Fault	c.2	98	1	@1+	0	1	0	1
Reserved	c.3	99	1	@1+	0	1	0	1
VisionCo_Init_Timeout_Fault	c.4	100	1	@1+	0	1	0	1
VisionCo_Core_Dump_Fault	c.5	101	1	@1+	0	1	0	1
Parameters_Invalid	c.6	102	1	@1+	0	1	0	1
Unreasonable_Radar_Data	c.7	103	1	@1+	0	1	0	1
FWC_Camera_Wrong_Assembly_Fault	d.0	104	1	@1+	0	1	0	1
AEB_shutdown_Invalid_Reason	d.1	105	1	@1+	0	1	0	1
AEB_shutdown_7_POS_DIFF_KAN_VAL	d.2	106	1	@1+	0	1	0	1
AEB_Shutdown_9_ROI_DIFF_VC_FCV	d.3	107	1	@1+	0	1	0	1
AEB_Shutdown_6_FCV_FCW_TTC	d.4	108	1	@1+	0	1	0	1
AEB_Shutdown_25_CO_FOF_DIFF	d.5	109	1	@1+	0	1	0	1
VISIONCO_GEN_30_DT_ERR	d.6	110	1	@1+	0	1	0	1
VISIONCO_GEN_18_IMAGER_DATA_ROWS_COLS	d.7	111	1	@1+	0	1	0	1
VISIONCO_GEN_4_YAW_HORIZON_DEVIATION	e.0	112	1	@1+	0	1	0	1

VISIONCO_GEN_26_CODE_CRC	e.1	113	1	@1+	0	1	0	1

VISIONCO_GEN_13_OLD_EGODATA_LAT	e.2	114	1	@1+	0	1	0	1
Reserved	e.3	115	1	@1+	0	1	0	1
Reserved	e.4	116	1	@1+	0	1	0	1
Reserved	e.5	117	1	@1+	0	1	0	1
VISIONCO_GEN_31_AEB_PARAM_FFI	e.6	118	1	@1+	0	1	0	1
VISIONCO_GEN_29_RFC_ERR	e.7	119	1	@1+	0	1	0	1
VISIONCO_GEN_11_INPUT_CRC_MISMATCH	f.0	120	1	@1+	0	1	0	1
VISIONCO_GEN_34_Corrupted_Vehicle_Info_Input_Signals	f.1	121	1	@1+	0	1	0	1
AEB_Shutdown_27_Fcv_Sig_Ffi	f.2	122	1	@1+	0	1	0	1
VISIONCO_GEN_54_ACC_STL_EXECUTION_FAILED	f.3	123	1	@1+	0	1	0	1
Reserved	f.4	124	1	@1+	0	1	0	1
Missing_Camera_Callibration_Fault	f.5	125	1	@1+	0	1	0	1
FWC_Camera_Type_Fault	f.6	126	1	@1+	0	1	0	1
Reserved	f.7	127	1	@1+	0	1	0	1
NTSC_Encoder_amplifier_ShortToBatt	10	128	1	@1+	0	1	0	1
Reserved	10.1	129	1	@1+	0	1	0	1
Reserved	10.2	130	1	@1+	0	1	0	1
Reserved	10.3	131	1	@1+	0	1	0	1
Full_Blockage_Fault	10.4	132	1	@1+	0	1	0	1
Partial_Blockage_Fault	10.5	133	1	@1+	0	1	0	1
Blurred_Image_Fault	10.6	134	1	@1+	0	1	0	1
Splashes_Fault	10.7	135	1	@1+	0	1	0	1
Low_Sun_Fault	11	136	1	@1+	0	1	0	1
Sun_Rays_Fault	11.1	137	1	@1+	0	1	0	1
Foggy_Spots_Fault	11.2	138	1	@1+	0	1	0	1
Spot_Rays_Fault	11.3	139	1	@1+	0	1	0	1
Smeared_Spots_Fault	11.4	140	1	@1+	0	1	0	1
SelfGlare_Fault	11.5	141	1	@1+	0	1	0	1
Low_Visibility_Fault	11.6	142	1	@1+	0	1	0	1
Vision_OutofFocus_Fault	11.7	143	1	@1+	0	1	0	1
Vision_OutofCalibration_Fault	12	144	1	@1+	0	1	0	1
TD_BlockageTest	12.1	145	1	@1+	0	1	0	1
VisionCo_Fatal_Error_APP_CPS_STL_FAILED	12.2	146	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Error_Fault	12.3	147	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Fs_Error_Fault	12.4	148	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Calibration_Error_Fault	12.5	149	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Init_Failed_Fault	12.6	150	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Init_Camera_Init_Fault	12.7	151	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_I2C_Video_Grab_Failed_Fault	13	152	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_I2C_Camera_Self_Reset_Fault	13.1	153	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_I2C_Timeout_Error_Fault	13.2	154	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Pattern_Test_Fault	13.3	155	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Cam_Params_Ccft_Crc_Failed_Fault	13.4	156	1	@1+	0	1	0	1
VisionCo_Fatal_Error_Pll_Comparison_Error_Fault	13.5	157	1	@1+	0	1	0	1
VisionCo_Fatal_Error_Pv_General_Error_Fault	13.6	158	1	@1+	0	1	0	1

VisionCo_Fatal_Error_Pv_Verificattion_Error_Fault	13.7	159	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Error_Fault	14	160	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Error_Fault	14.1	161	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Setting_Fault	14.2	162	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Registry_Fault	14.3	163	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Registry_Fault	14.4	164	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Fault	14.5	165	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Fault	14.6	166	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Reg_Fault	14.7	167	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Open_Blackbox_Fault	15	168	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Ep_Fault	15.1	169	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Ep_Fault	15.2	170	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Create_Logger_Fault	15.3	171	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_IL_Fault	15.4	172	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Check_Reg_Versions_Fault	15.5	173	1	@1+	0	1	0	1
VISIONCO_GEN_C_R_Error	15.6	174	1	@1+	0	1	0	1
Init_Vdi_Failure	15.7	175	1	@1+	0	1	0	1
Init_I2C_Failure	16	176	1	@1+	0	1	0	1
Grab_No_Free_Header_Failure	16.1	177	1	@1+	0	1	0	1
Grab_Issue_Failure	16.2	178	1	@1+	0	1	0	1
Grab_Acquire_Failure	16.3	179	1	@1+	0	1	0	1
Grab_Release_Failure	16.4	180	1	@1+	0	1	0	1
VisionCo_Cam_Mal_Func	16.5	181	1	@1+	0	1	0	1
VisionCo_Cam_Ctrl_Flt	16.6	182	1	@1+	0	1	0	1
VisionCo_Cam_Vdo_Flt	16.7	183	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Video_Error_Fault	17	184	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_I2C_Not_On_Time_Video_Error_Fault	17.1	185	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Logical_Video_Error_Fault	17.2	186	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Inconsistent_Frames_Video_Error_Fault	17.3	187	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Verification_Failure_Video_Error_Fault	17.4	188	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Invalid_Video_Header_Fault	17.5	189	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Error_Reg_Flag_Fault	17.6	190	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Reg_Write_Hil_Failure_Error_Fault	17.7	191	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Zero_Histogram_Error_Fault	18	192	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Sum_Not_Match_Histogram_Error_Fault	18.1	193	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Internal_Error_Fault	18.2	194	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_No_Buffer_Fault	18.3	195	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Buffer_Invalid_Format_Fault	18.4	196	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Timeout_Fault	18.5	197	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Overflow_Fault	18.6	198	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Underflow_Fault	18.7	199	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Fifo_Fault	19	200	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Weights_Fault	19.1	201	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Gamma_Fault	19.2	202	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Hist_Fault	19.3	203	1	@1+	0	1	0	1

VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Start_After_Start_Fault	19.4	204	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Core_Id_Fault	19.5	205	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Frame_Count_Read_Fail_Fault	19.6	206	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Histograms_Mismatch_Fault	19.7	207	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Ccft_Decisions_Mismatch_Fault	1a.0	208	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Header_Verification_Error_Fault	1a.1	209	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Shutdown_Fault	1a.2	210	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Shot_Out_Of_Sync_Fault	1a.3	211	1	@1+	0	1	0	1
VisionCo_Camera1_temperature_Fault	1a.4	212	1	@1+	0	1	0	1
VisionCo_Error_Out_Signal_Internal_Fault	1a.5	213	1	@1+	0	1	0	1
VisionCo_Error_Out_Temperature_Fault	1a.6	214	1	@1+	0	1	0	1
PCM_LostCom	1a.7	215	1	@1+	0	1	0	1
TCM_LostCom	1b.0	216	1	@1+	0	1	0	1
GSM_LostCom	1b.1	217	1	@1+	0	1	0	1
ABS_LostCom	1b.2	218	1	@1+	0	1	0	1
SECM_LostCom	1b.3	219	1	@1+	0	1	0	1
PSCM_LostCom	1b.4	220	1	@1+	0	1	0	1
BCM_LostCom	1b.5	221	1	@1+	0	1	0	1
RCM_LostCom	1b.6	222	1	@1+	0	1	0	1
IPC_LostCom	1b.7	223	1	@1+	0	1	0	1
HUD_LostCom	1c.0	224	1	@1+	0	1	0	1
SCCM_LostCom	1c.1	225	1	@1+	0	1	0	1
ACM_LostCom	1c.2	226	1	@1+	0	1	0	1
DDM_LostCom	1c.3	227	1	@1+	0	1	0	1
PDM_LostCom	1c.4	228	1	@1+	0	1	0	1
APIM_LostCom	1c.5	229	1	@1+	0	1	0	1
Reserved	1c.6	230	1	@1+	0	1	0	1
PCMH_LostCom	1c.7	231	1	@1+	0	1	0	1
VDM_LostCom	1d.0	232	1	@1+	0	1	0	1
PCM_InvalidData	1d.1	233	1	@1+	0	1	0	1
TCM_InvalidData	1d.2	234	1	@1+	0	1	0	1
GSM_InvalidData	1d.3	235	1	@1+	0	1	0	1
ABS_InvalidData	1d.4	236	1	@1+	0	1	0	1
SECM_InvalidData	1d.5	237	1	@1+	0	1	0	1
PSCM_InvalidData	1d.6	238	1	@1+	0	1	0	1
BCM_InvalidData	1d.7	239	1	@1+	0	1	0	1
IPC_InvaildData	1e.0	240	1	@1+	0	1	0	1
SCCM_InvalidData	1e.1	241	1	@1+	0	1	0	1
RCM_InvalidData	1e.2	242	1	@1+	0	1	0	1
DDM_InvalidData	1e.3	243	1	@1+	0	1	0	1
PDM_InvalidData	1e.4	244	1	@1+	0	1	0	1
HVAC_InvalidData	1e.5	245	1	@1+	0	1	0	1
APIM_InvalidData	1e.6	246	1	@1+	0	1	0	1
Reserved	1e.7	247	1	@1+	0	1	0	1
Reserved	1f.0	248	1	@1+	0	1	0	1
PCMH_InvalidData	1f.1	249	1	@1+	0	1	0	1

VDM_InvalidData	1f.2	250	1	@1+	0	1	0	1
IPC_InvaildRollCnt	1f.3	251	1	@1+	0	1	0	1
RCM_InvalidRollCnt	1f.4	252	1	@1+	0	1	0	1
RCM_InvalidChkSum	1f.5	253	1	@1+	0	1	0	1
reserved	1f.6	254	1	@1+	0	1	0	1
VCAN_Bus_Off_Fault	1f.7	255	1	@1+	0	1	0	1
FLRCAN_Bus_Off_Fault	20	256	1	@1+	0	1	0	1
SIDERADAR_RF_LF_CAN_Bus_Off_Fault	20.1	257	1	@1+	0	1	0	1
SIDERADAR_RR_LR_CAN_Bus_Off_Fault	20.2	258	1	@1+	0	1	0	1
MAINRADAR_PCAN_Missing_Message_Fault	20.3	259	1	@1+	0	1	0	1
MAINRADAR_PCAN_Range_Check_Fault	20.4	260	1	@1+	0	1	0	1
Reserved	20.5	261	1	@1+	0	1	0	1
SIDERADAR_FR_PCAN_Range_Check_Fault	20.6	262	1	@1+	0	1	0	1
SIDERADAR_FL_PCAN_Range_Check_Fault	20.7	263	1	@1+	0	1	0	1
SIDERADAR_RL_PCAN_Range_Check_Fault	21	264	1	@1+	0	1	0	1
SIDERADAR_RR_PCAN_Range_Check_Fault	21.1	265	1	@1+	0	1	0	1
CarCfg_Implausible_Denied_Param_23	21.2	266	1	@1+	0	1	0	1
CarCfg_Implausible_Denied_Param_96	21.3	267	1	@1+	0	1	0	1
SIDERADAR_Software_Compatibility_Fault	21.4	268	1	@1+	0	1	0	1
MAINRADAR_Software_Compatibility_Fault	21.5	269	1	@1+	0	1	0	1
GCCConfigurationNotconfigured_CarCfg_Undefined_Denied	21.6	270	1	@1+	0	1	0	1
SIDERADAR_Invalid_Hardware_configuration	21.7	271	1	@1+	0	1	0	1
ABS_InvalidRollCnt	22	272	1	@1+	0	1	0	1
ABS_InvalidChkSum	22.1	273	1	@1+	0	1	0	1
Brake_Temperature_Too_High	22.2	274	1	@1+	0	1	0	1
Camera_Module_Alignment_or_Adjustment_Incorrect_RearCamera_	22.3	275	1	@1+	0	1	0	1
DPAC_Vision_System_Camera_Alignment_or_Adjustment_Incorrect	22.4	276	1	@1+	0	1	0	1
Internal_Control_Module_Software_Incompatibility	22.5	277	1	@1+	0	1	0	1
Initial_Configuration_Not_Complete	22.6	278	1	@1+	0	1	0	1
Control_Module_Configuration_Incompatible	22.7	279	1	@1+	0	1	0	1
LIN_Rear_Camera_Bus_Failure	23	280	1	@1+	0	1	0	1
LIN_Rear_Camera_Not_Configured	23.1	281	1	@1+	0	1	0	1
LIN_Rear_Camera_Fault	23.2	282	1	@1+	0	1	0	1
Rear_Camera2_Bus_Failure	23.3	283	1	@1+	0	1	0	1
Rear_Camera2_Not_Configured	23.4	284	1	@1+	0	1	0	1
Rear_Camera2_Fault	23.5	285	1	@1+	0	1	0	1
Front_USS_HSD_FailurontUSS_Ckt_ShortToGnd	23.6	286	1	@1+	0	1	0	1
Rear_USS_HSD_FailureRearUSS_Ckt_ShortToGnd	23.7	287	1	@1+	0	1	0	1
Front_USS_HSD_ShortToBattFrontUSS_Ckt_ShortToBatt	24	288	1	@1+	0	1	0	1
Rear_USS_HSD_ShortToBattRearUSS_Ckt_ShortTobBatt	24.1	289	1	@1+	0	1	0	1
Ignition_Status_Unknown_Fault	24.2	290	1	@1+	0	1	0	1
Ignition_Status_Missing_Fault	24.3	291	1	@1+	0	1	0	1
Veh_V_ActlEng_Missing_Fault	24.4	292	1	@1+	0	1	0	1
VehVActlEng_D_Qf_Missing_Fault	24.5	293	1	@1+	0	1	0	1
VehVActlEng_D_Qf_Fault	24.6	294	1	@1+	0	1	0	1

VehVActlEng_D_Qf_No_data_exists_Fault	24.7	295	1	@1+	0	1	0	1
VISIONCO_MES_49_Data_Path_Corruption	25	296	1	@1+	0	1	0	1
VISIONCO_MES_50_VMP_SANITY_FAILED	25.1	297	1	@1+	0	1	0	1
Reserved	25.2	298	1	@1+	0	1	0	1
VISIONCO_40_PED_SANITY_FAULT	25.3	299	1	@1+	0	1	0	1
VISIONCO_39_PED_OVERLAP_FAULT	25.4	300	1	@1+	0	1	0	1
Reserved	25.5	301	1	@1+	0	1	0	1
Reserved	25.6	302	1	@1+	0	1	0	1
Reserved	25.7	303	1	@1+	0	1	0	1
Reserved	26	304	1	@1+	0	1	0	1
Reserved	26.1	305	1	@1+	0	1	0	1
VISIONCO_48_ALGO_PITCH_MISMATCH	26.2	306	1	@1+	0	1	0	1
VISIONCO_45_ROAD_SFCONF_VALIDATION	26.3	307	1	@1+	0	1	0	1
VISIONCO_42_ROAD_PREDICTION_VALIDATION	26.4	308	1	@1+	0	1	0	1
VISIONCO_33_ROAD_LANE_DISQUALIFICATINO_FAILED	26.5	309	1	@1+	0	1	0	1
TrlrLampCnnct_B_Actl_Missing_Fault	26.6	310	1	@1+	0	1	0	1
BLISLEDStatDriverSide_MissingMsg_Fault	26.7	311	1	@1+	0	1	0	1
BLISLEDStatPassSide_MissingMsg_Fault	27	312	1	@1+	0	1	0	1
BLISLEDStatDriverSide_DCUFault	27.1	313	1	@1+	0	1	0	1
BLISLEDStatPassSide_DCUFault	27.2	314	1	@1+	0	1	0	1
BLISLEDStatDriverSide_ON_OFF_Mismatch_Fault	27.3	315	1	@1+	0	1	0	1
BLISLEDStatPassSide_ON_OFF_Mismatch_Fault	27.4	316	1	@1+	0	1	0	1
IgnKeyType_D_Actl_Unknown_Fault	27.5	317	1	@1+	0	1	0	1
IgnKeyType_D_ActlMissing_Fault	27.6	318	1	@1+	0	1	0	1
Parklamp_Status_Missing_Fault	27.7	319	1	@1+	0	1	0	1
TR_Application_CRC_Fault	28	320	1	@1+	0	1	0	1
Reserved	28.1	321	1	@1+	0	1	0	1
TR_DDR_RAM_CRC_Fault	28.2	322	1	@1+	0	1	0	1
Reserved	28.3	323	1	@1+	0	1	0	1
Reserved	28.4	324	1	@1+	0	1	0	1
Reserved	28.5	325	1	@1+	0	1	0	1
Reserved	28.6	326	1	@1+	0	1	0	1
Reserved	28.7	327	1	@1+	0	1	0	1
Reserved	29	328	1	@1+	0	1	0	1
Reserved	29.1	329	1	@1+	0	1	0	1
Reserved	29.2	330	1	@1+	0	1	0	1
Reserved	29.3	331	1	@1+	0	1	0	1
Reserved	29.4	332	1	@1+	0	1	0	1
Reserved	29.5	333	1	@1+	0	1	0	1
Reserved	29.6	334	1	@1+	0	1	0	1
Reserved	29.7	335	1	@1+	0	1	0	1
TR_INT_OSC_TEST_FAIL	2a.0	336	1	@1+	0	1	0	1
Perf_DPAC_Process_Fault	2a.1	337	1	@1+	0	1	0	1
Perf_FWC_Process_Fault	2a.2	338	1	@1+	0	1	0	1
Reserved	2a.3	339	1	@1+	0	1	0	1

TR_Stack_Overflow_Fault	2a.4	340	1	@1+	0	1	0	1
Reserved	2a.5	341	1	@1+	0	1	0	1
Reserved	2a.6	342	1	@1+	0	1	0	1
Reserved	2a.7	343	1	@1+	0	1	0	1
Reserved	2b.0	344	1	@1+	0	1	0	1
TR_DDR_Fault	2b.1	345	1	@1+	0	1	0	1
TR_Boot_Fault	2b.2	346	1	@1+	0	1	0	1
Reserved	2b.3	347	1	@1+	0	1	0	1
TR_Appl_Complete_And_Compatible	2b.4	348	1	@1+	0	1	0	1
TR_Core_Exception_Fault	2b.5	349	1	@1+	0	1	0	1
TR_Control_ConfigReg_Fault	2b.6	350	1	@1+	0	1	0	1
reserved	2b.7	351	1	@1+	0	1	0	1
TR_ENET_Message_CRC_Err	2c.0	352	1	@1+	0	1	0	1
TR_ENET_Missing_Message_Err	2c.1	353	1	@1+	0	1	0	1
TR_ENET_Missing_Data_Err	2c.2	354	1	@1+	0	1	0	1
TR_ENET_MessageReception_Err	2c.3	355	1	@1+	0	1	0	1
TR_ENET_VersionMismatch_Err	2c.4	356	1	@1+	0	1	0	1
TR_ENET_Communication_Drv_Err	2c.5	357	1	@1+	0	1	0	1
Reserved	2c.6	358	1	@1+	0	1	0	1
Reserved	2c.7	359	1	@1+	0	1	0	1
Reserved	2d.0	360	1	@1+	0	1	0	1
QSPI_Flash_Illegal_Instruction_Err	2d.1	361	1	@1+	0	1	0	1
QSPI_Communication_Drv_Err	2d.2	362	1	@1+	0	1	0	1
Reserved	2d.3	363	1	@1+	0	1	0	1
Reserved	2d.4	364	1	@1+	0	1	0	1
Reserved	2d.5	365	1	@1+	0	1	0	1
DTC_F00042_PERFPROC_IntMemFail	2d.6	366	1	@1+	0	1	0	1
DTC_F00049_PERFPROC_IntElectronicFail	2d.7	367	1	@1+	0	1	0	1
LEFT_CAMERA_FAULT_CKT_ShortToBatt	2e.0	368	1	@1+	0	1	0	1
RIGHT_CAMERA_FAULT_CKT_ShortToBatt	2e.1	369	1	@1+	0	1	0	1
BA_Switch_Stuck_On_Fault	2e.2	370	1	@1+	0	1	0	1
Reserved	2e.3	371	1	@1+	0	1	0	1
Reserved	2e.4	372	1	@1+	0	1	0	1
Reserved	2e.5	373	1	@1+	0	1	0	1
Reserved	2e.6	374	1	@1+	0	1	0	1
Reserved	2e.7	375	1	@1+	0	1	0	1
Reserved	2f.0	376	1	@1+	0	1	0	1
Reserved	2f.1	377	1	@1+	0	1	0	1
Reserved	2f.2	378	1	@1+	0	1	0	1
Reserved	2f.3	379	1	@1+	0	1	0	1
Reserved	2f.4	380	1	@1+	0	1	0	1
Reserved	2f.5	381	1	@1+	0	1	0	1
Reserved	2f.6	382	1	@1+	0	1	0	1
Reserved	2f.7	383	1	@1+	0	1	0	1
VISIONCO_GEN_21_CAMERA_HEIGHT_LIMIT_EXCEEDED	30	384	1	@1+	0	1	0	1
VISIONCO_44_ROAD_SANITY_CORRECTION_VALIDATION	30.1	385	1	@1+	0	1	0	1

VISIONCO_53_DECELERATION_DBS_OVERLAP_FAULT	30.2	386	1	@1+	0	1	0	1
VISIONCO_GEN_24_CCFT_DIFF_RESULTS	30.3	387	1	@1+	0	1	0	1
GearRvrse_D_Actl_Missing_Fault	30.4	388	1	@1+	0	1	0	1
SOD_TCM_LostComm	30.5	389	1	@1+	0	1	0	1
WheelData_Missing_Fault	30.6	390	1	@1+	0	1	0	1
SOD_BCM_LostComm	30.7	391	1	@1+	0	1	0	1
SOD_RCM_LostCom	31	392	1	@1+	0	1	0	1
SOD_ABS_InvalidData	31.1	393	1	@1+	0	1	0	1
SOD_TCM_InvalidData	31.2	394	1	@1+	0	1	0	1
GearRvrse_D_Actl_invalid_Fault	31.3	395	1	@1+	0	1	0	1
SOD_IPC_LostCom	31.4	396	1	@1+	0	1	0	1
AEB_Inhibited_Fault_Flag	31.5	397	1	@1+	0	1	0	1
AEB_Validity_Fault_Flag	31.6	398	1	@1+	0	1	0	1
LaneFusion_Validity_Fault_Flag	31.7	399	1	@1+	0	1	0	1
Reserved	32	400	1	@1+	0	1	0	1
Reserved	32.1	401	1	@1+	0	1	0	1
Reserved	32.2	402	1	@1+	0	1	0	1
FWC_CAMERA_FAULT_CKT_ShortToBatt	32.3	403	1	@1+	0	1	0	1
LeftRearSOD_Ckt_ShortToBatt	32.4	404	1	@1+	0	1	0	1
LeftRearSOD_Ckt_ShortToGnd	32.5	405	1	@1+	0	1	0	1
RightRearSOD_Ckt_ShortToBatt	32.6	406	1	@1+	0	1	0	1
RightRearSOD_Ckt_ShortToGnd	32.7	407	1	@1+	0	1	0	1
LeftFrontSOD_Ckt_ShortToBatt	33	408	1	@1+	0	1	0	1
LeftFrontSOD_Ckt_ShortToGnd	33.1	409	1	@1+	0	1	0	1
RightFrontSOD_Ckt_ShortToBatt	33.2	410	1	@1+	0	1	0	1
RightFrontSOD_Ckt_ShortToGnd	33.3	411	1	@1+	0	1	0	1
FRONT_CAMERA_FAULT_CKT_ShortToBatt	33.4	412	1	@1+	0	1	0	1
REAR_CAMERA_FAULT_CKT_ShortToBatt	33.5	413	1	@1+	0	1	0	1
Reserved	33.6	414	1	@1+	0	1	0	1
DTC_915E31_PerfProcessor	33.7	415	1	@1+	0	1	0	1
DTC_9B4008_FAULT_CONDITION	34	416	1	@1+	0	1	0	1
DTC_9B4096_FAULT_CONDITION	34.1	417	1	@1+	0	1	0	1
DTC_9B4098_FAULT_CONDITION	34.2	418	1	@1+	0	1	0	1
DTC_9B4208_FAULT_CONDITION	34.3	419	1	@1+	0	1	0	1
DTC_9B4296_FAULT_CONDITION	34.4	420	1	@1+	0	1	0	1
DTC_9B4298_FAULT_CONDITION	34.5	421	1	@1+	0	1	0	1
DTC_929C08_FAULT_CONDITION	34.6	422	1	@1+	0	1	0	1
DTC_929C96_FAULT_CONDITION	34.7	423	1	@1+	0	1	0	1
DTC_929C98_FAULT_CONDITION	35	424	1	@1+	0	1	0	1
DTC_9B4808_FAULT_CONDITION	35.1	425	1	@1+	0	1	0	1
DTC_9B4896_FAULT_CONDITION	35.2	426	1	@1+	0	1	0	1
DTC_9B4898_FAULT_CONDITION	35.3	427	1	@1+	0	1	0	1
DTC_9B5008_FAULT_CONDITION	35.4	428	1	@1+	0	1	0	1
DTC_9B5096_FAULT_CONDITION	35.5	429	1	@1+	0	1	0	1
DTC_9B5098_FAULT_CONDITION	35.6	430	1	@1+	0	1	0	1

DTC_93F308_FAULT_CONDITION	35.7	431	1	@1+	0	1	0	1
DTC_93F396_FAULT_CONDITION	36	432	1	@1+	0	1	0	1
DTC_93F398_FAULT_CONDITION	36.1	433	1	@1+	0	1	0	1
DTC_9B3608_FAULT_CONDITION	36.2	434	1	@1+	0	1	0	1
DTC_9B3696_FAULT_CONDITION	36.3	435	1	@1+	0	1	0	1
DTC_9B3698_FAULT_CONDITION	36.4	436	1	@1+	0	1	0	1
DTC_9B3808_FAULT_CONDITION	36.5	437	1	@1+	0	1	0	1
DTC_9B3896_FAULT_CONDITION	36.6	438	1	@1+	0	1	0	1
DTC_9B3898_FAULT_CONDITION	36.7	439	1	@1+	0	1	0	1
DTC_929D08_FAULT_CONDITION	37	440	1	@1+	0	1	0	1
DTC_929D96_FAULT_CONDITION	37.1	441	1	@1+	0	1	0	1
DTC_929D98_FAULT_CONDITION	37.2	442	1	@1+	0	1	0	1
DTC_9B4408_FAULT_CONDITION	37.3	443	1	@1+	0	1	0	1
DTC_9B4496_FAULT_CONDITION	37.4	444	1	@1+	0	1	0	1
DTC_9B4498_FAULT_CONDITION	37.5	445	1	@1+	0	1	0	1
DTC_9B4608_FAULT_CONDITION	37.6	446	1	@1+	0	1	0	1
DTC_9B4696_FAULT_CONDITION	37.7	447	1	@1+	0	1	0	1
DTC_9B4698_FAULT_CONDITION	38	448	1	@1+	0	1	0	1
DTC_93F408_FAULT_CONDITION	38.1	449	1	@1+	0	1	0	1
DTC_93F496_FAULT_CONDITION	38.2	450	1	@1+	0	1	0	1
DTC_93F498_FAULT_CONDITION	38.3	451	1	@1+	0	1	0	1
DTC_93039E_FAULT_CONDITION	38.4	452	1	@1+	0	1	0	1
DTC_915E31_FAULT_CONDITION	38.5	453	1	@1+	0	1	0	1
DTC_948E31_FAULT_CONDITION	38.6	454	1	@1+	0	1	0	1
DTC_92BE31_FAULT_CONDITION	38.7	455	1	@1+	0	1	0	1
DTC_92BF31_FAULT_CONDITION	39	456	1	@1+	0	1	0	1
DTC_F00042_IMAGE2	39.1	457	1	@1+	0	1	0	1
DTC_F00089_IMAGE2	39.2	458	1	@1+	0	1	0	1
SIDERADAR_FL_PCAN_Missing_Message_Fault	39.3	459	1	@1+	0	1	0	1
SIDERADAR_FR_PCAN_Missing_Message_Fault	39.4	460	1	@1+	0	1	0	1
SIDERADAR_RL_PCAN_Missing_Message_Fault	39.5	461	1	@1+	0	1	0	1
SIDERADAR_RR_PCAN_Missing_Message_Fault	39.6	462	1	@1+	0	1	0	1
DSCM_InvalidData	39.7	463	1	@1+	0	1	0	1
DSCM_InvalidRollCnt	3a.0	464	1	@1+	0	1	0	1
DSCM_InvalidChkSum	3a.1	465	1	@1+	0	1	0	1
TRM_InvalidData	3a.2	466	1	@1+	0	1	0	1
DSCM_LostComm	3a.3	467	1	@1+	0	1	0	1
HVAC_LostComm	3a.4	468	1	@1+	0	1	0	1
TRM_LostComm	3a.5	469	1	@1+	0	1	0	1
ATCM_LostComm	3a.6	470	1	@1+	0	1	0	1
HCM_LostComm	3a.7	471	1	@1+	0	1	0	1
Parklamp_Status_Unknown_Fault	3b.0	472	1	@1+	0	1	0	1
Litval_Unknown_Fault	3b.1	473	1	@1+	0	1	0	1
Litval_Missing_Fault	3b.2	474	1	@1+	0	1	0	1
TurnLghtSwitch_D_Stat_Missing_Fault	3b.3	475	1	@1+	0	1	0	1

Btt_L_Actl_MissingMsg_Fault	3b.4	476	1	@1+	0	1	0	1
ATD_Mismatch_Fault	3b.5	477	1	@1+	0	1	0	1
TLM_Mismatch_Fault	3b.6	478	1	@1+	0	1	0	1
TBM_Mismatch_Fault	3b.7	479	1	@1+	0	1	0	1
Reserved	3c.0	480	1	@1+	0	1	0	1
WhlDirRr_D_Actl_Missing	3c.1	481	1	@1+	0	1	0	1
WhlDirRl_D_Actl_Missing	3c.2	482	1	@1+	0	1	0	1
WhlDirFl_D_Actl_Missing	3c.3	483	1	@1+	0	1	0	1
WhlDirFr_D_Actl_Missing	3c.4	484	1	@1+	0	1	0	1
WhlDirRr_D_Actl_Failed_Fault	3c.5	485	1	@1+	0	1	0	1
WhlDirRl_D_Actl_Failed_Fault	3c.6	486	1	@1+	0	1	0	1
WhlDirFr_D_Actl_Failed_Fault	3c.7	487	1	@1+	0	1	0	1
WhlDirFl_D_Actl_Failed_Fault	3d.0	488	1	@1+	0	1	0	1
CtaBrk_D_Stat_Denied_Fault	3d.1	489	1	@1+	0	1	0	1
CtaBrk_D_Stat_Missing_Fault	3d.2	490	1	@1+	0	1	0	1
Isig_brake_timeout_fault	3d.3	491	1	@1+	0	1	0	1
Reserved	3d.4	492	1	@1+	0	1	0	1
CtaYBrkEnbl_B_Rq_Mismatch_Fault	3d.5	493	1	@1+	0	1	0	1
DVR_SELECT_STAT_Fault	3d.6	494	1	@1+	0	1	0	1
DTC_F00049_IMAGE2	3d.7	495	1	@1+	0	1	0	1
CMbBDeniedByABS	3e.0	496	1	@1+	0	1	0	1
CMbBDeniedByPCM	3e.1	497	1	@1+	0	1	0	1
CMbBImplausByABS	3e.2	498	1	@1+	0	1	0	1
CMbBImplausByPCM	3e.3	499	1	@1+	0	1	0	1
AGENT_InvalidData	3e.4	500	1	@1+	0	1	0	1
unused	3e.5	501	1	@1+	0	1	0	1
ACCImplausByABS	3e.6	502	1	@1+	0	1	0	1
ACCImplausByPCM	3e.7	503	1	@1+	0	1	0	1
ACCImplausAccelByABS	3f.0	504	1	@1+	0	1	0	1
ACCFailToCancelByPCM	3f.1	505	1	@1+	0	1	0	1
ACCDeniedbyABS	3f.2	506	1	@1+	0	1	0	1
SetDTC_ACCDeniedByDIM	3f.3	507	1	@1+	0	1	0	1
SetDTCImplausibleCMbB_ASILB	3f.4	508	1	@1+	0	1	0	1
TJAImplausByPSCM	3f.5	509	1	@1+	0	1	0	1
SetDTCImplausibleHA_FSM	3f.6	510	1	@1+	0	1	0	1
SetDTCImplausibleFAPA_FSM	3f.7	511	1	@1+	0	1	0	1

Latch_Fault_Status

Latch_Fault_Status-STNumber(76)-STVersion(1) **This message is used to report fault status information.**

Message Latch_Fault_Status shall be implemented as fault table defined in [5].

Signal Name	Description	StartBit	Size	Sign	Offset	Factor	Min	Max
VisionCo_3_3V_Supply_Out_Of_Range	0	0	1	@1+	0	1	0	1
VisionCo_1_8V_Supply_Out_Of_Range	0.1	1	1	@1+	0	1	0	1
VisionCo_1_1V_Supply_Out_Of_Range	0.2	2	1	@1+	0	1	0	1
VisionCo_1_0V_Supply_Out_Of_Range	0.3	3	1	@1+	0	1	0	1

TR_3_3V_Supply_Out_Of_Range	0.4	4	1	@1+	0	1	0	1
TR_1_8V_Supply_Out_Of_Range	0.5	5	1	@1+	0	1	0	1
TR_1_35V_Supply_Out_Of_Range	0.6	6	1	@1+	0	1	0	1
TR_1_0V_Supply_Out_Of_Range	0.7	7	1	@1+	0	1	0	1
IMAGE2P_PGOOD_Fail_Fault	1	8	1	@1+	0	1	0	1
FVC_CAMPWR_LDO_FailFRONT_CAMERA_FAULT_CKT_ShortToGnd	1.1	9	1	@1+	0	1	0	1
RVC_CAMPWR_LDO_FailREAR_CAMERA_FAULT_CKT_ShortToGnd	1.2	10	1	@1+	0	1	0	1
LH_Mirror_CAMPWR_LDO_FailLEFT_CAMERA_FAULT_CKT_ShortToGnd	1.3	11	1	@1+	0	1	0	1
RH_Mirror_CAMPWR_LDO_FailRIGHT_CAMERA_FAULT_CKT_ShortToGnd	1.4	12	1	@1+	0	1	0	1
FWC_CAMPWR_LDO_FailFWC_CAMERA_FAULT_CKT_ShortToGnd	1.5	13	1	@1+	0	1	0	1
Reserved	1.6	14	1	@1+	0	1	0	1
Reserved	1.7	15	1	@1+	0	1	0	1
Reserved	2	16	1	@1+	0	1	0	1
Reserved	2.1	17	1	@1+	0	1	0	1
Reserved	2.2	18	1	@1+	0	1	0	1
MAINRADAR_Output_Open_Circuit	2.3	19	1	@1+	0	1	0	1
FET_Fault_Across_cycles	2.4	20	1	@1+	0	1	0	1
FET_Fail_Across_Cycles	2.5	21	1	@1+	0	1	0	1
TR_SerialFlash_PowerUp_Fail	2.6	22	1	@1+	0	1	0	1
System_Over_Temperature	2.7	23	1	@1+	0	1	0	1
CPYR agent_On_Board_Thermistor_Over_Temp	3	24	1	@1+	0	1	0	1
VisionCo_On_Board_Thermistor_Over_Temp	3.1	25	1	@1+	0	1	0	1
TR_On_Board_Thermistor_Over_Temp	3.2	26	1	@1+	0	1	0	1
IMAGE2P_On_Board_Thermistor_Over_Temp	3.3	27	1	@1+	0	1	0	1
VisionCo_MIPS_Over_Temp	3.4	28	1	@1+	0	1	0	1
VisionCo_VMP_Over_Temp	3.5	29	1	@1+	0	1	0	1
VisionCo_Imager_Over_Temp	3.6	30	1	@1+	0	1	0	1
VisionCo_Boot_OverTemp_Fault	3.7	31	1	@1+	0	1	0	1
VisionCo_DDR_Over_Temp	4	32	1	@1+	0	1	0	1
CPYR agent_On_Board_Thermistor_Out_Of_Range	4.1	33	1	@1+	0	1	0	1
VisionCo_On_Board_Thermistor_Out_Of_Range	4.2	34	1	@1+	0	1	0	1
IMAGE2P_On_Board_Thermistor_Out_Of_Range	4.3	35	1	@1+	0	1	0	1
TR_On_Board_Thermistor_Out_Of_Range	4.4	36	1	@1+	0	1	0	1
BatteryLowFault	4.5	37	1	@1+	0	1	0	1
Reserved	4.6	38	1	@1+	0	1	0	1
Reserved	4.7	39	1	@1+	0	1	0	1
DTC_F00049_AURIX_Fault	5	40	1	@1+	0	1	0	1
DTC_F00042_AURIX_Fault	5.1	41	1	@1+	0	1	0	1
Reserved	5.2	42	1	@1+	0	1	0	1
Heater_Short_Across_Cycles	5.3	43	1	@1+	0	1	0	1
Heater_Output_Short_Circuit	5.4	44	1	@1+	0	1	0	1
Heater_Output_Open_Circuit	5.5	45	1	@1+	0	1	0	1
Heater_Short_Fail_Across_Cycles	5.6	46	1	@1+	0	1	0	1
CHMSL_camera_DEC_amplifier_ShortToBatt	5.7	47	1	@1+	0	1	0	1
AUX_camera_DEC_amplifier_ShortToBatt	6	48	1	@1+	0	1	0	1
LKS_Switch_Stuck_On_Fault	6.1	49	1	@1+	0	1	0	1

APA_Switch_Stuck_On_Fault	6.2	50	1	@1+	0	1	0	1
_360_Switch_Stuck_On_Fault	6.3	51	1	@1+	0	1	0	1
BPA_Switch_Stuck_On_Fault	6.4	52	1	@1+	0	1	0	1
L_BLIS_LED_ShortToBatt	6.5	53	1	@1+	0	1	0	1
R_BLIS_LED_ShortToBatt	6.6	54	1	@1+	0	1	0	1
PARK_AID_LED_ShortToBatt	6.7	55	1	@1+	0	1	0	1
L_BLIS_LED_ShortToGnd	7	56	1	@1+	0	1	0	1
R_BLIS_LED_ShortToGnd	7.1	57	1	@1+	0	1	0	1
PARK_AID_LED_ShortToGnd	7.2	58	1	@1+	0	1	0	1
Reserved	7.3	59	1	@1+	0	1	0	1
AlgoRunner_Error_Fault	7.4	60	1	@1+	0	1	0	1
Reserved	7.5	61	1	@1+	0	1	0	1
Reserved	7.6	62	1	@1+	0	1	0	1
Pmic_Image2_Reset_Fault	7.7	63	1	@1+	0	1	0	1
CPYR agent_ENET_Message_CRC_Err	8	64	1	@1+	0	1	0	1
CPYR agent_ENET_Missing_Message_Err	8.1	65	1	@1+	0	1	0	1
CPYR agent_ENET_Missing_Data_Err	8.2	66	1	@1+	0	1	0	1
CPYR agent_ENET_MessageReception_Err	8.3	67	1	@1+	0	1	0	1
CPYR agent_ENET_VersionMismatch_Err	8.4	68	1	@1+	0	1	0	1
CPYR agent_Application_CRC_Fault	8.5	69	1	@1+	0	1	0	1
CPYR agent_Callibration_CRC_Fault	8.6	70	1	@1+	0	1	0	1
CPYR agent_Cal_Complete_And_Compatible	8.7	71	1	@1+	0	1	0	1
VisionCo_Cal_Complete_And_Compatible_SWP3	9	72	1	@1+	0	1	0	1
VisionCo_MESP_Coding_failure	9.1	73	1	@1+	0	1	0	1
VisionCo_Application_Complete_And_Compatible	9.2	74	1	@1+	0	1	0	1
VisionCo_Application_Checksum	9.3	75	1	@1+	0	1	0	1
VisionCo_Calibration_Checksum	9.4	76	1	@1+	0	1	0	1
IMAGE2P_Appl_Complete_And_Compatible	9.5	77	1	@1+	0	1	0	1
CPYR agent_Spi_Frame_Format_Error	9.6	78	1	@1+	0	1	0	1
CPYR agent_Spi_Msg_Format_Error	9.7	79	1	@1+	0	1	0	1
CPYR agent_spi_msg_timeout	a.0	80	1	@1+	0	1	0	1
CPYR agent_Spi_Tx_Error	a.1	81	1	@1+	0	1	0	1
CPYR agent_Spi_CRC_Error	a.2	82	1	@1+	0	1	0	1
Spi_version_mismatch	a.3	83	1	@1+	0	1	0	1
Spi_Ipc_Comm_Reset	a.4	84	1	@1+	0	1	0	1
VisionCo_Spi_Frame_Format_Error	a.5	85	1	@1+	0	1	0	1
VISIONCO_Spi_Msg_Format_Error	a.6	86	1	@1+	0	1	0	1
VisionCo_Spi_Msg_Timeout	a.7	87	1	@1+	0	1	0	1
VisionCo_SPI_CRC_ERROR	b.0	88	1	@1+	0	1	0	1
VisionCo_Serial_Flash_Power_Up_Fail	b.1	89	1	@1+	0	1	0	1
TR_Aurix_FirstTimePowerUpFlashCheck	b.2	90	1	@1+	0	1	0	1
Aurix_OTA_Flash_Fail	b.3	91	1	@1+	0	1	0	1
KAM_Design_Read_Fail_Fault	b.4	92	1	@1+	0	1	0	1
KAM_Design_Write_Fail_Fault	b.5	93	1	@1+	0	1	0	1
VisionCo_DDR_Fault	b.6	94	1	@1+	0	1	0	1

Reserved	b.7	95	1	@1+	0	1	0	1
VisionCo_Boot_Fault	c.0	96	1	@1+	0	1	0	1
VisionCo_Core_Fault	c.1	97	1	@1+	0	1	0	1
VisionCo_Bist_Memory_Fault	c.2	98	1	@1+	0	1	0	1
Reserved	c.3	99	1	@1+	0	1	0	1
VisionCo_Init_Timeout_Fault	c.4	100	1	@1+	0	1	0	1
VisionCo_Core_Dump_Fault	c.5	101	1	@1+	0	1	0	1
Parameters_Invalid	c.6	102	1	@1+	0	1	0	1
Unreasonable_Radar_Data	c.7	103	1	@1+	0	1	0	1
FWC_Camera_Wrong_Assembly_Fault	d.0	104	1	@1+	0	1	0	1
AEB_shutdown_Invalid_Reason	d.1	105	1	@1+	0	1	0	1
AEB_shutdown_7_POS_DIFF_KAN_VAL	d.2	106	1	@1+	0	1	0	1
AEB_Shutdown_9_ROI_DIFF_VC_FCV	d.3	107	1	@1+	0	1	0	1
AEB_Shutdown_6_FCV_FCW_TTC	d.4	108	1	@1+	0	1	0	1
AEB_Shutdown_25_CO_FOF_DIFF	d.5	109	1	@1+	0	1	0	1
VISIONCO_GEN_30_DT_ERR	d.6	110	1	@1+	0	1	0	1
VISIONCO_GEN_18_IMAGER_DATA_ROWS_COLS	d.7	111	1	@1+	0	1	0	1
VISIONCO_GEN_4_YAW_HORIZON_DEVIATION	e.0	112	1	@1+	0	1	0	1
VISIONCO_GEN_26_CODE_CRC	e.1	113	1	@1+	0	1	0	1
VISIONCO_GEN_13_OLD_EGODATA_LAT	e.2	114	1	@1+	0	1	0	1
Reserved	e.3	115	1	@1+	0	1	0	1
Reserved	e.4	116	1	@1+	0	1	0	1
Reserved	e.5	117	1	@1+	0	1	0	1
VISIONCO_GEN_31_AEB_PARAM_FFI	e.6	118	1	@1+	0	1	0	1
VISIONCO_GEN_29_RFC_ERR	e.7	119	1	@1+	0	1	0	1
VISIONCO_GEN_11_INPUT_CRC_MISMATCH	f.0	120	1	@1+	0	1	0	1
VISIONCO_GEN_34_Corrupted_Vehicle_Info_Input_Signals	f.1	121	1	@1+	0	1	0	1
AEB_Shutdown_27_Fcv_Sig_Ffi	f.2	122	1	@1+	0	1	0	1
VISIONCO_GEN_54_ACC_STL_EXECUTION_FAILED	f.3	123	1	@1+	0	1	0	1
Reserved	f.4	124	1	@1+	0	1	0	1
Missing_Camera_Callibration_Fault	f.5	125	1	@1+	0	1	0	1
FWC_Camera_Type_Fault	f.6	126	1	@1+	0	1	0	1
Reserved	f.7	127	1	@1+	0	1	0	1
NTSC_Encoder_amplifier_ShortToBatt	10	128	1	@1+	0	1	0	1
Reserved	10.1	129	1	@1+	0	1	0	1
Reserved	10.2	130	1	@1+	0	1	0	1
Reserved	10.3	131	1	@1+	0	1	0	1
Full_Blockage_Fault	10.4	132	1	@1+	0	1	0	1
Partial_Blockage_Fault	10.5	133	1	@1+	0	1	0	1
Blurred_Image_Fault	10.6	134	1	@1+	0	1	0	1
Splashes_Fault	10.7	135	1	@1+	0	1	0	1
Low_Sun_Fault	11	136	1	@1+	0	1	0	1
Sun_Rays_Fault	11.1	137	1	@1+	0	1	0	1
Foggy_Spots_Fault	11.2	138	1	@1+	0	1	0	1
Spot_Rays_Fault	11.3	139	1	@1+	0	1	0	1

Smeared_Spots_Fault	11.4	140	1	@1+	0	1	0	1
SelfGlare_Fault	11.5	141	1	@1+	0	1	0	1
Low_Visibility_Fault	11.6	142	1	@1+	0	1	0	1
Vision_OutofFocus_Fault	11.7	143	1	@1+	0	1	0	1
Vision_OutofCalibration_Fault	12	144	1	@1+	0	1	0	1
TD_BlockageTest	12.1	145	1	@1+	0	1	0	1
VisionCo_Fatal_Error_APP_CPS_STL_FAILED	12.2	146	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Error_Fault	12.3	147	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Fs_Error_Fault	12.4	148	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Calibration_Error_Fault	12.5	149	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Init_Failed_Fault	12.6	150	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Init_Camera_Init_Fault	12.7	151	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_I2C_Video_Grab_Failed_Fault	13	152	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_I2C_Camera_Self_Reset_Fault	13.1	153	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_I2C_Timeout_Error_Fault	13.2	154	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Pattern_Test_Fault	13.3	155	1	@1+	0	1	0	1
VisionCo_Fatal_Error_App_Cam_Params_Ccft_Crc_Failed_Fault	13.4	156	1	@1+	0	1	0	1
VisionCo_Fatal_Error_Pll_Comparison_Error_Fault	13.5	157	1	@1+	0	1	0	1
VisionCo_Fatal_Error_Pv_General_Error_Fault	13.6	158	1	@1+	0	1	0	1
VisionCo_Fatal_Error_Pv_Verificattion_Error_Fault	13.7	159	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Error_Fault	14	160	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Error_Fault	14.1	161	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Setting_Fault	14.2	162	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Registry_Fault	14.3	163	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Registry_Fault	14.4	164	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Fault	14.5	165	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Fault	14.6	166	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Reg_Fault	14.7	167	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Open_Blackbox_Fault	15	168	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Ep_Fault	15.1	169	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Ep_Fault	15.2	170	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Create_Logger_Fault	15.3	171	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_IL_Fault	15.4	172	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Check_Reg_Versions_Fault	15.5	173	1	@1+	0	1	0	1
VISIONCO_GEN_C_R_Error	15.6	174	1	@1+	0	1	0	1
Init_Vdi_Failure	15.7	175	1	@1+	0	1	0	1
Init_I2C_Failure	16	176	1	@1+	0	1	0	1
Grab_No_Free_Header_Failure	16.1	177	1	@1+	0	1	0	1
Grab_Issue_Failure	16.2	178	1	@1+	0	1	0	1
Grab_Acquire_Failure	16.3	179	1	@1+	0	1	0	1
Grab_Release_Failure	16.4	180	1	@1+	0	1	0	1
VisionCo_Cam_Mal_Func	16.5	181	1	@1+	0	1	0	1
VisionCo_Cam_Ctrl_Flt	16.6	182	1	@1+	0	1	0	1
VisionCo_Cam_Vdo_Flt	16.7	183	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Video_Error_Fault	17	184	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_I2C_Not_On_Time_Video_Error_Fault	17.1	185	1	@1+	0	1	0	1

VisionCo_Camera1_VideoErrorFlags_Logical_Video_Error_Fault	17.2	186	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Inconsistent_Frames_Video_Error_Fault	17.3	187	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Verification_Failure_Video_Error_Fault	17.4	188	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Invalid_Video_Header_Fault	17.5	189	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Error_Reg_Flag_Fault	17.6	190	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Reg_Write_Hil_Failure_Error_Fault	17.7	191	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Zero_Histogram_Error_Fault	18	192	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Sum_Not_Match_Histogram_Error_Fault	18.1	193	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Internal_Error_Fault	18.2	194	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_No_Buffer_Fault	18.3	195	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Buffer_Invalid_Format_Fault	18.4	196	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Timeout_Fault	18.5	197	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Overflow_Fault	18.6	198	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Underflow_Fault	18.7	199	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Fifo_Fault	19	200	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Weights_Fault	19.1	201	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Gamma_Fault	19.2	202	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Hist_Fault	19.3	203	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Start_After_Start_Fault	19.4	204	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Core_Id_Fault	19.5	205	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Frame_Count_Read_Fail_Fault	19.6	206	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Histograms_Mismatch_Fault	19.7	207	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Ccft_Decisions_Mismatch_Fault	1a.0	208	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Header_Verification_Error_Fault	1a.1	209	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Shutdown_Fault	1a.2	210	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Shot_Out_Of_Sync_Fault	1a.3	211	1	@1+	0	1	0	1
VisionCo_Camera1_temperature_Fault	1a.4	212	1	@1+	0	1	0	1
VisionCo_Error_Out_Signal_Internal_Fault	1a.5	213	1	@1+	0	1	0	1
VisionCo_Error_Out_Temperature_Fault	1a.6	214	1	@1+	0	1	0	1
PCM_LostCom	1a.7	215	1	@1+	0	1	0	1
TCM_LostCom	1b.0	216	1	@1+	0	1	0	1
GSM_LostCom	1b.1	217	1	@1+	0	1	0	1
ABS_LostCom	1b.2	218	1	@1+	0	1	0	1
SECM_LostCom	1b.3	219	1	@1+	0	1	0	1
PSCM_LostCom	1b.4	220	1	@1+	0	1	0	1
BCM_LostCom	1b.5	221	1	@1+	0	1	0	1
RCM_LostCom	1b.6	222	1	@1+	0	1	0	1
IPC_LostCom	1b.7	223	1	@1+	0	1	0	1
HUD_LostCom	1c.0	224	1	@1+	0	1	0	1
SCCM_LostCom	1c.1	225	1	@1+	0	1	0	1
ACM_LostCom	1c.2	226	1	@1+	0	1	0	1
DDM_LostCom	1c.3	227	1	@1+	0	1	0	1
PDM_LostCom	1c.4	228	1	@1+	0	1	0	1
APIM_LostCom	1c.5	229	1	@1+	0	1	0	1
Reserved	1c.6	230	1	@1+	0	1	0	1

PCMH_LostCom	1c.7	231	1	@1+	0	1	0	1
VDM_LostCom	1d.0	232	1	@1+	0	1	0	1
PCM_InvalidData	1d.1	233	1	@1+	0	1	0	1
TCM_InvalidData	1d.2	234	1	@1+	0	1	0	1
GSM_InvalidData	1d.3	235	1	@1+	0	1	0	1
ABS_InvalidData	1d.4	236	1	@1+	0	1	0	1
SECM_InvalidData	1d.5	237	1	@1+	0	1	0	1
PSCM_InvalidData	1d.6	238	1	@1+	0	1	0	1
BCM_InvalidData	1d.7	239	1	@1+	0	1	0	1
IPC_InvaildData	1e.0	240	1	@1+	0	1	0	1
SCCM_InvalidData	1e.1	241	1	@1+	0	1	0	1
RCM_InvalidData	1e.2	242	1	@1+	0	1	0	1
DDM_InvalidData	1e.3	243	1	@1+	0	1	0	1
PDM_InvalidData	1e.4	244	1	@1+	0	1	0	1
HVAC_InvalidData	1e.5	245	1	@1+	0	1	0	1
APIM_InvalidData	1e.6	246	1	@1+	0	1	0	1
Reserved	1e.7	247	1	@1+	0	1	0	1
Reserved	1f.0	248	1	@1+	0	1	0	1
PCMH_InvalidData	1f.1	249	1	@1+	0	1	0	1
VDM_InvalidData	1f.2	250	1	@1+	0	1	0	1
IPC_InvaildRollCnt	1f.3	251	1	@1+	0	1	0	1
RCM_InvalidRollCnt	1f.4	252	1	@1+	0	1	0	1
RCM_InvalidChkSum	1f.5	253	1	@1+	0	1	0	1
reserved	1f.6	254	1	@1+	0	1	0	1
VCAN_Bus_Off_Fault	1f.7	255	1	@1+	0	1	0	1
FLRCAN_Bus_Off_Fault	20	256	1	@1+	0	1	0	1
SIDERADAR_RF_LF_CAN_Bus_Off_Fault	20.1	257	1	@1+	0	1	0	1
SIDERADAR_RR_LR_CAN_Bus_Off_Fault	20.2	258	1	@1+	0	1	0	1
MAINRADAR_PCAN_Missing_Message_Fault	20.3	259	1	@1+	0	1	0	1
MAINRADAR_PCAN_Range_Check_Fault	20.4	260	1	@1+	0	1	0	1
Reserved	20.5	261	1	@1+	0	1	0	1
SIDERADAR_FR_PCAN_Range_Check_Fault	20.6	262	1	@1+	0	1	0	1
SIDERADAR_FL_PCAN_Range_Check_Fault	20.7	263	1	@1+	0	1	0	1
SIDERADAR_RL_PCAN_Range_Check_Fault	21	264	1	@1+	0	1	0	1
SIDERADAR_RR_PCAN_Range_Check_Fault	21.1	265	1	@1+	0	1	0	1
CarCfg_Implausible_Denied_Param_23	21.2	266	1	@1+	0	1	0	1
CarCfg_Implausible_Denied_Param_96	21.3	267	1	@1+	0	1	0	1
SIDERADAR_Software_Compatibility_Fault	21.4	268	1	@1+	0	1	0	1
MAINRADAR_Software_Compatibility_Fault	21.5	269	1	@1+	0	1	0	1
GCCConfigurationNotconfigured_CarCfg_Undefined_Denied	21.6	270	1	@1+	0	1	0	1
SIDERADAR_Invalid_Hardware_configuration	21.7	271	1	@1+	0	1	0	1
ABS_InvalidRollCnt	22	272	1	@1+	0	1	0	1
ABS_InvalidChkSum	22.1	273	1	@1+	0	1	0	1
Brake_Temperature_Too_High	22.2	274	1	@1+	0	1	0	1
Camera_Module_Alignment_or_Adjustment_Incorrect_RearCamera_	22.3	275	1	@1+	0	1	0	1

DPAC_Vision_System_Camera_Alignment_or_Adjustment_Incorrect	22.4	276	1	@1+	0	1	0	1
Internal_Control_Module_Software_Incompatibility	22.5	277	1	@1+	0	1	0	1
Initial_Configuration_Not_Complete	22.6	278	1	@1+	0	1	0	1
Control_Module_Configuration_Incompatible	22.7	279	1	@1+	0	1	0	1
LIN_Rear_Camera_Bus_Failure	23	280	1	@1+	0	1	0	1
LIN_Rear_Camera_Not_Configured	23.1	281	1	@1+	0	1	0	1
LIN_Rear_Camera_Fault	23.2	282	1	@1+	0	1	0	1
Rear_Camera2_Bus_Failure	23.3	283	1	@1+	0	1	0	1
Rear_Camera2_Not_Configured	23.4	284	1	@1+	0	1	0	1
Rear_Camera2_Fault	23.5	285	1	@1+	0	1	0	1
Front_USS_HSD_FailurontUSS_Ckt_ShortToGnd	23.6	286	1	@1+	0	1	0	1
Rear_USS_HSD_FailureRearUSS_Ckt_ShortToGnd	23.7	287	1	@1+	0	1	0	1
Front_USS_HSD_ShortToBattFrontUSS_Ckt_ShortToBatt	24	288	1	@1+	0	1	0	1
Rear_USS_HSD_ShortToBattRearUSS_Ckt_ShortTobBatt	24.1	289	1	@1+	0	1	0	1
Ignition_Status_Unknown_Fault	24.2	290	1	@1+	0	1	0	1
Ignition_Status_Missing_Fault	24.3	291	1	@1+	0	1	0	1
Veh_V_ActlEng_Missing_Fault	24.4	292	1	@1+	0	1	0	1
VehVActlEng_D_Qf_Missing_Fault	24.5	293	1	@1+	0	1	0	1
VehVActlEng_D_Qf_Fault	24.6	294	1	@1+	0	1	0	1
VehVActlEng_D_Qf_No_data_exists_Fault	24.7	295	1	@1+	0	1	0	1
VISIONCO_MES_49_Data_Path_Corruption	25	296	1	@1+	0	1	0	1
VISIONCO_MES_50_VMP_SANITY_FAILED	25.1	297	1	@1+	0	1	0	1
Reserved	25.2	298	1	@1+	0	1	0	1
VISIONCO_40_PED_SANITY_FAULT	25.3	299	1	@1+	0	1	0	1
VISIONCO_39_PED_OVERLAP_FAULT	25.4	300	1	@1+	0	1	0	1
Reserved	25.5	301	1	@1+	0	1	0	1
Reserved	25.6	302	1	@1+	0	1	0	1
Reserved	25.7	303	1	@1+	0	1	0	1
Reserved	26	304	1	@1+	0	1	0	1
Reserved	26.1	305	1	@1+	0	1	0	1
VISIONCO_48_ALGO_PITCH_MISMATCH	26.2	306	1	@1+	0	1	0	1
VISIONCO_45_ROAD_SFCONF_VALIDATION	26.3	307	1	@1+	0	1	0	1
VISIONCO_42_ROAD_PREDICTION_VALIDATION	26.4	308	1	@1+	0	1	0	1
VISIONCO_33_ROAD_LANE_DISQUALIFICATINO_FAILED	26.5	309	1	@1+	0	1	0	1
TrlrLampCnnct_B_Actl_Missing_Fault	26.6	310	1	@1+	0	1	0	1
BLISLEDStatDriverSide_MissingMsg_Fault	26.7	311	1	@1+	0	1	0	1
BLISLEDStatPassSide_MissingMsg_Fault	27	312	1	@1+	0	1	0	1
BLISLEDStatDriverSide_DCUFault	27.1	313	1	@1+	0	1	0	1
BLISLEDStatPassSide_DCUFault	27.2	314	1	@1+	0	1	0	1
BLISLEDStatDriverSide_ON_OFF_Mismatch_Fault	27.3	315	1	@1+	0	1	0	1
BLISLEDStatPassSide_ON_OFF_Mismatch_Fault	27.4	316	1	@1+	0	1	0	1
IgnKeyType_D_Actl_Unknown_Fault	27.5	317	1	@1+	0	1	0	1
IgnKeyType_D_ActlMissing_Fault	27.6	318	1	@1+	0	1	0	1
Parklamp_Status_Missing_Fault	27.7	319	1	@1+	0	1	0	1
TR_Application_CRC_Fault	28	320	1	@1+	0	1	0	1
Reserved	28.1	321	1	@1+	0	1	0	1

TR_DDR_RAM_CRC_Fault	28.2	322	1	@1+	0	1	0	1
Reserved	28.3	323	1	@1+	0	1	0	1
Reserved	28.4	324	1	@1+	0	1	0	1
Reserved	28.5	325	1	@1+	0	1	0	1
Reserved	28.6	326	1	@1+	0	1	0	1
Reserved	28.7	327	1	@1+	0	1	0	1
Reserved	29	328	1	@1+	0	1	0	1
Reserved	29.1	329	1	@1+	0	1	0	1
Reserved	29.2	330	1	@1+	0	1	0	1
Reserved	29.3	331	1	@1+	0	1	0	1
Reserved	29.4	332	1	@1+	0	1	0	1
Reserved	29.5	333	1	@1+	0	1	0	1
Reserved	29.6	334	1	@1+	0	1	0	1
Reserved	29.7	335	1	@1+	0	1	0	1
TR_INT_OSC_TEST_FAIL	2a.0	336	1	@1+	0	1	0	1
Perf_DPAC_Process_Fault	2a.1	337	1	@1+	0	1	0	1
Perf_FWC_Process_Fault	2a.2	338	1	@1+	0	1	0	1
Reserved	2a.3	339	1	@1+	0	1	0	1
TR_Stack_Overflow_Fault	2a.4	340	1	@1+	0	1	0	1
Reserved	2a.5	341	1	@1+	0	1	0	1
Reserved	2a.6	342	1	@1+	0	1	0	1
Reserved	2a.7	343	1	@1+	0	1	0	1
Reserved	2b.0	344	1	@1+	0	1	0	1
TR_DDR_Fault	2b.1	345	1	@1+	0	1	0	1
TR_Boot_Fault	2b.2	346	1	@1+	0	1	0	1
Reserved	2b.3	347	1	@1+	0	1	0	1
TR_Appl_Complete_And_Compatible	2b.4	348	1	@1+	0	1	0	1
TR_Core_Exception_Fault	2b.5	349	1	@1+	0	1	0	1
TR_Control_ConfigReg_Fault	2b.6	350	1	@1+	0	1	0	1
reserved	2b.7	351	1	@1+	0	1	0	1
TR_ENET_Message_CRC_Err	2c.0	352	1	@1+	0	1	0	1
TR_ENET_Missing_Message_Err	2c.1	353	1	@1+	0	1	0	1
TR_ENET_Missing_Data_Err	2c.2	354	1	@1+	0	1	0	1
TR_ENET_MessageReception_Err	2c.3	355	1	@1+	0	1	0	1
TR_ENET_VersionMismatch_Err	2c.4	356	1	@1+	0	1	0	1
TR_ENET_Communication_Drv_Err	2c.5	357	1	@1+	0	1	0	1
Reserved	2c.6	358	1	@1+	0	1	0	1
Reserved	2c.7	359	1	@1+	0	1	0	1
Reserved	2d.0	360	1	@1+	0	1	0	1
QSPI_Flash_Illegal_Instruction_Err	2d.1	361	1	@1+	0	1	0	1
QSPI_Communication_Drv_Err	2d.2	362	1	@1+	0	1	0	1
Reserved	2d.3	363	1	@1+	0	1	0	1
Reserved	2d.4	364	1	@1+	0	1	0	1
Reserved	2d.5	365	1	@1+	0	1	0	1
DTC_F00042_PERFPROC_IntMemFail	2d.6	366	1	@1+	0	1	0	1

DTC_F00049_PERFPROC_IntElectronicFail	2d.7	367	1	@1+	0	1	0	1
LEFT_CAMERA_FAULT_CKT_ShortToBatt	2e.0	368	1	@1+	0	1	0	1
RIGHT_CAMERA_FAULT_CKT_ShortToBatt	2e.1	369	1	@1+	0	1	0	1
BA_Switch_Stuck_On_Fault	2e.2	370	1	@1+	0	1	0	1
Reserved	2e.3	371	1	@1+	0	1	0	1
Reserved	2e.4	372	1	@1+	0	1	0	1
Reserved	2e.5	373	1	@1+	0	1	0	1
Reserved	2e.6	374	1	@1+	0	1	0	1
Reserved	2e.7	375	1	@1+	0	1	0	1
Reserved	2f.0	376	1	@1+	0	1	0	1
Reserved	2f.1	377	1	@1+	0	1	0	1
Reserved	2f.2	378	1	@1+	0	1	0	1
Reserved	2f.3	379	1	@1+	0	1	0	1
Reserved	2f.4	380	1	@1+	0	1	0	1
Reserved	2f.5	381	1	@1+	0	1	0	1
Reserved	2f.6	382	1	@1+	0	1	0	1
Reserved	2f.7	383	1	@1+	0	1	0	1
VISIONCO_GEN_21_CAMERA_HEIGHT_LIMIT_EXCEEDED	30	384	1	@1+	0	1	0	1
VISIONCO_44_ROAD_SANITY_CORRECTION_VALIDATION	30.1	385	1	@1+	0	1	0	1
VISIONCO_53_DECELERATION_DBS_OVERLAP_FAULT	30.2	386	1	@1+	0	1	0	1
VISIONCO_GEN_24_CCFT_DIFF_RESULTS	30.3	387	1	@1+	0	1	0	1
GearRvrse_D_Actl_Missing_Fault	30.4	388	1	@1+	0	1	0	1
SOD_TCM_LostComm	30.5	389	1	@1+	0	1	0	1
WheelData_Missing_Fault	30.6	390	1	@1+	0	1	0	1
SOD_BCM_LostComm	30.7	391	1	@1+	0	1	0	1
SOD_RCM_LostCom	31	392	1	@1+	0	1	0	1
SOD_ABS_InvalidData	31.1	393	1	@1+	0	1	0	1
SOD_TCM_InvalidData	31.2	394	1	@1+	0	1	0	1
GearRvrse_D_Actl_invalid_Fault	31.3	395	1	@1+	0	1	0	1
SOD_IPC_LostCom	31.4	396	1	@1+	0	1	0	1
AEB_Inhibited_Fault_Flag	31.5	397	1	@1+	0	1	0	1
AEB_Validity_Fault_Flag	31.6	398	1	@1+	0	1	0	1
LaneFusion_Validity_Fault_Flag	31.7	399	1	@1+	0	1	0	1
Reserved	32	400	1	@1+	0	1	0	1
Reserved	32.1	401	1	@1+	0	1	0	1
Reserved	32.2	402	1	@1+	0	1	0	1
FWC_CAMERA_FAULT_CKT_ShortToBatt	32.3	403	1	@1+	0	1	0	1
LeftRearSOD_Ckt_ShortToBatt	32.4	404	1	@1+	0	1	0	1
LeftRearSOD_Ckt_ShortToGnd	32.5	405	1	@1+	0	1	0	1
RightRearSOD_Ckt_ShortToBatt	32.6	406	1	@1+	0	1	0	1
RightRearSOD_Ckt_ShortToGnd	32.7	407	1	@1+	0	1	0	1
LeftFrontSOD_Ckt_ShortToBatt	33	408	1	@1+	0	1	0	1
LeftFrontSOD_Ckt_ShortToGnd	33.1	409	1	@1+	0	1	0	1
RightFrontSOD_Ckt_ShortToBatt	33.2	410	1	@1+	0	1	0	1
RightFrontSOD_Ckt_ShortToGnd	33.3	411	1	@1+	0	1	0	1

FRONT_CAMERA_FAULT_CKT_ShortToBatt	33.4	412	1	@1+	0	1	0	1
REAR_CAMERA_FAULT_CKT_ShortToBatt	33.5	413	1	@1+	0	1	0	1
Reserved	33.6	414	1	@1+	0	1	0	1
DTC_915E31_PerfProcessor	33.7	415	1	@1+	0	1	0	1
DTC_9B4008_FAULT_CONDITION	34	416	1	@1+	0	1	0	1
DTC_9B4096_FAULT_CONDITION	34.1	417	1	@1+	0	1	0	1
DTC_9B4098_FAULT_CONDITION	34.2	418	1	@1+	0	1	0	1
DTC_9B4208_FAULT_CONDITION	34.3	419	1	@1+	0	1	0	1
DTC_9B4296_FAULT_CONDITION	34.4	420	1	@1+	0	1	0	1
DTC_9B4298_FAULT_CONDITION	34.5	421	1	@1+	0	1	0	1
DTC_929C08_FAULT_CONDITION	34.6	422	1	@1+	0	1	0	1
DTC_929C96_FAULT_CONDITION	34.7	423	1	@1+	0	1	0	1
DTC_929C98_FAULT_CONDITION	35	424	1	@1+	0	1	0	1
DTC_9B4808_FAULT_CONDITION	35.1	425	1	@1+	0	1	0	1
DTC_9B4896_FAULT_CONDITION	35.2	426	1	@1+	0	1	0	1
DTC_9B4898_FAULT_CONDITION	35.3	427	1	@1+	0	1	0	1
DTC_9B5008_FAULT_CONDITION	35.4	428	1	@1+	0	1	0	1
DTC_9B5096_FAULT_CONDITION	35.5	429	1	@1+	0	1	0	1
DTC_9B5098_FAULT_CONDITION	35.6	430	1	@1+	0	1	0	1
DTC_93F308_FAULT_CONDITION	35.7	431	1	@1+	0	1	0	1
DTC_93F396_FAULT_CONDITION	36	432	1	@1+	0	1	0	1
DTC_93F398_FAULT_CONDITION	36.1	433	1	@1+	0	1	0	1
DTC_9B3608_FAULT_CONDITION	36.2	434	1	@1+	0	1	0	1
DTC_9B3696_FAULT_CONDITION	36.3	435	1	@1+	0	1	0	1
DTC_9B3698_FAULT_CONDITION	36.4	436	1	@1+	0	1	0	1
DTC_9B3808_FAULT_CONDITION	36.5	437	1	@1+	0	1	0	1
DTC_9B3896_FAULT_CONDITION	36.6	438	1	@1+	0	1	0	1
DTC_9B3898_FAULT_CONDITION	36.7	439	1	@1+	0	1	0	1
DTC_929D08_FAULT_CONDITION	37	440	1	@1+	0	1	0	1
DTC_929D96_FAULT_CONDITION	37.1	441	1	@1+	0	1	0	1
DTC_929D98_FAULT_CONDITION	37.2	442	1	@1+	0	1	0	1
DTC_9B4408_FAULT_CONDITION	37.3	443	1	@1+	0	1	0	1
DTC_9B4496_FAULT_CONDITION	37.4	444	1	@1+	0	1	0	1
DTC_9B4498_FAULT_CONDITION	37.5	445	1	@1+	0	1	0	1
DTC_9B4608_FAULT_CONDITION	37.6	446	1	@1+	0	1	0	1
DTC_9B4696_FAULT_CONDITION	37.7	447	1	@1+	0	1	0	1
DTC_9B4698_FAULT_CONDITION	38	448	1	@1+	0	1	0	1
DTC_93F408_FAULT_CONDITION	38.1	449	1	@1+	0	1	0	1
DTC_93F496_FAULT_CONDITION	38.2	450	1	@1+	0	1	0	1
DTC_93F498_FAULT_CONDITION	38.3	451	1	@1+	0	1	0	1
DTC_93039E_FAULT_CONDITION	38.4	452	1	@1+	0	1	0	1
DTC_915E31_FAULT_CONDITION	38.5	453	1	@1+	0	1	0	1
DTC_948E31_FAULT_CONDITION	38.6	454	1	@1+	0	1	0	1
DTC_92BE31_FAULT_CONDITION	38.7	455	1	@1+	0	1	0	1
DTC_92BF31_FAULT_CONDITION	39	456	1	@1+	0	1	0	1
DTC_F00042_IMAGE2	39.1	457	1	@1+	0	1	0	1

DTC_F00089_IMAGE2	39.2	458	1	@1+	0	1	0	1
SIDERADAR_FL_PCAN_Missing_Message_Fault	39.3	459	1	@1+	0	1	0	1
SIDERADAR_FR_PCAN_Missing_Message_Fault	39.4	460	1	@1+	0	1	0	1
SIDERADAR_RL_PCAN_Missing_Message_Fault	39.5	461	1	@1+	0	1	0	1
SIDERADAR_RR_PCAN_Missing_Message_Fault	39.6	462	1	@1+	0	1	0	1
DSCM_InvalidData	39.7	463	1	@1+	0	1	0	1
DSCM_InvalidRollCnt	3a.0	464	1	@1+	0	1	0	1
DSCM_InvalidChkSum	3a.1	465	1	@1+	0	1	0	1
TRM_InvalidData	3a.2	466	1	@1+	0	1	0	1
DSCM_LostComm	3a.3	467	1	@1+	0	1	0	1
HVAC_LostComm	3a.4	468	1	@1+	0	1	0	1
TRM_LostComm	3a.5	469	1	@1+	0	1	0	1
ATCM_LostComm	3a.6	470	1	@1+	0	1	0	1
HCM_LostComm	3a.7	471	1	@1+	0	1	0	1
Parklamp_Status_Unknown_Fault	3b.0	472	1	@1+	0	1	0	1
Litval_Unknown_Fault	3b.1	473	1	@1+	0	1	0	1
Litval_Missing_Fault	3b.2	474	1	@1+	0	1	0	1
TurnLghtSwitch_D_Stat_Missing_Fault	3b.3	475	1	@1+	0	1	0	1
Btt_L_Actl_MissingMsg_Fault	3b.4	476	1	@1+	0	1	0	1
ATD_Mismatch_Fault	3b.5	477	1	@1+	0	1	0	1
TLM_Mismatch_Fault	3b.6	478	1	@1+	0	1	0	1
TBM_Mismatch_Fault	3b.7	479	1	@1+	0	1	0	1
Reserved	3c.0	480	1	@1+	0	1	0	1
WhlDirRr_D_Actl_Missing	3c.1	481	1	@1+	0	1	0	1
WhlDirRl_D_Actl_Missing	3c.2	482	1	@1+	0	1	0	1
WhlDirFl_D_Actl_Missing	3c.3	483	1	@1+	0	1	0	1
WhlDirFr_D_Actl_Missing	3c.4	484	1	@1+	0	1	0	1
WhlDirRr_D_Actl_Failed_Fault	3c.5	485	1	@1+	0	1	0	1
WhlDirRl_D_Actl_Failed_Fault	3c.6	486	1	@1+	0	1	0	1
WhlDirFr_D_Actl_Failed_Fault	3c.7	487	1	@1+	0	1	0	1
WhlDirFl_D_Actl_Failed_Fault	3d.0	488	1	@1+	0	1	0	1
CtaBrk_D_Stat_Denied_Fault	3d.1	489	1	@1+	0	1	0	1
CtaBrk_D_Stat_Missing_Fault	3d.2	490	1	@1+	0	1	0	1
Isig_brake_timeout_fault	3d.3	491	1	@1+	0	1	0	1
Reserved	3d.4	492	1	@1+	0	1	0	1
CtaYBrkEnbl_B_Rq_Mismatch_Fault	3d.5	493	1	@1+	0	1	0	1
DVR_SELECT_STAT_Fault	3d.6	494	1	@1+	0	1	0	1
DTC_F00049_IMAGE2	3d.7	495	1	@1+	0	1	0	1
CMbBDeniedByABS	3e.0	496	1	@1+	0	1	0	1
CMbBDeniedByPCM	3e.1	497	1	@1+	0	1	0	1
CMbBImplausByABS	3e.2	498	1	@1+	0	1	0	1
CMbBImplausByPCM	3e.3	499	1	@1+	0	1	0	1
AGENT_InvalidData	3e.4	500	1	@1+	0	1	0	1
unused	3e.5	501	1	@1+	0	1	0	1
ACCImplausByABS	3e.6	502	1	@1+	0	1	0	1

ACCImplausByPCM	3e.7	503	1	@1+	0	1	0	1
ACCImplausAccelByABS	3f.0	504	1	@1+	0	1	0	1
ACCFailToCancelByPCM	3f.1	505	1	@1+	0	1	0	1
ACCDeniedbyABS	3f.2	506	1	@1+	0	1	0	1
SetDTC_ACCDeniedByDIM	3f.3	507	1	@1+	0	1	0	1
SetDTCImplausibleCMbB_ASILB	3f.4	508	1	@1+	0	1	0	1
TJAImplausByPSCM	3f.5	509	1	@1+	0	1	0	1
SetDTCImplausibleHA_FSM	3f.6	510	1	@1+	0	1	0	1
SetDTCImplausibleFAPA_FSM	3f.7	511	1	@1+	0	1	0	1

History_Fault_Status

History_Fault_Status-STNumber(77)-STVersion(1) **This message is used to report fault status information.**

Message History_Fault_Status shall be implemented as fault table defined in [5].

Signal Name	Description	StartBit	Size	Sign	Offset	Factor	Min	Max
VisionCo_3_3V_Supply_Out_Of_Range	0	0	1	@1+	0	1	0	1
VisionCo_1_8V_Supply_Out_Of_Range	0.1	1	1	@1+	0	1	0	1
VisionCo_1_1V_Supply_Out_Of_Range	0.2	2	1	@1+	0	1	0	1
VisionCo_1_0V_Supply_Out_Of_Range	0.3	3	1	@1+	0	1	0	1
TR_3_3V_Supply_Out_Of_Range	0.4	4	1	@1+	0	1	0	1
TR_1_8V_Supply_Out_Of_Range	0.5	5	1	@1+	0	1	0	1
TR_1_35V_Supply_Out_Of_Range	0.6	6	1	@1+	0	1	0	1
TR_1_0V_Supply_Out_Of_Range	0.7	7	1	@1+	0	1	0	1
IMAGE2P_PGOOD_Fail_Fault	1	8	1	@1+	0	1	0	1
FVC_CAMPWR_LDO_FailFRONT_CAMERA_FAULT_CKT_ShortToGnd	1.1	9	1	@1+	0	1	0	1
RVC_CAMPWR_LDO_FailREAR_CAMERA_FAULT_CKT_ShortToGnd	1.2	10	1	@1+	0	1	0	1
LH_Mirror_CAMPWR_LDO_FailLEFT_CAMERA_FAULT_CKT_ShortToGnd	1.3	11	1	@1+	0	1	0	1
RH_Mirror_CAMPWR_LDO_FailRIGHT_CAMERA_FAULT_CKT_ShortToGnd	1.4	12	1	@1+	0	1	0	1
FWC_CAMPWR_LDO_FailFWC_CAMERA_FAULT_CKT_ShortToGnd	1.5	13	1	@1+	0	1	0	1
Reserved	1.6	14	1	@1+	0	1	0	1
Reserved	1.7	15	1	@1+	0	1	0	1
Reserved	2	16	1	@1+	0	1	0	1
Reserved	2.1	17	1	@1+	0	1	0	1
Reserved	2.2	18	1	@1+	0	1	0	1
MAINRADAR_Output_Open_Circuit	2.3	19	1	@1+	0	1	0	1
FET_Fault_Across_cycles	2.4	20	1	@1+	0	1	0	1
FET_Fail_Across_Cycles	2.5	21	1	@1+	0	1	0	1
TR_SerialFlash_PowerUp_Fail	2.6	22	1	@1+	0	1	0	1
System_Over_Temperature	2.7	23	1	@1+	0	1	0	1
CPYR agent_On_Board_Thermistor_Over_Temp	3	24	1	@1+	0	1	0	1
VisionCo_On_Board_Thermistor_Over_Temp	3.1	25	1	@1+	0	1	0	1
TR_On_Board_Thermistor_Over_Temp	3.2	26	1	@1+	0	1	0	1
IMAGE2P_On_Board_Thermistor_Over_Temp	3.3	27	1	@1+	0	1	0	1
VisionCo_MIPS_Over_Temp	3.4	28	1	@1+	0	1	0	1
VisionCo_VMP_Over_Temp	3.5	29	1	@1+	0	1	0	1
VisionCo_Imager_Over_Temp	3.6	30	1	@1+	0	1	0	1

VisionCo_Boot_OverTemp_Fault	3.7	31	1	@1+	0	1	0	1
VisionCo_DDR_Over_Temp	4	32	1	@1+	0	1	0	1
CPYR agent_On_Board_Thermistor_Out_Of_Range	4.1	33	1	@1+	0	1	0	1
VisionCo_On_Board_Thermistor_Out_Of_Range	4.2	34	1	@1+	0	1	0	1
IMAGE2P_On_Board_Thermistor_Out_Of_Range	4.3	35	1	@1+	0	1	0	1
TR_On_Board_Thermistor_Out_Of_Range	4.4	36	1	@1+	0	1	0	1
BatteryLowFault	4.5	37	1	@1+	0	1	0	1
Reserved	4.6	38	1	@1+	0	1	0	1
Reserved	4.7	39	1	@1+	0	1	0	1
DTC_F00049_AURIX_Fault	5	40	1	@1+	0	1	0	1
DTC_F00042_AURIX_Fault	5.1	41	1	@1+	0	1	0	1
Reserved	5.2	42	1	@1+	0	1	0	1
Heater_Short_Across_Cycles	5.3	43	1	@1+	0	1	0	1
Heater_Output_Short_Circuit	5.4	44	1	@1+	0	1	0	1
Heater_Output_Open_Circuit	5.5	45	1	@1+	0	1	0	1
Heater_Short_Fail_Across_Cycles	5.6	46	1	@1+	0	1	0	1
CHMSL_camera_DEC_amplifier_ShortToBatt	5.7	47	1	@1+	0	1	0	1
AUX_camera_DEC_amplifier_ShortToBatt	6	48	1	@1+	0	1	0	1
LKS_Switch_Stuck_On_Fault	6.1	49	1	@1+	0	1	0	1
APA_Switch_Stuck_On_Fault	6.2	50	1	@1+	0	1	0	1
_360_Switch_Stuck_On_Fault	6.3	51	1	@1+	0	1	0	1
BPA_Switch_Stuck_On_Fault	6.4	52	1	@1+	0	1	0	1
L_BLIS_LED_ShortToBatt	6.5	53	1	@1+	0	1	0	1
R_BLIS_LED_ShortToBatt	6.6	54	1	@1+	0	1	0	1
PARK_AID_LED_ShortToBatt	6.7	55	1	@1+	0	1	0	1
L_BLIS_LED_ShortToGnd	7	56	1	@1+	0	1	0	1
R_BLIS_LED_ShortToGnd	7.1	57	1	@1+	0	1	0	1
PARK_AID_LED_ShortToGnd	7.2	58	1	@1+	0	1	0	1
Reserved	7.3	59	1	@1+	0	1	0	1
AlgoRunner_Error_Fault	7.4	60	1	@1+	0	1	0	1
Reserved	7.5	61	1	@1+	0	1	0	1
Reserved	7.6	62	1	@1+	0	1	0	1
Pmic_Image2_Reset_Fault	7.7	63	1	@1+	0	1	0	1
CPYR agent_ENET_Message_CRC_Err	8	64	1	@1+	0	1	0	1
CPYR agent_ENET_Missing_Message_Err	8.1	65	1	@1+	0	1	0	1
CPYR agent_ENET_Missing_Data_Err	8.2	66	1	@1+	0	1	0	1
CPYR agent_ENET_MessageReception_Err	8.3	67	1	@1+	0	1	0	1
CPYR agent_ENET_VersionMismatch_Err	8.4	68	1	@1+	0	1	0	1
CPYR agent_Application_CRC_Fault	8.5	69	1	@1+	0	1	0	1
CPYR agent_Callibration_CRC_Fault	8.6	70	1	@1+	0	1	0	1
CPYR agent_Cal_Complete_And_Compatible	8.7	71	1	@1+	0	1	0	1
VisionCo_Cal_Complete_And_Compatible_SWP3	9	72	1	@1+	0	1	0	1
VisionCo_MESP_Coding_failure	9.1	73	1	@1+	0	1	0	1
VisionCo_Application_Complete_And_Compatible	9.2	74	1	@1+	0	1	0	1
VisionCo_Application_Checksum	9.3	75	1	@1+	0	1	0	1
VisionCo_Calibration_Checksum	9.4	76	1	@1+	0	1	0	1

IMAGE2P_Appl_Complete_And_Compatible	9.5	77	1	@1+	0	1	0	1
CPYR agent_Spi_Frame_Format_Error	9.6	78	1	@1+	0	1	0	1
CPYR agent_Spi_Msg_Format_Error	9.7	79	1	@1+	0	1	0	1
CPYR agent_spi_msg_timeout	a.0	80	1	@1+	0	1	0	1
CPYR agent_Spi_Tx_Error	a.1	81	1	@1+	0	1	0	1
CPYR agent_Spi_CRC_Error	a.2	82	1	@1+	0	1	0	1
Spi_version_mismatch	a.3	83	1	@1+	0	1	0	1
Spi_Ipc_Comm_Reset	a.4	84	1	@1+	0	1	0	1
VisionCo_Spi_Frame_Format_Error	a.5	85	1	@1+	0	1	0	1
VISIONCO_Spi_Msg_Format_Error	a.6	86	1	@1+	0	1	0	1
VisionCo_Spi_Msg_Timeout	a.7	87	1	@1+	0	1	0	1
VisionCo_SPI_CRC_ERROR	b.0	88	1	@1+	0	1	0	1
VisionCo_Serial_Flash_Power_Up_Fail	b.1	89	1	@1+	0	1	0	1
TR_Aurix_FirstTimePowerUpFlashCheck	b.2	90	1	@1+	0	1	0	1
Aurix_OTA_Flash_Fail	b.3	91	1	@1+	0	1	0	1
KAM_Design_Read_Fail_Fault	b.4	92	1	@1+	0	1	0	1
KAM_Design_Write_Fail_Fault	b.5	93	1	@1+	0	1	0	1
VisionCo_DDR_Fault	b.6	94	1	@1+	0	1	0	1
Reserved	b.7	95	1	@1+	0	1	0	1
VisionCo_Boot_Fault	c.0	96	1	@1+	0	1	0	1
VisionCo_Core_Fault	c.1	97	1	@1+	0	1	0	1
VisionCo_Bist_Memory_Fault	c.2	98	1	@1+	0	1	0	1
Reserved	c.3	99	1	@1+	0	1	0	1
VisionCo_Init_Timeout_Fault	c.4	100	1	@1+	0	1	0	1
VisionCo_Core_Dump_Fault	c.5	101	1	@1+	0	1	0	1
Parameters_Invalid	c.6	102	1	@1+	0	1	0	1
Unreasonable_Radar_Data	c.7	103	1	@1+	0	1	0	1
FWC_Camera_Wrong_Assembly_Fault	d.0	104	1	@1+	0	1	0	1
AEB_shutdown_Invalid_Reason	d.1	105	1	@1+	0	1	0	1
AEB_shutdown_7_POS_DIFF_KAN_VAL	d.2	106	1	@1+	0	1	0	1
AEB_Shutdown_9_ROI_DIFF_VC_FCV	d.3	107	1	@1+	0	1	0	1
AEB_Shutdown_6_FCV_FCW_TTC	d.4	108	1	@1+	0	1	0	1
AEB_Shutdown_25_CO_FOF_DIFF	d.5	109	1	@1+	0	1	0	1
VISIONCO_GEN_30_DT_ERR	d.6	110	1	@1+	0	1	0	1
VISIONCO_GEN_18_IMAGER_DATA_ROWS_COLS	d.7	111	1	@1+	0	1	0	1
VISIONCO_GEN_4_YAW_HORIZON_DEVIATION	e.0	112	1	@1+	0	1	0	1
VISIONCO_GEN_26_CODE_CRC	e.1	113	1	@1+	0	1	0	1
VISIONCO_GEN_13_OLD_EGODATA_LAT	e.2	114	1	@1+	0	1	0	1
Reserved	e.3	115	1	@1+	0	1	0	1
Reserved	e.4	116	1	@1+	0	1	0	1
Reserved	e.5	117	1	@1+	0	1	0	1
VISIONCO_GEN_31_AEB_PARAM_FFI	e.6	118	1	@1+	0	1	0	1
VISIONCO_GEN_29_RFC_ERR	e.7	119	1	@1+	0	1	0	1
VISIONCO_GEN_11_INPUT_CRC_MISMATCH	f.0	120	1	@1+	0	1	0	1
VISIONCO_GEN_34_Corrupted_Vehicle_Info_Input_Signals	f.1	121	1	@1+	0	1	0	1

AEB_Shutdown_27_Fcv_Sig_Ffi	f.2	122	1	@1+	0	1	0	1	
VISIONCO_GEN_54_ACC_STL_EXECUTION_FAILED	f.3	123	1	@1+	0	1	0	1	
Reserved	f.4	124	1	@1+	0	1	0	1	
Missing_Camera_Callibration_Fault	f.5	125	1	@1+	0	1	0	1	
FWC_Camera_Type_Fault	f.6	126	1	@1+	0	1	0	1	
Reserved	f.7	127	1	@1+	0	1	0	1	
NTSC_Encoder_amplifier_ShortToBatt	10	128	1	@1+	0	1	0	1	
Reserved	10.1	129	1	@1+	0	1	0	1	
Reserved	10.2	130	1	@1+	0	1	0	1	
Reserved	10.3	131	1	@1+	0	1	0	1	
Full_Blockage_Fault	10.4	132	1	@1+	0	1	0	1	
Partial_Blockage_Fault	10.5	133	1	@1+	0	1	0	1	
Blurred_Image_Fault	10.6	134	1	@1+	0	1	0	1	
Splashes_Fault	10.7	135	1	@1+	0	1	0	1	
Low_Sun_Fault	11	136	1	@1+	0	1	0	1	
Sun_Rays_Fault	11.1	137	1	@1+	0	1	0	1	
Foggy_Spots_Fault	11.2	138	1	@1+	0	1	0	1	
Spot_Rays_Fault	11.3	139	1	@1+	0	1	0	1	
Smeared_Spots_Fault	11.4	140	1	@1+	0	1	0	1	
SelfGlare_Fault	11.5	141	1	@1+	0	1	0	1	
Low_Visibility_Fault	11.6	142	1	@1+	0	1	0	1	
Vision_OutofFocus_Fault	11.7	143	1	@1+	0	1	0	1	
Vision_OutofCalibration_Fault	12	144	1	@1+	0	1	0	1	
TD_BlockageTest	12.1	145	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_APP_CPS_STL_FAILED	12.2	146	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_Error_Fault	12.3	147	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_Fs_Error_Fault	12.4	148	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_Calibration_Error_Fault	12.5	149	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_Init_Failed_Fault	12.6	150	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_Init_Camera_Init_Fault	12.7	151	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_I2C_Video_Grab_Failed_Fault	13	152	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_I2C_Camera_Self_Reset_Fault	13.1	153	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_I2C_Timeout_Error_Fault	13.2	154	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_Pattern_Test_Fault	13.3	155	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_App_Cam_Params_Ccft_Crc_Failed_Fault	13.4	156	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_Pll_Comparison_Error_Fault	13.5	157	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_Pv_General_Error_Fault	13.6	158	1	@1+	0	1	0	1	
VisionCo_Fatal_Error_Pv_Verificattion_Error_Fault	13.7	159	1	@1+	0	1	0	1	
VisionCo_Minor_Error_Bm_Error_Fault	14	160	1	@1+	0	1	0	1	
VisionCo_Minor_Error_Bm_Em_Error_Fault	14.1	161	1	@1+	0	1	0	1	
VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Setting_Fault	14.2	162	1	@1+	0	1	0	1	
VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Registry_Fault	14.3	163	1	@1+	0	1	0	1	
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Registry_Fault	14.4	164	1	@1+	0	1	0	1	
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Fault	14.5	165	1	@1+	0	1	0	1	
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Fault	14.6	166	1	@1+	0	1	0	1	

VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Reg_Fault	14.7	167	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Open_Blackbox_Fault	15	168	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Ep_Fault	15.1	169	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Ep_Fault	15.2	170	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Create_Logger_Fault	15.3	171	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_IL_Fault	15.4	172	1	@1+	0	1	0	1
VisionCo_Minor_Error_Bm_Em_Err_Failed_Check_Reg_Versions_Fault	15.5	173	1	@1+	0	1	0	1
VISIONCO_GEN_C_R_Error	15.6	174	1	@1+	0	1	0	1
Init_Vdi_Failure	15.7	175	1	@1+	0	1	0	1
Init_I2C_Failure	16	176	1	@1+	0	1	0	1
Grab_No_Free_Header_Failure	16.1	177	1	@1+	0	1	0	1
Grab_Issue_Failure	16.2	178	1	@1+	0	1	0	1
Grab_Acquire_Failure	16.3	179	1	@1+	0	1	0	1
Grab_Release_Failure	16.4	180	1	@1+	0	1	0	1
VisionCo_Cam_Mal_Func	16.5	181	1	@1+	0	1	0	1
VisionCo_Cam_Ctrl_Flt	16.6	182	1	@1+	0	1	0	1
VisionCo_Cam_Vdo_Flt	16.7	183	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Video_Error_Fault	17	184	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_I2C_Not_On_Time_Video_Error_Fault	17.1	185	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Logical_Video_Error_Fault	17.2	186	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Inconsistent_Frames_Video_Error_Fault	17.3	187	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Verification_Failure_Video_Error_Fault	17.4	188	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Invalid_Video_Header_Fault	17.5	189	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Error_Reg_Flag_Fault	17.6	190	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Reg_Write_Hil_Failure_Error_Fault	17.7	191	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Zero_Histogram_Error_Fault	18	192	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Sum_Not_Match_Histogram_Error_Fault	18.1	193	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Internal_Error_Fault	18.2	194	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_No_Buffer_Fault	18.3	195	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Buffer_Invalid_Format_Fault	18.4	196	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Timeout_Fault	18.5	197	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Overflow_Fault	18.6	198	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Underflow_Fault	18.7	199	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Fifo_Fault	19	200	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Weights_Fault	19.1	201	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Gamma_Fault	19.2	202	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Hist_Fault	19.3	203	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Start_After_Start_Fault	19.4	204	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Core_Id_Fault	19.5	205	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Frame_Count_Read_Fail_Fault	19.6	206	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Histograms_Mismatch_Fault	19.7	207	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Ccft_Decisions_Mismatch_Fault	1a.0	208	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Header_Verification_Error_Fault	1a.1	209	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Shutdown_Fault	1a.2	210	1	@1+	0	1	0	1
VisionCo_Camera1_VideoErrorFlags_Shot_Out_Of_Sync_Fault	1a.3	211	1	@1+	0	1	0	1
VisionCo_Camera1_temperature_Fault	1a.4	212	1	@1+	0	1	0	1

VisionCo_Error_Out_Signal_Internal_Fault	1a.5	213	1	@1+	0	1	0	1
VisionCo_Error_Out_Temperature_Fault	1a.6	214	1	@1+	0	1	0	1
PCM_LostCom	1a.7	215	1	@1+	0	1	0	1
TCM_LostCom	1b.0	216	1	@1+	0	1	0	1
GSM_LostCom	1b.1	217	1	@1+	0	1	0	1
ABS_LostCom	1b.2	218	1	@1+	0	1	0	1
SECM_LostCom	1b.3	219	1	@1+	0	1	0	1
PSCM_LostCom	1b.4	220	1	@1+	0	1	0	1
BCM_LostCom	1b.5	221	1	@1+	0	1	0	1
RCM_LostCom	1b.6	222	1	@1+	0	1	0	1
IPC_LostCom	1b.7	223	1	@1+	0	1	0	1
HUD_LostCom	1c.0	224	1	@1+	0	1	0	1
SCCM_LostCom	1c.1	225	1	@1+	0	1	0	1
ACM_LostCom	1c.2	226	1	@1+	0	1	0	1
DDM_LostCom	1c.3	227	1	@1+	0	1	0	1
PDM_LostCom	1c.4	228	1	@1+	0	1	0	1
APIM_LostCom	1c.5	229	1	@1+	0	1	0	1
Reserved	1c.6	230	1	@1+	0	1	0	1
PCMH_LostCom	1c.7	231	1	@1+	0	1	0	1
VDM_LostCom	1d.0	232	1	@1+	0	1	0	1
PCM_InvalidData	1d.1	233	1	@1+	0	1	0	1
TCM_InvalidData	1d.2	234	1	@1+	0	1	0	1
GSM_InvalidData	1d.3	235	1	@1+	0	1	0	1
ABS_InvalidData	1d.4	236	1	@1+	0	1	0	1
SECM_InvalidData	1d.5	237	1	@1+	0	1	0	1
PSCM_InvalidData	1d.6	238	1	@1+	0	1	0	1
BCM_InvalidData	1d.7	239	1	@1+	0	1	0	1
IPC_InvaildData	1e.0	240	1	@1+	0	1	0	1
SCCM_InvalidData	1e.1	241	1	@1+	0	1	0	1
RCM_InvalidData	1e.2	242	1	@1+	0	1	0	1
DDM_InvalidData	1e.3	243	1	@1+	0	1	0	1
PDM_InvalidData	1e.4	244	1	@1+	0	1	0	1
HVAC_InvalidData	1e.5	245	1	@1+	0	1	0	1
APIM_InvalidData	1e.6	246	1	@1+	0	1	0	1
Reserved	1e.7	247	1	@1+	0	1	0	1
Reserved	1f.0	248	1	@1+	0	1	0	1
PCMH_InvalidData	1f.1	249	1	@1+	0	1	0	1
VDM_InvalidData	1f.2	250	1	@1+	0	1	0	1
IPC_InvaildRollCnt	1f.3	251	1	@1+	0	1	0	1
RCM_InvalidRollCnt	1f.4	252	1	@1+	0	1	0	1
RCM_InvalidChkSum	1f.5	253	1	@1+	0	1	0	1
reserved	1f.6	254	1	@1+	0	1	0	1
VCAN_Bus_Off_Fault	1f.7	255	1	@1+	0	1	0	1
FLRCAN_Bus_Off_Fault	20	256	1	@1+	0	1	0	1
SIDERADAR_RF_LF_CAN_Bus_Off_Fault	20.1	257	1	@1+	0	1	0	1

SIDERADAR_RR_LR_CAN_Bus_Off_Fault	20.2	258	1	@1+	0	1	0	1
MAINRADAR_PCAN_Missing_Message_Fault	20.3	259	1	@1+	0	1	0	1
MAINRADAR_PCAN_Range_Check_Fault	20.4	260	1	@1+	0	1	0	1
Reserved	20.5	261	1	@1+	0	1	0	1
SIDERADAR_FR_PCAN_Range_Check_Fault	20.6	262	1	@1+	0	1	0	1
SIDERADAR_FL_PCAN_Range_Check_Fault	20.7	263	1	@1+	0	1	0	1
SIDERADAR_RL_PCAN_Range_Check_Fault	21	264	1	@1+	0	1	0	1
SIDERADAR_RR_PCAN_Range_Check_Fault	21.1	265	1	@1+	0	1	0	1
CarCfg_Implausible_Denied_Param_23	21.2	266	1	@1+	0	1	0	1
CarCfg_Implausible_Denied_Param_96	21.3	267	1	@1+	0	1	0	1
SIDERADAR_Software_Compatibility_Fault	21.4	268	1	@1+	0	1	0	1
MAINRADAR_Software_Compatibility_Fault	21.5	269	1	@1+	0	1	0	1
GCCConfigurationNotconfigured_CarCfg_Undefined_Denied	21.6	270	1	@1+	0	1	0	1
SIDERADAR_Invalid_Hardware_configuration	21.7	271	1	@1+	0	1	0	1
ABS_InvalidRollCnt	22	272	1	@1+	0	1	0	1
ABS_InvalidChkSum	22.1	273	1	@1+	0	1	0	1
Brake_Temperature_Too_High	22.2	274	1	@1+	0	1	0	1
Camera_Module_Alignment_or_Adjustment_Incorrect_RearCamera_	22.3	275	1	@1+	0	1	0	1
DPAC_Vision_System_Camera_Alignment_or_Adjustment_Incorrect	22.4	276	1	@1+	0	1	0	1
Internal_Control_Module_Software_Incompatibility	22.5	277	1	@1+	0	1	0	1
Initial_Configuration_Not_Complete	22.6	278	1	@1+	0	1	0	1
Control_Module_Configuration_Incompatible	22.7	279	1	@1+	0	1	0	1
LIN_Rear_Camera_Bus_Failure	23	280	1	@1+	0	1	0	1
LIN_Rear_Camera_Not_Configured	23.1	281	1	@1+	0	1	0	1
LIN_Rear_Camera_Fault	23.2	282	1	@1+	0	1	0	1
Rear_Camera2_Bus_Failure	23.3	283	1	@1+	0	1	0	1
Rear_Camera2_Not_Configured	23.4	284	1	@1+	0	1	0	1
Rear_Camera2_Fault	23.5	285	1	@1+	0	1	0	1
Front_USS_HSD_FailurontUSS_Ckt_ShortToGnd	23.6	286	1	@1+	0	1	0	1
Rear_USS_HSD_FailureRearUSS_Ckt_ShortToGnd	23.7	287	1	@1+	0	1	0	1
Front_USS_HSD_ShortToBattFrontUSS_Ckt_ShortToBatt	24	288	1	@1+	0	1	0	1
Rear_USS_HSD_ShortToBattRearUSS_Ckt_ShortTobBatt	24.1	289	1	@1+	0	1	0	1
Ignition_Status_Unknown_Fault	24.2	290	1	@1+	0	1	0	1
Ignition_Status_Missing_Fault	24.3	291	1	@1+	0	1	0	1
Veh_V_ActlEng_Missing_Fault	24.4	292	1	@1+	0	1	0	1
VehVActlEng_D_Qf_Missing_Fault	24.5	293	1	@1+	0	1	0	1
VehVActlEng_D_Qf_Fault	24.6	294	1	@1+	0	1	0	1
VehVActlEng_D_Qf_No_data_exists_Fault	24.7	295	1	@1+	0	1	0	1
VISIONCO_MES_49_Data_Path_Corruption	25	296	1	@1+	0	1	0	1
VISIONCO_MES_50_VMP_SANITY_FAILED	25.1	297	1	@1+	0	1	0	1
Reserved	25.2	298	1	@1+	0	1	0	1
VISIONCO_40_PED_SANITY_FAULT	25.3	299	1	@1+	0	1	0	1
VISIONCO_39_PED_OVERLAP_FAULT	25.4	300	1	@1+	0	1	0	1
Reserved	25.5	301	1	@1+	0	1	0	1
Reserved	25.6	302	1	@1+	0	1	0	1

Reserved	25.7	303	1	@1+	0	1	0	1
Reserved	26	304	1	@1+	0	1	0	1
Reserved	26.1	305	1	@1+	0	1	0	1
VISIONCO_48_ALGO_PITCH_MISMATCH	26.2	306	1	@1+	0	1	0	1
VISIONCO_45_ROAD_SFCONF_VALIDATION	26.3	307	1	@1+	0	1	0	1
VISIONCO_42_ROAD_PREDICTION_VALIDATION	26.4	308	1	@1+	0	1	0	1
VISIONCO_33_ROAD_LANE_DISQUALIFICATINO_FAILED	26.5	309	1	@1+	0	1	0	1
TrlrLampCnnct_B_Actl_Missing_Fault	26.6	310	1	@1+	0	1	0	1
BLISLEDStatDriverSide_MissingMsg_Fault	26.7	311	1	@1+	0	1	0	1
BLISLEDStatPassSide_MissingMsg_Fault	27	312	1	@1+	0	1	0	1
BLISLEDStatDriverSide_DCUFault	27.1	313	1	@1+	0	1	0	1
BLISLEDStatPassSide_DCUFault	27.2	314	1	@1+	0	1	0	1
BLISLEDStatDriverSide_ON_OFF_Mismatch_Fault	27.3	315	1	@1+	0	1	0	1
BLISLEDStatPassSide_ON_OFF_Mismatch_Fault	27.4	316	1	@1+	0	1	0	1
IgnKeyType_D_Actl_Unknown_Fault	27.5	317	1	@1+	0	1	0	1
IgnKeyType_D_ActlMissing_Fault	27.6	318	1	@1+	0	1	0	1
Parklamp_Status_Missing_Fault	27.7	319	1	@1+	0	1	0	1
TR_Application_CRC_Fault	28	320	1	@1+	0	1	0	1
Reserved	28.1	321	1	@1+	0	1	0	1
TR_DDR_RAM_CRC_Fault	28.2	322	1	@1+	0	1	0	1
Reserved	28.3	323	1	@1+	0	1	0	1
Reserved	28.4	324	1	@1+	0	1	0	1
Reserved	28.5	325	1	@1+	0	1	0	1
Reserved	28.6	326	1	@1+	0	1	0	1
Reserved	28.7	327	1	@1+	0	1	0	1
Reserved	29	328	1	@1+	0	1	0	1
Reserved	29.1	329	1	@1+	0	1	0	1
Reserved	29.2	330	1	@1+	0	1	0	1
Reserved	29.3	331	1	@1+	0	1	0	1
Reserved	29.4	332	1	@1+	0	1	0	1
Reserved	29.5	333	1	@1+	0	1	0	1
Reserved	29.6	334	1	@1+	0	1	0	1
Reserved	29.7	335	1	@1+	0	1	0	1
TR_INT_OSC_TEST_FAIL	2a.0	336	1	@1+	0	1	0	1
Perf_DPAC_Process_Fault	2a.1	337	1	@1+	0	1	0	1
Perf_FWC_Process_Fault	2a.2	338	1	@1+	0	1	0	1
Reserved	2a.3	339	1	@1+	0	1	0	1
TR_Stack_Overflow_Fault	2a.4	340	1	@1+	0	1	0	1
Reserved	2a.5	341	1	@1+	0	1	0	1
Reserved	2a.6	342	1	@1+	0	1	0	1
Reserved	2a.7	343	1	@1+	0	1	0	1
Reserved	2b.0	344	1	@1+	0	1	0	1
TR_DDR_Fault	2b.1	345	1	@1+	0	1	0	1
TR_Boot_Fault	2b.2	346	1	@1+	0	1	0	1
Reserved	2b.3	347	1	@1+	0	1	0	1
TR_Appl_Complete_And_Compatible	2b.4	348	1	@1+	0	1	0	1

TR_Core_Exception_Fault	2b.5	349	1	@1+	0	1	0	1
TR_Control_ConfigReg_Fault	2b.6	350	1	@1+	0	1	0	1
reserved	2b.7	351	1	@1+	0	1	0	1
TR_ENET_Message_CRC_Err	2c.0	352	1	@1+	0	1	0	1
TR_ENET_Missing_Message_Err	2c.1	353	1	@1+	0	1	0	1
TR_ENET_Missing_Data_Err	2c.2	354	1	@1+	0	1	0	1
TR_ENET_MessageReception_Err	2c.3	355	1	@1+	0	1	0	1
TR_ENET_VersionMismatch_Err	2c.4	356	1	@1+	0	1	0	1
TR_ENET_Communication_Drv_Err	2c.5	357	1	@1+	0	1	0	1
Reserved	2c.6	358	1	@1+	0	1	0	1
Reserved	2c.7	359	1	@1+	0	1	0	1
Reserved	2d.0	360	1	@1+	0	1	0	1
QSPI_Flash_Illegal_Instruction_Err	2d.1	361	1	@1+	0	1	0	1
QSPI_Communication_Drv_Err	2d.2	362	1	@1+	0	1	0	1
Reserved	2d.3	363	1	@1+	0	1	0	1
Reserved	2d.4	364	1	@1+	0	1	0	1
Reserved	2d.5	365	1	@1+	0	1	0	1
DTC_F00042_PERFPROC_IntMemFail	2d.6	366	1	@1+	0	1	0	1
DTC_F00049_PERFPROC_IntElectronicFail	2d.7	367	1	@1+	0	1	0	1
LEFT_CAMERA_FAULT_CKT_ShortToBatt	2e.0	368	1	@1+	0	1	0	1
RIGHT_CAMERA_FAULT_CKT_ShortToBatt	2e.1	369	1	@1+	0	1	0	1
BA_Switch_Stuck_On_Fault	2e.2	370	1	@1+	0	1	0	1
Reserved	2e.3	371	1	@1+	0	1	0	1
Reserved	2e.4	372	1	@1+	0	1	0	1
Reserved	2e.5	373	1	@1+	0	1	0	1
Reserved	2e.6	374	1	@1+	0	1	0	1
Reserved	2e.7	375	1	@1+	0	1	0	1
Reserved	2f.0	376	1	@1+	0	1	0	1
Reserved	2f.1	377	1	@1+	0	1	0	1
Reserved	2f.2	378	1	@1+	0	1	0	1
Reserved	2f.3	379	1	@1+	0	1	0	1
Reserved	2f.4	380	1	@1+	0	1	0	1
Reserved	2f.5	381	1	@1+	0	1	0	1
Reserved	2f.6	382	1	@1+	0	1	0	1
Reserved	2f.7	383	1	@1+	0	1	0	1
VISIONCO_GEN_21_CAMERA_HEIGHT_LIMIT_EXCEEDED	30	384	1	@1+	0	1	0	1
VISIONCO_44_ROAD_SANITY_CORRECTION_VALIDATION	30.1	385	1	@1+	0	1	0	1
VISIONCO_53_DECELERATION_DBS_OVERLAP_FAULT	30.2	386	1	@1+	0	1	0	1
VISIONCO_GEN_24_CCFT_DIFF_RESULTS	30.3	387	1	@1+	0	1	0	1
GearRvrse_D_Actl_Missing_Fault	30.4	388	1	@1+	0	1	0	1
SOD_TCM_LostComm	30.5	389	1	@1+	0	1	0	1
WheelData_Missing_Fault	30.6	390	1	@1+	0	1	0	1
SOD_BCM_LostComm	30.7	391	1	@1+	0	1	0	1
SOD_RCM_LostCom	31	392	1	@1+	0	1	0	1
SOD_ABS_InvalidData	31.1	393	1	@1+	0	1	0	1

SOD_TCM_InvalidData	31.2	394	1	@1+	0	1	0	1
GearRvrse_D_Actl_invalid_Fault	31.3	395	1	@1+	0	1	0	1
SOD_IPC_LostCom	31.4	396	1	@1+	0	1	0	1
AEB_Inhibited_Fault_Flag	31.5	397	1	@1+	0	1	0	1
AEB_Validity_Fault_Flag	31.6	398	1	@1+	0	1	0	1
LaneFusion_Validity_Fault_Flag	31.7	399	1	@1+	0	1	0	1
Reserved	32	400	1	@1+	0	1	0	1
Reserved	32.1	401	1	@1+	0	1	0	1
Reserved	32.2	402	1	@1+	0	1	0	1
FWC_CAMERA_FAULT_CKT_ShortToBatt	32.3	403	1	@1+	0	1	0	1
LeftRearSOD_Ckt_ShortToBatt	32.4	404	1	@1+	0	1	0	1
LeftRearSOD_Ckt_ShortToGnd	32.5	405	1	@1+	0	1	0	1
RightRearSOD_Ckt_ShortToBatt	32.6	406	1	@1+	0	1	0	1
RightRearSOD_Ckt_ShortToGnd	32.7	407	1	@1+	0	1	0	1
LeftFrontSOD_Ckt_ShortToBatt	33	408	1	@1+	0	1	0	1
LeftFrontSOD_Ckt_ShortToGnd	33.1	409	1	@1+	0	1	0	1
RightFrontSOD_Ckt_ShortToBatt	33.2	410	1	@1+	0	1	0	1
RightFrontSOD_Ckt_ShortToGnd	33.3	411	1	@1+	0	1	0	1
FRONT_CAMERA_FAULT_CKT_ShortToBatt	33.4	412	1	@1+	0	1	0	1
REAR_CAMERA_FAULT_CKT_ShortToBatt	33.5	413	1	@1+	0	1	0	1
Reserved	33.6	414	1	@1+	0	1	0	1
DTC_915E31_PerfProcessor	33.7	415	1	@1+	0	1	0	1
DTC_9B4008_FAULT_CONDITION	34	416	1	@1+	0	1	0	1
DTC_9B4096_FAULT_CONDITION	34.1	417	1	@1+	0	1	0	1
DTC_9B4098_FAULT_CONDITION	34.2	418	1	@1+	0	1	0	1
DTC_9B4208_FAULT_CONDITION	34.3	419	1	@1+	0	1	0	1
DTC_9B4296_FAULT_CONDITION	34.4	420	1	@1+	0	1	0	1
DTC_9B4298_FAULT_CONDITION	34.5	421	1	@1+	0	1	0	1
DTC_929C08_FAULT_CONDITION	34.6	422	1	@1+	0	1	0	1
DTC_929C96_FAULT_CONDITION	34.7	423	1	@1+	0	1	0	1
DTC_929C98_FAULT_CONDITION	35	424	1	@1+	0	1	0	1
DTC_9B4808_FAULT_CONDITION	35.1	425	1	@1+	0	1	0	1
DTC_9B4896_FAULT_CONDITION	35.2	426	1	@1+	0	1	0	1
DTC_9B4898_FAULT_CONDITION	35.3	427	1	@1+	0	1	0	1
DTC_9B5008_FAULT_CONDITION	35.4	428	1	@1+	0	1	0	1
DTC_9B5096_FAULT_CONDITION	35.5	429	1	@1+	0	1	0	1
DTC_9B5098_FAULT_CONDITION	35.6	430	1	@1+	0	1	0	1
DTC_93F308_FAULT_CONDITION	35.7	431	1	@1+	0	1	0	1
DTC_93F396_FAULT_CONDITION	36	432	1	@1+	0	1	0	1
DTC_93F398_FAULT_CONDITION	36.1	433	1	@1+	0	1	0	1
DTC_9B3608_FAULT_CONDITION	36.2	434	1	@1+	0	1	0	1
DTC_9B3696_FAULT_CONDITION	36.3	435	1	@1+	0	1	0	1
DTC_9B3698_FAULT_CONDITION	36.4	436	1	@1+	0	1	0	1
DTC_9B3808_FAULT_CONDITION	36.5	437	1	@1+	0	1	0	1
DTC_9B3896_FAULT_CONDITION	36.6	438	1	@1+	0	1	0	1

DTC_9B3898_FAULT_CONDITION	36.7	439	1	@1+	0	1	0	1
DTC_929D08_FAULT_CONDITION	37	440	1	@1+	0	1	0	1
DTC_929D96_FAULT_CONDITION	37.1	441	1	@1+	0	1	0	1
DTC_929D98_FAULT_CONDITION	37.2	442	1	@1+	0	1	0	1
DTC_9B4408_FAULT_CONDITION	37.3	443	1	@1+	0	1	0	1
DTC_9B4496_FAULT_CONDITION	37.4	444	1	@1+	0	1	0	1
DTC_9B4498_FAULT_CONDITION	37.5	445	1	@1+	0	1	0	1
DTC_9B4608_FAULT_CONDITION	37.6	446	1	@1+	0	1	0	1
DTC_9B4696_FAULT_CONDITION	37.7	447	1	@1+	0	1	0	1
DTC_9B4698_FAULT_CONDITION	38	448	1	@1+	0	1	0	1
DTC_93F408_FAULT_CONDITION	38.1	449	1	@1+	0	1	0	1
DTC_93F496_FAULT_CONDITION	38.2	450	1	@1+	0	1	0	1
DTC_93F498_FAULT_CONDITION	38.3	451	1	@1+	0	1	0	1
DTC_93039E_FAULT_CONDITION	38.4	452	1	@1+	0	1	0	1
DTC_915E31_FAULT_CONDITION	38.5	453	1	@1+	0	1	0	1
DTC_948E31_FAULT_CONDITION	38.6	454	1	@1+	0	1	0	1
DTC_92BE31_FAULT_CONDITION	38.7	455	1	@1+	0	1	0	1
DTC_92BF31_FAULT_CONDITION	39	456	1	@1+	0	1	0	1
DTC_F00042_IMAGE2	39.1	457	1	@1+	0	1	0	1
DTC_F00089_IMAGE2	39.2	458	1	@1+	0	1	0	1
SIDERADAR_FL_PCAN_Missing_Message_Fault	39.3	459	1	@1+	0	1	0	1
SIDERADAR_FR_PCAN_Missing_Message_Fault	39.4	460	1	@1+	0	1	0	1
SIDERADAR_RL_PCAN_Missing_Message_Fault	39.5	461	1	@1+	0	1	0	1
SIDERADAR_RR_PCAN_Missing_Message_Fault	39.6	462	1	@1+	0	1	0	1
DSCM_InvalidData	39.7	463	1	@1+	0	1	0	1
DSCM_InvalidRollCnt	3a.0	464	1	@1+	0	1	0	1
DSCM_InvalidChkSum	3a.1	465	1	@1+	0	1	0	1
TRM_InvalidData	3a.2	466	1	@1+	0	1	0	1
DSCM_LostComm	3a.3	467	1	@1+	0	1	0	1
HVAC_LostComm	3a.4	468	1	@1+	0	1	0	1
TRM_LostComm	3a.5	469	1	@1+	0	1	0	1
ATCM_LostComm	3a.6	470	1	@1+	0	1	0	1
HCM_LostComm	3a.7	471	1	@1+	0	1	0	1
Parklamp_Status_Unknown_Fault	3b.0	472	1	@1+	0	1	0	1
Litval_Unknown_Fault	3b.1	473	1	@1+	0	1	0	1
Litval_Missing_Fault	3b.2	474	1	@1+	0	1	0	1
TurnLghtSwitch_D_Stat_Missing_Fault	3b.3	475	1	@1+	0	1	0	1
Btt_L_Actl_MissingMsg_Fault	3b.4	476	1	@1+	0	1	0	1
ATD_Mismatch_Fault	3b.5	477	1	@1+	0	1	0	1
TLM_Mismatch_Fault	3b.6	478	1	@1+	0	1	0	1
TBM_Mismatch_Fault	3b.7	479	1	@1+	0	1	0	1
Reserved	3c.0	480	1	@1+	0	1	0	1
WhlDirRr_D_Actl_Missing	3c.1	481	1	@1+	0	1	0	1
WhlDirRl_D_Actl_Missing	3c.2	482	1	@1+	0	1	0	1
WhlDirFl_D_Actl_Missing	3c.3	483	1	@1+	0	1	0	1
WhlDirFr_D_Actl_Missing	3c.4	484	1	@1+	0	1	0	1

WhlDirRr_D_Actl_Failed_Fault	3c.5	485	1	@1+	0	1	0	1
WhlDirRl_D_Actl_Failed_Fault	3c.6	486	1	@1+	0	1	0	1
WhlDirFr_D_Actl_Failed_Fault	3c.7	487	1	@1+	0	1	0	1
WhlDirFl_D_Actl_Failed_Fault	3d.0	488	1	@1+	0	1	0	1
CtaBrk_D_Stat_Denied_Fault	3d.1	489	1	@1+	0	1	0	1
CtaBrk_D_Stat_Missing_Fault	3d.2	490	1	@1+	0	1	0	1
Isig_brake_timeout_fault	3d.3	491	1	@1+	0	1	0	1
Reserved	3d.4	492	1	@1+	0	1	0	1
CtaYBrkEnbl_B_Rq_Mismatch_Fault	3d.5	493	1	@1+	0	1	0	1
DVR_SELECT_STAT_Fault	3d.6	494	1	@1+	0	1	0	1
DTC_F00049_IMAGE2	3d.7	495	1	@1+	0	1	0	1
CMbBDeniedByABS	3e.0	496	1	@1+	0	1	0	1
CMbBDeniedByPCM	3e.1	497	1	@1+	0	1	0	1
CMbBImplausByABS	3e.2	498	1	@1+	0	1	0	1
CMbBImplausByPCM	3e.3	499	1	@1+	0	1	0	1
AGENT_InvalidData	3e.4	500	1	@1+	0	1	0	1
unused	3e.5	501	1	@1+	0	1	0	1
ACCImplausByABS	3e.6	502	1	@1+	0	1	0	1
ACCImplausByPCM	3e.7	503	1	@1+	0	1	0	1
ACCImplausAccelByABS	3f.0	504	1	@1+	0	1	0	1
ACCFailToCancelByPCM	3f.1	505	1	@1+	0	1	0	1
ACCDeniedbyABS	3f.2	506	1	@1+	0	1	0	1
SetDTC_ACCDeniedByDIM	3f.3	507	1	@1+	0	1	0	1
SetDTCImplausibleCMbB_ASILB	3f.4	508	1	@1+	0	1	0	1
TJAImplausByPSCM	3f.5	509	1	@1+	0	1	0	1
SetDTCImplausibleHA_FSM	3f.6	510	1	@1+	0	1	0	1
SetDTCImplausibleFAPA_FSM	3f.7	511	1	@1+	0	1	0	1

CPYR AGENT_SPI_Debug

CPYR AGENT_SPI_Debug(0x203)-STNumber(78)-STVersion(1) **Message CPYR AGENT_SPI_Debug (0X203) shall be implemented as follows:**

Signal Name	Description	Startbit	Length [Bit]	Value Type	Factor	Offset	Minimum	Maximum	Unit
CPYR agentSpiDllRxBadHdrCtr	CPYR agent receives frame with "D" nibble not equal to 8, 9, or A.	0	8	Unsigned	1	0	0	255	NA
CPYR agentSpiDllRxLongFrameCtr	SPI driver indicates incorrect frame (not 128 bytes)	8	8	Unsigned	1	0	0	255	NA
CPYR agentSpiDllRxShortFrameCtr	SPI driver indicates incorrect frame (not 128 bytes)	16	8	Unsigned	1	0	0	255	NA
CPYR agentSpiTpRxBadHdrCtr	The transport layer on the CPYR agent detects an unexpected header or length.	24	8	Unsigned	1	0	0	255	NA
CPYR agentSpiTpRxBadSeqCtr	The transport layer on the CPYR agent receives an unexpected frame.	32	8	Unsigned	1	0	0	255	NA

Temperature

Temperature(0x206)-STNumber(79)-STVersion(1) **Message _Temperature (0x206) shall be implemented as follows:**

Signal Name	Description	Startbit	Length [Bit]	Value Type	Factor	Offset	Minimum	Maximum	Unit
CPYR agentOnBdT	CPYR agent Temperature	0	8	Signed	1	0	-40	125	C
VisionCo4OnBdT	VisionCo4 Temperature	8	8	Signed	1	0	-40	125	C
IMAGE2POnBdT	IMAGE2P Temperature	16	8	Signed	1	0	-40	125	C
AlgorunnerOnBdT	Algorunner Temperature	24	8	Signed	1	0	-40	125	C
imagerTemperature	Imager temperature 0x80 = Unknown	32	8	Signed	1	0	-40	125	C
mipsTemperature	MIPS temperature 0x80 = Unknown	40	8	Signed	1	0	-40	125	C
vmpTemperature	VMP temperature 0x80 = Unknown	48	8	Signed	1	0	-40	125	C
ddrTemperature	DDR Temperature 0 = Range OK 1 = Below Spec 2 = Above Spec 3 = Above 85C 255 = Error (if the read failed.)	56	8	Signed	1	0	-40	125	C
CPYR agent_ThermSts	00 = NORMAL 01 = OVERTEMP 10 = INVALID	64	2	Signed	1	0	-2	1	NA
VisionCo_ThermSts	00 = NORMAL 01 = OVERTEMP 10 = INVALID	66	2	Signed	1	0	-2	1	NA
Algorunner_ThermSts	00 = NORMAL 01 = OVERTEMP 10 = INVALID	68	2	Signed	1	0	-2	1	NA
IMAGE2P_ThermSts	00 = NORMAL 01 = OVERTEMP 10 = INVALID	70	2	Signed	1	0	-2	1	NA

OBJECTSENSE_Signals

Message OBJECTSENSE_Signals shall be implemented as

ST Number (80)

SIGNALS	Description	StartBit	Length [Bit]	Value Type	Offset	Factor	Minimum	Maximum
OBJECTSENSEYawRate	Yaw Rate	0	32	@1-	0	0.0002	-2	2
OBJECTSENSEYawRate_Est	Wheel-Speed based YawRate	32	32	@1-	0	0.0002	-2	2
OBJECTSENSE_FiltVehicleSpeedOverGround	Vehicle Speed over ground	64	32	@1-	0	0.01	0	191.13
OBJECTSENSE_V_CPYR agent_Long_front	CPYR agent Vehicle Longitudinal Velocity relative to the ground	96	32	@1-	0	1	-20	100
OBJECTSENSE_V_CPYR agent_Lat_front	CPYR agent Vehicle	128	32	@1-	0	1	-20	20

	Lateral Velocity relative to the ground							
StePinComp_An_Est	Steering Wheel Angle	160	32	@1-	0	0.1	0.1	25.59
VehicleCfg_SteeringRatio	Steering Gear Ratio	192	32	@1-	0	0.1	0.1	25.59
VehicleCfg_Wheelbase	Wheelbase	224	32	@1-	0	1	0	65535
VISION_TIMESTAMP	Timestamp of FWC vision data to determine delay / to compensate latency (synchronized with CPYR agent timer)	256	32	@1+	0	1	0	4.29E+09
CPYR AGENT_CURRENT_TIME	Current timer value of CPYR agent µController to determine delay / to compensate latency	288	32	@1+	0	1	0	4.29E+09
OBJECTSENSE_FiltVehicleSpeedOverGround_signed	Signed Vehicle Speed	320	32	@1-	0	0.01	-100	100
OBJECTSENSE_FUS_LookIndex	OBJECTSE NSE Scan Index	352	16	@1+	0	1	0	65535
OBJECTSENSE_VIS_GrabIndex	Vision Frame Index	368	16	@1+	0	1	0	65535
OBJECTSENSE_DrivingDirection	Driving Direction UNKNOWN (0x0) BACKWARD (0x1) FORWARD (0x2)	384	2	@1+	0	1	0	2
OBJECTSENSE_DrivingReverse	Not_Reverse (0x0) Reverse (0x1)	386	1	@1+	0	1	0	1
RadarSystemNotOperational_FrCtr	OBJECTSENSE Status	387	1	@1+	0	1	0	1
RadarSystemNotOperational_FrLft	OBJECTSENSE Status	388	1	@1+	0	1	0	1
RadarSystemNotOperational_FrRt	OBJECTSENSE Status	389	1	@1+	0	1	0	1
RadarSystemNotOperational_RrLft	OBJECTSENSE Status	390	1	@1+	0	1	0	1
RadarSystemNotOperational_RrRt	OBJECTSENSE Status	391	1	@1+	0	1	0	1
VisionSystemNotOperational_FWC	OBJECTSENSE Status	392	1	@1+	0	1	0	1
PerfSOCNotOperational	OBJECTSENSE Status	393	1	@1+	0	1	0	1
VisSystemNotOperational_RDPAC	OBJECTSENSE Status	394	1	@1+	0	1	0	1

VisSystemNotOperational_360	OBJECTSENSE Status	395	1	@1+	0	1	0	1
Perf_DPAC_ProcessNotOperational	OBJECTSENSE Status	396	1	@1+	0	1	0	1
Perf_FWC_ProcessNotOperational	OBJECTSENSE Status	397	1	@1+	0	1	0	1
USLSystemNotOperational	OBJECTSENSE Status	398	1	@1+	0	1	0	1
ParkAidLEDNotOperational	OBJECTSENSE Status	399	1	@1+	0	1	0	1

BLISLEDNotOperational	OBJECTSENSE Status	400	1	@1+	0	1	0	1
APA_SwitchNotOperational	OBJECTSENSE Status	401	1	@1+	0	1	0	1
BA_SwitchNotOperational	OBJECTSENSE Status	402	1	@1+	0	1	0	1
BPA_SwitchNotOperational	OBJECTSENSE Status	403	1	@1+	0	1	0	1
LKS_SwitchNotOperational	OBJECTSENSE Status	404	1	@1+	0	1	0	1
View_SwitchNotOperational	OBJECTSENSE Status	405	1	@1+	0	1	0	1
VisSystemNotOperational_LINRVC	OBJECTSENSE Status	406	1	@1+	0	1	0	1
VisSystemNotOperational_CHMSL	OBJECTSENSE Status	407	1	@1+	0	1	0	1
RadarAlignmentOutOfRange_FrCtr	OBJECTSENSE Status	408	1	@1+	0	1	0	1
RadarAlignmentOutOfRange_FrLft	OBJECTSENSE Status	409	1	@1+	0	1	0	1
RadarAlignmentOutOfRange_FrRt	OBJECTSENSE Status	410	1	@1+	0	1	0	1
RadarAlignmentOutOfRange_RrLft	OBJECTSENSE Status	411	1	@1+	0	1	0	1
RadarAlignmentOutOfRange_RrRt	OBJECTSENSE Status	412	1	@1+	0	1	0	1
VisionAlignmentOutOfRange_FWC	OBJECTSENSE Status	413	1	@1+	0	1	0	1
RadarBlockageDetected_FrCtr	OBJECTSENSE Status	414	1	@1+	0	1	0	1
RadarBlockageDetected_FrLft	OBJECTSENSE Status	415	1	@1+	0	1	0	1
RadarBlockageDetected_FrRt	OBJECTSENSE Status	416	1	@1+	0	1	0	1
RadarBlockageDetected_RrLft	OBJECTSENSE Status	417	1	@1+	0	1	0	1
RadarBlockageDetected_RrRt	OBJECTSENSE Status	418	1	@1+	0	1	0	1
VisionBlockageDetected_FWC	OBJECTSENSE Status	419	1	@1+	0	1	0	1
USLBlockageDetected_Front	OBJECTSENSE Status	420	1	@1+	0	1	0	1
USLBlockageDetected_Rear	OBJECTSENSE Status	421	1	@1+	0	1	0	1
RadarExtConditionsNotOK_FrCtr	OBJECTSENSE Status	422	1	@1+	0	1	0	1
RadarExtConditionsNotOK_FrLft	OBJECTSENSE Status	423	1	@1+	0	1	0	1
RadarExtConditionsNotOK_FrRt	OBJECTSENSE Status	424	1	@1+	0	1	0	1
RadarExtConditionsNotOK_RrLft	OBJECTSENSE Status	425	1	@1+	0	1	0	1
RadarExtConditionsNotOK_RrRt	OBJECTSENSE Status	426	1	@1+	0	1	0	1
VisionExtConditionsNotOK_FWC	OBJECTSENSE Status	427	1	@1+	0	1	0	1
SOCExtConditionsNotOK_Perf	OBJECTSENSE Status	428	1	@1+	0	1	0	1

SOCExtConditionsNotOK_DPAC	OBJECTSENSE Status	429	1	@1+	0	1	0	1
RadarAlignmentIncomplete_FrCtr	OBJECTSENSE Status	430	1	@1+	0	1	0	1
RadarAlignmentIncomplete_FrLft	OBJECTSENSE Status	431	1	@1+	0	1	0	1
RadarAlignmentIncomplete_FrRt	OBJECTSENSE Status	432	1	@1+	0	1	0	1
RadarAlignmentIncomplete_RrLft	OBJECTSENSE Status	433	1	@1+	0	1	0	1
RadarAlignmentIncomplete_RrRt	OBJECTSENSE Status	434	1	@1+	0	1	0	1
VisionAlignmentIncomplete_FWC	OBJECTSENSE Status	435	1	@1+	0	1	0	1
VisionAlignmentIncomplete_360	OBJECTSENSE Status	436	1	@1+	0	1	0	1
VisionAlignmentIncomplete_DRVC	OBJECTSENSE Status	437	1	@1+	0	1	0	1
VisionAlignmentNotStarted_FWC	OBJECTSENSE Status	438	1	@1+	0	1	0	1
RadarAlignmentNotStarted_FrCtr	OBJECTSENSE Status	439	1	@1+	0	1	0	1
RadarAlignmentNotStarted_FrLft	OBJECTSENSE Status	440	1	@1+	0	1	0	1
RadarAlignmentNotStarted_FrRt	OBJECTSENSE Status	441	1	@1+	0	1	0	1
RadarAlignmentNotStarted_RrLft	OBJECTSENSE Status	442	1	@1+	0	1	0	1
RadarAlignmentNotStarted_RrRt	OBJECTSENSE Status	443	1	@1+	0	1	0	1
VisionSensorDataUpdating_FWC	OBJECTSENSE Status	444	1	@1+	0	1	0	1

PerfSOCDataUpdating	OBJECTSENSE Status	445	1	@1+	0	1	0	1
CentralConfigError	OBJECTSENSE Status	446	1	@1+	0	1	0	1
ModuleConfigError	OBJECTSENSE Status	447	1	@1+	0	1	0	1
APAImplausibleError	OBJECTSENSE Status	448	1	@1+	0	1	0	1
CmbbImplausibleError	OBJECTSENSE Status	449	1	@1+	0	1	0	1
LaneFusionImplausibleError	OBJECTSENSE Status	450	1	@1+	0	1	0	1
GearPosImplausibleError	OBJECTSENSE Status	451	1	@1+	0	1	0	1
YawRateImplausibleError	OBJECTSENSE Status	452	1	@1+	0	1	0	1
VehAccelImplausibleError	OBJECTSENSE Status	453	1	@1+	0	1	0	1
FirstRowBckDrvImplausibleError	OBJECTSENSE Status	454	1	@1+	0	1	0	1
VehSpdImplausibleError	OBJECTSENSE Status	455	1	@1+	0	1	0	1
DrvEngageLevelImplausibleError	OBJECTSENSE Status	456	1	@1+	0	1	0	1

SOD_Signals

SOD_Signals(0x400)-STNumber(81)-STVersion(1) **This message provides**

Message SOD_Signals (0x400) shall be implemented as follows:

Signal Name	Description	Startbit	Length [Bit]	Value Type	Factor	Offset	Minimum	Maximum	Unit
GearLvrPos_D_Actl	init value 14; 0x0: Park 0x1: Reverse 0x2: Neutral 0x3: Drive 0x4: Sport_DriveSport 0x5:low 0x6: 1st 0x7: 2nd 0x8:3rd 0x9: 4th 0xA: 5th 0xB: 6th 0xC:undefined_treat_as_fault 0xD: undefined_treat_as_fault 0xE:unknown pos 0xF: Fault	0	4	NA	1	0	0	0xF	N/A
GearRvrse_D_Actl	init value 7; 0x0 inactive not confirmed 0x1: inactive confirmed 0x2: Active not confirmed 0x3: Active confirmed 0x4 - 0x6 not used 0x7: Fault	4	3	NA	1	0	0	0x7	N/A
raw_long_accel		7	32	float	0.01	-40	-40	41.89	m/s^2
raw_lat_accel		39	32	float	0.01	-40	-40	41.89	m/s^2
SodSnsX_D_Stat	0 - Clear 1 - Blocked 2 - System Failure (fault) 3 - Second Warning Audio	71	2	NA	1	0	0	3	NA
SodX_D_Stat	0x0: Off 0x1: Trailer_Tow_Off 0x2: On 0x3: Disable 0x4: invalid 0x5-0x7: not used	73	2	NA	1	0	0	4	NA
BttX_D_Stat	0x0: NotDetermined 0x1: Connected 0x2: Pending 0x3: NotConnected 0x4: OffTemp 0x5: Off 0x6: Disable 0x7: not used	75	3	NA	1	0	0	7	NA
BttX_D_RqDrvr	0x0: Null 0x1: NoRequest 0x2: Request 0x3: not used 0	78	3	NA	1	0	0	3	NA
	bx0A-64 is valid length, bx7E (12.6) is no								

	trailer								
TrlrLampCnnct_B_Actl	Trailer connect signal from the Trailer Lighting Module (TLM): Connected / not connected	113	1	NA	1	0	0	1	NA
BTT_ENABLE_DISABLE	Enables and disables BLIS with Trailer Tow feature. Configured per program. Default is 0. Configured via DID 0xDE00	114	1	NA	1	0	0	1	NA
ATD_ENABLE_DISABLE	Enables and disables the ATD function. Default is 0.	115	1	NA	1	0	0	1	NA
BLIS_ENABLE_DISABLE	BLIS enable/disable state shall be set by the global parameter. 0x0: Enable 0x1: Disable	116	1	NA	1	0	0	1	NA
Sod_D_Rq	0 = OFF, 1 = BLIS On Secondary Warning OFF, 2 = BLIS ON Secondary Warning ON, 3 = unknown	117	2	NA	1	0	0	3	NA
SODX_ATD_Clutter	00 = PENDING 01 = CLUTTER DETECTED 02 = NO CLUTTER 03 = SEARCH TERMINATED	119	2	NA	1	0	0	3	NA
SodAlrtX_D_Stat	0 – Lamp Off 1 – Lamp On 2 – Flash 3 – Bulb Prove Out	121	2	NA	1	0	0	3	NA
TurnLghtSwtch_D_Stat	2 0 – Off 1 – Left 2 – Right 3 – Unused	123	2	NA	1	0	0	3	NA
raw_steering_angle_deg		125	32	float	1	0	-1600.00	1676.7	degree
BalrX_D_Stat	BAX_D_Stat denotes Boundary Alert feature On/Off status. Output of ON/OFF STD	157	1	NA	1	0	0	1	NA
BalrSnsX_D_Falt	BASnsLeft_D_Stat is used to denote the Boundary Alert Sensor Fault Status. States of signal are Clear(0x0), Blocked(0x1), SystemFailure(0x2) or Invalid(0x3)	158	2	NA	1	0	0	3	NA
RstrnImpactEvntStatus	This input signal sends the restraints impact event status to the SOD from the RCM. States of Signal are Normal(0x00), Not_Used_1(0x01), Not_Used_2(0x02), Threshold_1_Exceeded(0x03), Not_Used_3(0x04), Threshold_2_Exceeded(0x05), Not_Used_4(0x06) or Invalid(0x07)	160	3	NA	1	0	0	7	NA
BalrSwtch_D_Stat	Boundary Alert switch status for user. States of signal are Null(0x0), NotPressed(0x1) or Pressed(0x2)	163	2	NA	1	0	0	1	NA
DrStatDrv_B_Actl	Driver door open status. (Closed or Ajar)	165	1	NA	1	0	0	1	NA
DrStatPsngr_B_Actl	Passenger door open status. (Closed or Ajar)	166	1	NA	1	0	0	1	NA
DrStatRl_B_Actl	ear Left door open status. (Closed or Ajar	167	1	NA	1	0	0	1	NA
DrStatRr_B_Actl	Rear Right door open status. (Closed or Ajar)	168	1	NA	1	0	0	1	NA
WndwUpDrvFront_D_Falt	Driver Front Door Fault status. States of window fault signals are Null(0x0), No fault(0x1) or Fault(0x2)	169	2	NA	1	0	0	2	NA
WndwUpDrvRear_D_Falt	Driver Rear Door Fault status. States of window fault signals are Null(0x0), No fault(0x1) or Fault(0x2)	171	2	NA	1	0	0	2	NA
WndwUpPsngrFront_D_Falt	Passenger Rear Door Fault status. States of window fault signals are Null(0x0), No fault(0x1) or Fault(0x2)	173	2	NA	1	0	0	2	NA
BalrMde_D_Rq	Request from driver input to change BA mode. States of signal Null(0x00), Low(0x1), Med(0x2) or High(0x3)	175	2	NA	1	0	0	3	NA
CtaInnrX_D_Stat	0x0 = CTA Initializing 0x1 = CTA Standby State 0x2 = CTA Not Reporting State 0x3 = CTA Reporting State	177	2	NA	1	0	0	3	NA

VisionCo SPI Messages

VisionCo DASP Messages

The following table lists all Defined Serial Protocol Messages that shall be transmitted from the VisionCo and received by the CPYR agent. Implementation shall be according to the specifications given in the following subsections:

Data ID	Message Name	Modes	Periodic Rate (Hz)	Rate Tolerance (ms)	Description
0xA0	VISIONCO_BASE_DIAG_STATUS_MSG	APP / SPC / TAC	18	+/- 10	VisionCo4 base diagnostic status message.
0xA1	VISIONCO_APP_DIAG_STATUS_MSG	APP/ SPC	18	+/- 10	VisionCo4 application diagnostic status message.
0xA2	VISIONCO_DYNAMIC_CAL_MSG	APP/ SPC	18	+/- 10	VisionCo4 Service Point Calibration (SPC) and AutoFix output.
0xA3	VISIONCO_TAC_CAL_MSG	TAC	18	+/- 10	VisionCo4 Target Auto-Calibration (TAC) output.
0xE0	VISIONCO_VIS_ROAD_DATA_MSG	APP	18	+/- 10	VisionCo4 road detection data message.
0xE1	VISIONCO_VIS_LIGHT_SENSOR_DATA_MSG	APP	18	+/- 10	VisionCo4 AHBC function data message.
0xE2	VISIONCO_VIS_TRAFFIC_SIGNS_MSG	APP	18	+/- 10	Traffic signs as detected by vision subsystem.
0xE3	VISIONCO_VIS_AEB_INFO_MSG	APP	18	+/- 10	AEB decisions (for braking or warning) based on vision.
0xE4	VISIONCO_VIS_OBSTACLES_MSG	APP	18	+/- 10	Vision objects as detected by vision subsystem.
0xE5	VISIONCO_VIS_FAILSAFE_MSG	APP	18	+/- 10	Vision failsafes
0xE6	VISIONCO_VIS_SRP_MSG	APP	18	+/- 10	Suspension Road Preview
0xE7	VISIONCO_VIS_SAFETY_MSG	APP	18	+/- 10	SCFM and Challenge and Response

Note: Do not assign a Data ID to 0x5E.

All messages shall be transmitted periodically at the rate specified in attribute "Periodic Rate" with a tolerance as specified in "Rate Tolerance".

VISIONCO_APP_DIAG_STATUS_MSG (0xA1)

VISIONCO_APP_DIAG_STATUS_MSG(0xA1)-STNumber(41)-STVersion(1) **The DASP Data in the VisionCo App Diag Status Message shall be as shown below.**

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to $2^{32}-1$	ms
visionDiagnosticInfo	VisionCo_vision_diagnostic_info_t	n/a	This contains information about vision AGC control settings for day/night detection.	n/a	n/a

VISIONCO_DYNAMIC_CAL_MSG (0xA2)

103400, -VISIONCO_DYNAMIC_CAL_MSG(0xA2)-STNumber(42)-STVersion(1) **The VisionCo SPC and AutoFix Outputs Message shall be as shown below.**

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to 2^{32}-1	ms
dynamicCalParams	VisionCo_vision_calibration_dynamic_t	n/a	The SPC and AutoFix results with a 64 bit timestamp.	n/a	n/a

VISIONCO_TAC_CAL_MSG (0xA3)

VISIONCO_TAC_CAL_MSG(0xA3)-STNumber(43)-STVersion(1) **The VisionCo TAC Outputs Message shall be as shown below.**

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to 2^{32}-1	ms
tacCalParams	VisionCo_vision_calibration_static_t	n/a	The TAC output results with a 64 bit timestamp.	n/a	n/a

VISIONCO_VIS_ROAD_DATA_MSG (0xE0)

VISIONCO_VISROAD_DATA_MSG(0xE0)-STNumber(44)-STVersion(1) **The DASP Data in the VisionCo Road Data Message shall be as shown below.**

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to 2^{32}-1	ms
roadInfo	VisionCo_road_info_t	n/a	Information about the road including lane markers and road center with a 64 bit timestamp.	n/a	n/a

VISIONCO_VIS_LIGHT_SENSOR_DATA_MSG (0xE1)

VISIONCO_VIS_LIGHT_SENSOR_DATA_MSG (0xE1)- ST Number (45) - ST Version (1)

The DASP Data in the VisionCo Light Sensor Data Message shall be as shown below.

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to 2^{32}-1	ms
activeLightSensorInfo	visionq_active_light_sensor_info_t	n/a	The active light sensor information with a 64 bit timestamp.	n/a	n/a

VISIONCO_VIS_TRAFFIC_SIGNS_MSG (0xE2)

10VISIONCO_VIS_TRAFFIC_SIGNS_MSG (0xE2)- ST Number (46) - ST Version (1)

The DASP Data in the VisionCo Traffic Signs Message shall be as shown below.

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to 2^{32}-1	ms
tsrInfo	VisionCo_traffic_sign_info_t	n/a	Current traffic sign detections as received from vision subsystem with a 64 bit timestamp.	n/a	n/a

VISIONCO_VIS_AEB_INFO_MSG (0xE3)

VISIONCO_VIS_AEB_INFO_MSG(0xE3)-STNumber(47)-STVersion(1) **The DASP Data in the VisionCo Vision AEB Message shall be as shown below.**

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to 2^{32}-1	ms
visAEB	VisionCo_vision_AEB_info_t	n/a	Vision only AEB decisions (for warning or braking) with a 64 bit timestamp.	n/a	n/a

VISIONCO_VIS_OBSTACLES_MSG (0xE4)

VISIONCO_VIS_OBSTACLES_MSG(0xE4)-STNumber(48)-STVersion(1) **The DASP Data in the VisionCo Vision Obstacles Message shall be as shown below.**

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to 2^{32}-1	ms
visObjects	VisionCo_vision_obstacles_info_t	1	Vision only obstacles as received from Vision Processing subsystem with a 64 bit timestamp.	n/a	n/a

VISIONCO_VIS_FAILSAFE_MSG (0xE5)

VISIONCO_VIS_FAILSAFE_MSG(0xE5)-STNumber(49)-STVersion(1) **The DASP Data in the VisionCo Vision Failsafe Message shall be as shown below.**

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to 2^{32}-1	ms
visFailsafes	VisionCo_vision_failsafes_t	1	Vision Failsafes as defined in VisionCo_vision_failsafes_t	n/a	n/a

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VISIONCO_VIS_SRP_MSG (0xE6)

VISIONCO_VIS_SRP_MSG(0xE6)-STNumber(50)-STVersion(1) **The DASP Data in the VisionCo Vision SRP Message shall be as shown below.**

Member Name	Type	Array Size	Description	Range	Units
timestamp_ms	uint32_t	n/a	Timestamp to determine if message has been updated.	0 to 2^{32}-1	ms

VISIONCO_DIAGNOSTIC_MSG (0x41)

VISIONCO_DIAGNOSTIC_MSG (0x41)- ST Number (51) - ST Version (1)

Message 0x41 Fault Information from "Application Diagnostics Message.xlsx" document (previously named "BPP diagnostic protocol").

Name	Bit Offset	Byte Offset	Length	Type	Value	Resolution	Offset	Range From	Range To	Unit	Description
Reserved_1	0	0	8	uint		1	0	0	0	NA	
Application_Message_Version	8	1	8	uint	1	1	0	1	1	NA	
Main_State	16	2	8	uint	0x0 = UNKNOWN 0x1 = PENDING 0x2 = RUNNING_VISION 0x3 = BOOT 0x92 = PENDING_VISION 0x85 = PENDING_INIT_DV 0xAA = RUNNING_DV	1	0	0	255	Enum	Application main state.
Sub_State	24	3	8	uint		1	0	0	255	NA	Application sub-state.
VisionQ_Process_Index	32	4	32	uint		1	0	0	4.29E+09	NA	Frame number
VisionQ_Timestamp	64	8	32	uint		1	0	0	4.29E+09	millisec	The time when the image was grabbed. Used for synchronization with algorithmic

											output.
VisionCo_Current_Timestamp	96	12	32	uint		1	0	0	4.29E+09	millisec	The actual time read from the VisionCo. Used for synchronization with algorithmic output.
Application_Diagnostics_part_1	128	16	16	uint	**See description below.**	1	0	0	65535	bitwise	Application Diagnostics (first short)
Application_Diagnostics_part_2	144	18	16	uint	**See description below.**	1	0	0	65535	bitwise	Application Diagnostics (second short)
Fatal_Error	160	20	8	uint	0=APP_OK 1=APP_ERROR 10=APP_FS_ERROR 11=APP_CALIBRATION_ERROR 20=APP_INIT_FAILED 21=APP_INIT_CAMERA_INIT 50=APP_I2C_VIDEO_GRAB_FAILED 51=APP_I2C_CAMERA_SELF_RESET 52=APP_I2C_TIMEOUT_ERROR 70=APP_PATTERN_TEST 80=APP_CAM_PARAMS_CCFT_CRC_FAILED 81=PLL_COMPARISON_ERROR 90=PV_GENERAL_ERROR 91=PV_VERIFICATION_ERROR	1	0	0	91	Enum	Error that requires SW reset. VisionCo cannot continue to function. If happens continuously, needs to go to Service Point. Note - as long as this Enum is 0, there is no fatal error. In the event of any fatal errors, only message 0x41 will continue to be sent and no DASP messages will be sent.
Reserved_2	168	21	8	uint		1	0	0	0	NA	
Minor_Error	176	22	16	uint	0=BM_OK 1=BM_ERROR 5001=BM_EM_ERROR 5002=BM_EM_ERR_FAILED_LOAD_SETTING 5003=BM_EM_ERR_FAILED_LOAD_REGISTRY 5004=BM_EM_ERR_FAILED_INIT_REGISTRY 5005=BM_EM_ERR_FAILED_INIT 5006=BM_EM_ERR_FAILED_INIT_BB 5007=BM_EM_ERR_FAILED_INIT_BB_REG 5008=BM_EM_ERR_FAILED_OPEN_BLACKBOX 5009=BM_EM_ERR_FAILED_INIT_EP 5010=BM_EM_ERR_FAILED_POST_INIT_EP 5011=BM_EM_ERR_FAILED_INIT_CREATE_LOGGER 5012=BM_EM_ERR_FAILED_INIT_IL 5013=BM_EM_ERR_FAILED_CHECK_REG_VERSIONS	1	0	0	5013	Enum	BPP Minor error. problem exists in one of the elements (e.g - EDR problem). In this case, VisionCo main processing loop continue running as usual. NOTE: For now only mapped to EDR errors.
Temperature_MIPS	192	24	16	sint		1	0	-40	125	deg	MIPS temperature
Temperature_VMP	208	26	16	sint		1	0	-40	125	deg	VMP temperature
Temperature_DDR	224	28	8	uint	0 – temperature is 85°C; default resh. 1 – SDRAM low temperature operating limit exceeded. 2 – SDRAM high temperature operating limit exceeded. 3 – temperature is much greater than 85°C; 0.25x resh, with derating. 4 – temperature is much less than 85°C; default resh. 5 – temperature is less than 85°C; default resh. 6 – temperature is greater than 85°C; 0.5x resh. 7 – temperature is greater than 85°C; 0.25x resh, no derating	1	0	0	3	Enum	DDR Temperature
Reserved_3	232	29	8	uint		1	0	0	0	NA	
CFG_status	240	30	16	uint		1	0	0	65535	bitwise	TBD
SPI_status	256	32	16	uint		1	0	0	65535	bitwise	bit 0 - CRC SPI error bit 1 - CRC SPI continuos error (more than 3 times in a row) bit 2 - sequence SPI error bit 3 - sequence SPI continuos error (more than 3 times in a row) bits 4,5 - reserved bits 6-13 - SPI health, not functional yet
Reserved_4	272	34	8	uint		1	0	0	0	NA	
	280	35	8	uint	bit 0 - camera1 information valid bit 1 - camera2 information valid bit 2 - camera3 information valid bit 3 - camera4 information valid bit 4 - camera5 information valid bit 5 - camera6 information valid bit 6 - camera7 information valid bit 7 - camera8 information valid	1	0	0	7	bitwise	For systems with multiple cameras, the bits indicate which cameras are valid.
Camera1_temperature	288	36	8	sint		1	0	-40	125	deg	Camera1 temperature
Camera2_temperature	296	37	8	sint		1	0	-40	125	deg	Camera2 temperature
Camera3_temperature	304	38	8	sint		1	0	-40	125	deg	Camera3 temperature
Camera4_temperature	312	39	8	sint		1	0	-40	125	deg	Camera4 temperature
Camera5_temperature	320	40	8	sint		1	0	-40	125	deg	Camera5 temperature
Camera6_temperature	328	41	8	sint		1	0	-40	125	deg	Camera6 temperature
Camera7_temperature	336	42	8	sint		1	0	-40	125	deg	Camera7 temperature
Camera8_temperature	344	43	8	sint		1	0	-40	125	deg	Camera8 temperature

Camera1_VideoErrorRange	352	44	8	uint		1	0	0	255	bitwise	VideoErrorRange – This is **not implemented** yet! 4 MSB bits are how many video errors were in this frame. 4 LSB bits are when was the first error, it will be the last 4 bits of the frame index and not all the number.
Camera2_VideoErrorRange	360	45	8	uint		1	0	0	255	bitwise	Not functional yet
Camera3_VideoErrorRange	368	46	8	uint		1	0	0	255	bitwise	Not functional yet
Camera4_VideoErrorRange	376	47	8	uint		1	0	0	255	bitwise	Not functional yet
Camera5_VideoErrorRange	384	48	8	uint		1	0	0	255	bitwise	Not functional yet
Camera6_VideoErrorRange	392	49	8	uint		1	0	0	255	bitwise	Not functional yet
Camera8_VideoErrorRange	408	51	8	uint		1	0	0	255	bitwise	Not functional yet
Camera1_VideoErrorFlags	416	52	64	uint		1	0	0	1.84E+19	bitwise	NO_VIDEO_ERROR = 0x0, VDI_VIDEO_ERROR = 0x1, I2C_NOT_ON_TIME_VIDEO_ERROR = 0x2, LOGICAL_VIDEO_ERROR = 0x4, INCONSISTENT_FRAMES_VIDEO_ERROR = 0x8, VERIFICATION_FAILURE_VIDEO_ERROR = 0x10, INVALID_VIDEO_HEADER = 0x20, ERROR_REG_FLAG = 0x40, REG_WRITE_HIL_FAILURE_ERROR = 0x80, ZERO_HISTOGRAM_ERROR = 0x100, SUM_NOT_MATCH_HISTOGRAM_ERROR = 0x200, VDI_INTERNAL_ERROR = 0x400, VDI_ERR_NO_BUFFERS = 0x800, VDI_ERR_BUFFER_INVALID_FORMAT = 0x1000, VDI_ERR_TIMEOUT = 0x2000, VDI_ERR_FIFO_OVERFLOW = 0x4000, VDI_ERR_FIFO_UNDERFLOW = 0x8000, VDI_ERR_PARITY_FIFO = 0x10000, VDI_ERR_PARITY_WEIGHTS = 0x20000, VDI_ERR_PARITY_GAMMA = 0x40000, VDI_ERR_PARITY_HIST = 0x80000, VDI_ERR_START_AFTER_START = 0x100000, VDI_ERR_CORE_ID = 0x200000, FRAME_COUNT_READ_FAIL = 0x400000, HISTOGRAMS_MISMATCH = 0x800000, CCFT_DECISIONS_MISMATCH = 0x1000000, HEADER_VERIFICATION_ERROR = 0x2000000, VDI_ERR_SHUTDOWN = 0x4000000, SHOT_OUT_OF_SYNC = 0x8000000
Camera2_VideoErrorFlags	480	60	64	uint		1	0	0	1.84E+19	bitwise	TBD. Error Flags 2-8 are for systems with multiple cameras.
Camera3_VideoErrorFlags	544	68	64	uint		1	0	0	1.84E+19	bitwise	TBD
Camera4_VideoErrorFlags	608	76	64	uint		1	0	0	1.84E+19	bitwise	TBD
Camera5_VideoErrorFlags	672	84	64	uint		1	0	0	1.84E+19	bitwise	TBD
Camera6_VideoErrorFlags	736	92	64	uint		1	0	0	1.84E+19	bitwise	TBD
Camera7_VideoErrorFlags	800	100	64	uint		1	0	0	1.84E+19	bitwise	TBD
Camera8_VideoErrorFlags	864	108	64	uint		1	0	0	1.84E+19	bitwise	TBD
Total	928	116									

CPYR AGENT_VEH_STATE_MSG (0x20)

CPYR AGENT_VEH_STATE_MSG(0x20)-STNumber(53)-STVersion(1) **The DASP Data in the CPYR agent Vehicle State Message shall be as shown below.**

Member Name	Type	Array Size	Description	Range	Units
vehStateInfo	CPYR agent_vehicle_state_t	n/a	This contains information about the Ego vehicle state with 64 bit timestamp.	n/a	n/a
CPYR agent_sync_timestamp_ms	uint32_t	n/a	The time when the CPYR agent received the lastest VisionQ4 Clock Sync pulse.	0 to 2^{32}-1	ms
global_real_time	unt32_t	n/a	Global real time from the vehicle bus data to be populated and sent every SPI Transmission. This will be used in the EDR in the VisionQ. (This is not for	0 to 2^{32}-1	s

			is the date/time data received from the vehicle bus. Date/time data is stored with the EDR.)		
SPI_Rolling_Count	uint16_t	n/a	Rolling counter to be incremented by one on every message transmission	0 to 2^{16}-1	n/a
WheelSlipEvent	uint8_t	n/a	a Current wheel slip event: 0 = No Slip 1 = Brake Slip Control Active 2 = Traction Slip Control Active	0 to 2	n/a

VISIONCO_BOOT_STATUS_MSG (0x10)

VISIONCO_BOOT_STATUS_MSG(0x10)-STNumber52)-STVersion(1) VISIONCO_BOOT_STATUS_MSG (0x10)

Message Field	Byte	Start bit	Array Size(Bytes)	Description
Zero byte	0	0	1	Constant zero value
Protocol Version	1	0	1	The version of the message structure and data fields. Current protocol version is 2.
Reserved0	2	0	2	Reserved fields value is 0
Test results	4	0	2	It is a bit map where each bit represents single test result: 0 – OK, 1 – Failure. The bit values are as follows- Bit 0 = VisionCo Over Temperature Bit 1 = BIST Bit 2 = PLL init finished Bit 3 = Parity Check Bit 4 = Flash CRC of Boot Bit 5 = DDR init finished Bit 6 = DDR Temperature Bit 7 = DDR Test Bit 8 = Flash CRC of Application Bit 9 = Flash CRC of FFS Bit 10 = Unsupported VisionCo version Bit 11 = Authentication failure of the secure SW signature. Bit 12-15 = Reserved
Board project code ('s internal code)	6	0	1	Internal ' s code.
Board revision (based on GPIO)	7	0	1	Based on GPIO.
Boot version	8	0	4	4-bytes field: byte 0 (MSB) – first number of version; byte 1 – second number of version; byte 2 – third number of version; byte 3 – reserved; Example: Boot version 7.1.23 will be represented as the following byte sequence: byte#8 = 0x07, byte#9 = 0x01, byte#10 = 0x17, byte#11 = 0x00
Application version	12	0	4	
VisionCo Type	16	0	4	-Not Implemented
VisionCo ID	20	0	4	-Serial number of the specific VisionCo4 chip.
DDR manufacturer code	24	0	1	LPDDR4 Manufacturer ID according to JEDEC standard No. 209. For example: 0x08 – Winbond; 0xFF – Micron.
DDR size in MB	25	0	3	-

Flash manufacturer code	28	0	1	Each flash memory device has a manufacturer ID (which is assigned by JEDEC). For example: 0x01 – Spansion; 0x20 – Micron; Numonyx – 0x89.
Flash size in MB	29	0	3	-
VisionCo Temperature Sensor 1	32	0	1	The temperature measurement provided by two VisionCo4 internal sensors in degree Celsius.
VisionCo Temperature Sensor 2	33	0	1	
LPDDR4 Temperature and resh Status	34	0	1	LPDDR4 Temperature and resh Status 0 – temperature is 85°C; default resh. 1 – SDRAM low temperature operating limit exceeded. 2 – SDRAM high temperature operating limit exceeded. 3 – temperature is much greater than 85°C; 0.25x resh, with derating. 4 – temperature is much less than 85°C; default resh. 5 – temperature is less than 85°C; default resh. 6 – temperature is greater than 85°C; 0.5x resh. 7 – temperature is greater than 85°C; 0.25x resh, no derating
Reserved1	35	0	1	-
Flash 2 manufacturer code	36	0	1	-
Flash 2 size in MB	37	0	3	-
Boot stage completed (The sequential order is: 1. Boot Authentication, 2. BIST, 3.DRAM initialization, 4. DRAM test, 5. Application loading	40	0	1	0 – reserved (illegal value). 1 – BIST results. 2 – Boot authentication result. 3 – DRAM initialization & training result. 4 – DRAM test result. 5 – Application authentication and loading result.
Failure information	41	0	1	Reports only 0’s until TBD’s are defined. The failure information format is dependent on the boot stage and the failure type. 1. BIST result. Reports failures of the following types: - temperature issue; - VDI memory parity error; - BIST fault; Format: TBD 2. Boot authentication result. pass / fail 3. DRAM initialization & training result. - Memory controller initialization issue; - Memory PHY initialization issue; - DRAM training failure; Format: TBD 4. DRAM test result. - temperature issue; - DRAM test failure; Format: TBD 5. Application authentication and loading result. - authentication pass / fail; - loading failure; Format: TBD
Padding (0x00)	42	0	79	Padding by zeroes up to 128 bytes

1.1.1 Init_Params_Vision (0x92)

Init_Params_Vision (0x92) - ST Number 54) - ST Version (1)
Init_Params_Vision (0x92)

Feature Stream Messages

Algorunner Signals needed:

1. Bus: BT_TPP_SPP
   - Stream Number (60) ST Version (1) Length 48 Bytes
2. Bus: BT_TPP_TAD
   - Stream Number (61) ST Version (1) Length 460 Bytes
3. Bus: BT_TPP_AutoHitch
   - Stream Number (62) ST Version (1) Length 64 Bytes
4. Bus: BT_TPP_DriveHistory
   - Stream Number (63) ST Version (1) Length 40 Bytes
5. Bus: BT_TPP_Localization_Results
   - Stream Number (64) ST Version (1) - Length 32 Bytes